

O' Level
Chemistry
Integrated With Technology

LEARNER'S BOOK



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Integrated With Technology

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Contents

Preface	iii
Acknowledgement	iv
Introduction	v
Chapter 1: Introduction to Chemistry	1
Concept of chemistry	1
The Importance of Learning Chemistry	1
Laboratory Rules and Regulations	3
Common Apparatus in the Laboratory	4
Chapter 2: Laboratory Heating Apparatus	9
Examples of Heating Apparatus	9
Types of Flames	13
Chapter 3: Matter	17
States of Matter	17
Chapter 4: Mixtures and Compounds	27
Preparation of Mixtures	27
Chapter 5: The Atmosphere and Air	35
The Active Part of Air	37
Combustion of Elements in Oxygen	39
Chapter 6: Rusting of Iron	45
Factors Affecting Rusting	46
Prevention of Rusting	50
Chapter 7: Acids, Bases, Salts and Indicators	53
Reactions of Acids and Bases	53
Indicators	58
Chapter 8: Water and Life	63
The Water Cycle	64

Water Purification	68
Chapter 9: Hydrogen and Some of Its Sources	73
Reactions of Hydrogen with other Substances	74
Chapter 10: Atomic Structure	77
Electrical Charges	77
Chapter 11 : Structure and Bonding	85
Element Molecules and Compound Molecules	85
Chapter 12: Carbon and Its Compounds	95
Forms of Carbon, their Structure and Properties	95
Chemical Properties of Carbon	101
Carbonates and Hydrogen Carbonates	105
The Science Behind Bread Making.....	108
Chapter 13: Organic Chemistry	115
Soaps and Detergents	118
Liquid Soap	123
Chapter 14: Mole concept	129
Chapter 15: Energy Changes in Chemical Reactions	137
Chemical Reactions	140
Chapter 16: Electrochemistry	145
Chapter 17: Nitrogen and some of its Compounds	155
Nitrogen Gas	155
Ammonia Gas	158
Chapter 18: Sulphur and Some of Its Compounds	167
Sulphur dioxide	169
Chapter 19: Chlorine and its Compounds	173
References	179

Preface

The major reason for undertaking this new approach to teaching-learning Sciences and Mathematics at O level is to improve the academic achievements and interest of learners.

The idea of integrating technology into sciences and mathematics at lower secondary education was conceived after realising that Science and Mathematics teaching at this level hardly generated interest in the learners. In addition, these learners were lacking in practical skills and rarely applied science and mathematics to everyday life situations.

The National Curriculum Development Centre (NCDC) has developed this new approach of teaching Science and Mathematics by including practical activities that enable learners to come up with products or be able to offer services to the community leading to national sustainable development by generating their own income.

Acknowledgment

This book was developed and prepared in a series of writing and discussion workshops organised by the National Curriculum Development (NCDC) between 2007 and 2011.

I wish to acknowledge the contribution of the subject panel members, teachers and learners from various schools who participated at trial activity phase and other education stakeholders that provided input and direction of this material.

I am grateful to all those who worked behind the scenes for the commitment in ensuring the work is done and feedback from the field is incorporated. I also thank the African Development Bank (ADB) Project Phase-III for the financial support.

Last but not least, I wish to recognise agencies, companies and websites for the reference materials and pictures used in this book.



Connie Kateeba
Director
National Curriculum Development Centre (NCDC)

Introduction

This book has been written to help you to develop interest in Chemistry so that you can perform better in it. The concepts, ideas and activities in it are related to real life situations. As you go through the book, you will come across many interesting activities which will make learning science interesting.

Rationale

Throughout the book, an effort has been made to integrate Technology into Chemistry. This simply means that activities which will enable you to produce a product or products which are useful to you or to the community around you have been included in the book. In addition, as you carry on with the activities, you will develop new skills and a new way of learning and practising science.

How to use this book

Content in the book is in the form of learning points and activities. Each chapter has activities and each activity has the following features:

- a) **Title of activity**
- b) **What you need.** This is a list of materials and equipment you will need in order to carry out the activity. The list consists of ordinary objects and materials around your immediate environment. Some of these will have to be made by you.
- c) **What to do.** This is the procedure you will follow, step-by-step, in order to achieve the objectives of the activity. In some of the activities you will work alone, while in others you will work in groups or with your teacher. There are a few activities where you will need the assistance of a craftsman or other people in your community who are more experienced in the activity. In some activities, hints and precautions have been provided to enable you accomplish it successfully. Some activities are normal

laboratory experiments while others will take more time than is normally allocated to classroom/laboratory lessons. For the latter, you will have to plan how to accomplish them outside class time.

d) Learning points. This is the theory behind the activities you will be carrying out in the relevant topics. The theory has been written in simple language which will enable you to understand the content. The content has been accompanied with relevant pictures or diagrams which will enable you to understand the concepts clearly. Note that theory has only been provided on topics where activities in the book have been derived.

e) Follow-up activity

Follow-up activities are those additional activities that you are required to carry in addition to the main activity.

Chapter 1: Introduction to Chemistry

Inquiry question

- *What is chemistry?*

Concept of chemistry

In primary school, we learnt science as an integrated subject. At secondary level, science is divided into three main branches. These are Physics, Chemistry and Biology. Physics is the study of matter in relation to energy. We learnt some of this at primary level. Biology is the study of living things. We also learnt many Biology topics at primary school. Knowledge from these subjects can be applied in other subjects like agriculture and home economics.

Chemistry involves the study of the structure and composition of substances (matter) and the way they behave under different conditions. For example, what do you think happens during:

- a) the souring of milk?
- b) burning of wood, paper, kerosene and a candle?
- c) the decay of food?
- d) rusting of iron?

All the above can be explained by chemistry.

The Importance of Learning Chemistry

Look around you. Every material thing you can see or think of; be it furniture, cloth, paper, building materials, food and many more, are a result of one chemical process or another. Chemistry has contributed a lot to modern technology. Through laboratory experiments, chemists have discovered new materials and the chemical processes to produce them in industries. As a result, our homes are

made more comfortable today by the use of a variety of materials that were not known when our great grandparents were young. Chemistry has contributed very much to our understanding of material things which surround us. For example, cement, plastics, fertilizers and soap among others. The importance of chemistry in our society is almost endless. In order to see how chemistry is applied to our everyday life, see Figure. 1.1.

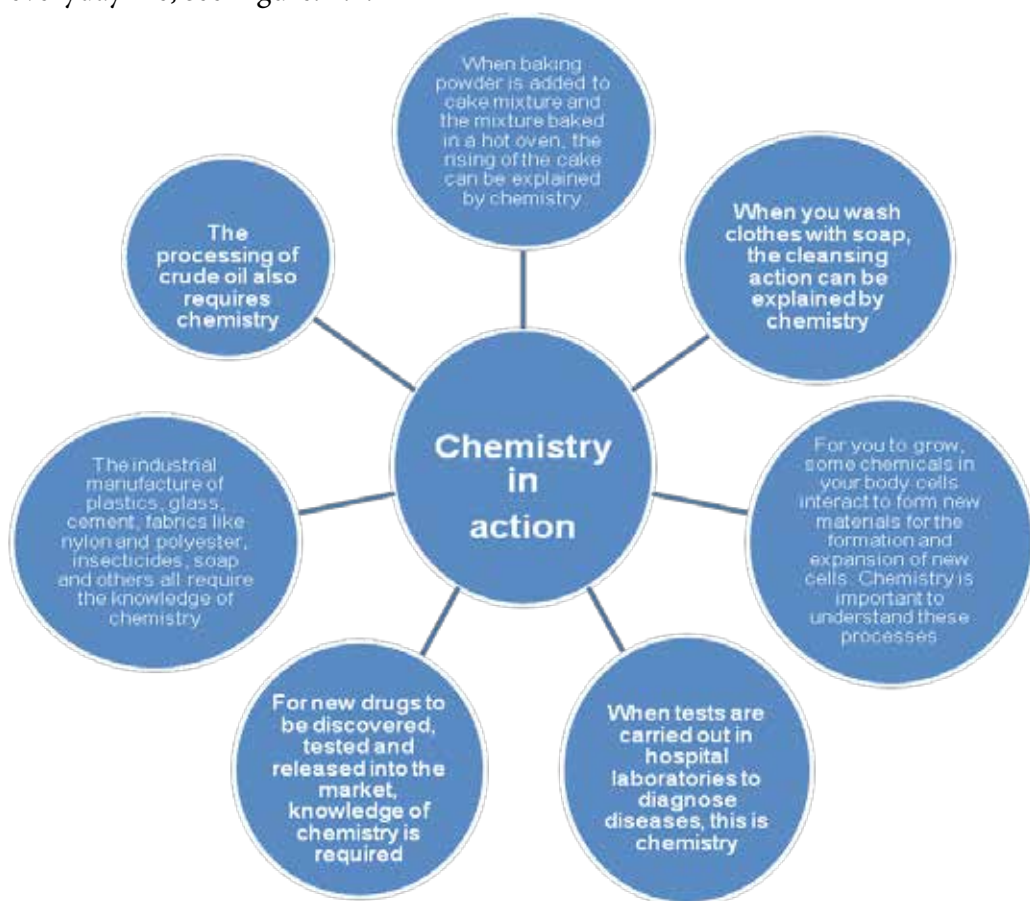


Figure 1.1: Showing Chemistry in action

Can you think of any other activity at home that involves chemistry?

From Figure.1.1, we can see that chemistry can offer a lot in our day-to-day lives. If you study Chemistry, there are many careers or professions that you can pursue

after leaving school, college or university and these include:

<i>Pharmacist</i>	<i>Chemical Engineer</i>
<i>Medical Doctor</i>	<i>Veterinary Officer</i>
<i>Chemistry Teacher</i>	<i>Government Chemist</i>
<i>Beautician</i>	<i>Analytical Chemist</i>
<i>Dentist</i>	<i>Forensic Chemist</i>
<i>Food Technologist</i>	<i>Farmer</i>

What other careers require knowledge of chemistry?

Laboratory Rules and Regulations

A laboratory is a room equipped with the necessary facilities for carrying out practical activities in science. In practical activities, we try to find out what happens when we do certain things. These activities are called experiments.

Chemistry experiments are done in order to learn about changes in substances. A laboratory is costly to build and maintain. Therefore, proper management and care of laboratory equipment is essential. It requires the cooperation of everyone who uses the laboratory.

Many chemicals are kept in the laboratory. Whereas some are harmless, others are poisonous. Some burn easily and can catch fire. Yet others have harmful effects on human beings or on the environment. All laboratory chemicals should therefore be handled carefully.

You should strictly follow the following rules and regulations used in the laboratory:

- Do not enter the laboratory unless a teacher gives you permission.
- Never run or play in the laboratory.
- Never eat or drink in the laboratory.

- Never carry out experiments without your teacher's approval.
- Do not throw solids in the sink.
- Before starting an experiment, put all bags, jackets and stools out of the way.
- Neckties, long-sleeved shirts, loose hair and loose clothing must all be tightly tucked away.
- Make sure you understand all instructions before you start any practical work.
- Always use small amounts of chemicals.
- When doing any practical work, ensure safety of yourself and other people. For example, when heating a test-tube, point the mouth away from people.
- Report all accidents and breakages immediately to the person in charge of the laboratory.
- Take your time and if you are not sure of what to do, ask!
- Clean and clear up all apparatus and chemicals and leave the laboratory clean and tidy. Dispose of all leftover chemicals as advised by the teacher or attendant.
- After use, immediately close water and gas taps and switch off power.

What other rules do you suggest for your safety in the laboratory?

Common Apparatus in the Laboratory

You must be aware that if the saucepans and other equipment used in cooking were not there, cooking would be very difficult. Similarly, the chemists have equipment in the laboratory which they can use to do their work. These equipment are called pieces of **apparatus**. They may be made of metal, wood, plastic or glass. However, most of the equipment used in the laboratory is made of glass. It is important to know what apparatus to choose and use when performing any practical work. You need to be aware of the purpose and accuracy of the common laboratory equipment.

Activity 1.1: Drawing and naming the various laboratory apparatus and giving their uses

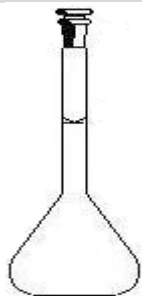
By the end this activity you should be able to:

1. identify different types of laboratory apparatus.
2. state the use of each apparatus.
3. draw the apparatus.

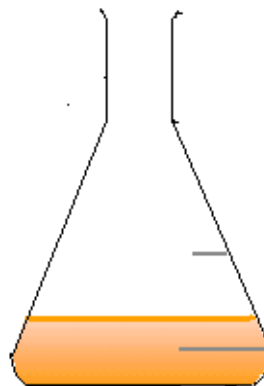
Draw each of the apparatus below and give its use.

Table 1.1: Some common apparatus

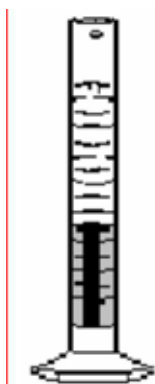
Apparatus



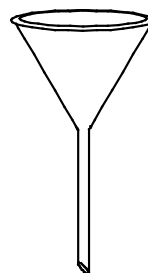
Volumetric flask



Conical flask

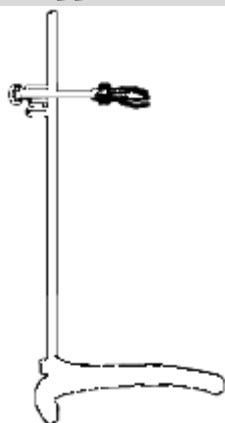


Measuring cylinder



Filter funnel

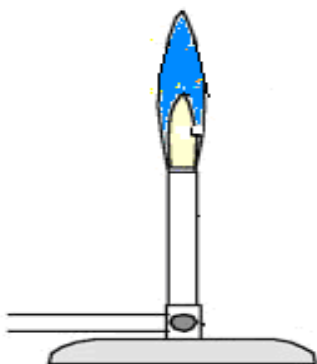
Apparatus



Retort stand

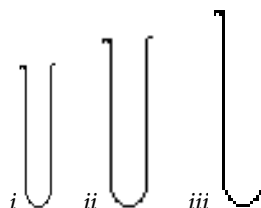


Liebig condenser



Bunsen burner

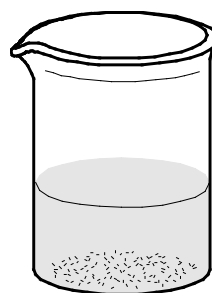
Test tubes



(i) Ignition tube (ii) Test-tube (iii) Boiling tube



Separating funnel

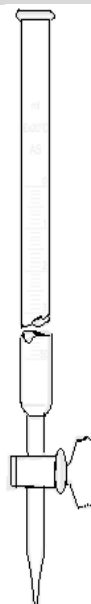


Beaker

Apparatus



Pipette



Burette



Gas jar



Bee-hive shelf



Evaporating basin



Mortar and pestle

1. What other apparatus can you identify in the laboratory?
2. Can you suggest why most of the apparatus are made of glass?

Chapter 2: Laboratory Heating Apparatus

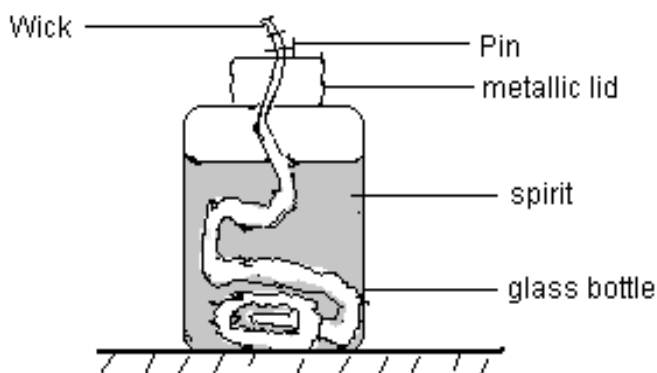
Inquiry question

- *What would you consider when choosing equipment for heating and lighting?*

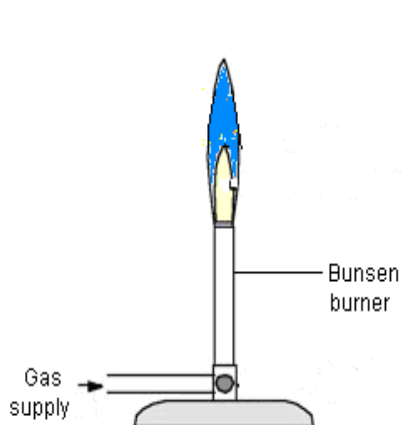
A heating apparatus is a tool that is used to produce heat energy. Heating apparatus are important tools in chemistry laboratories and in real life situations. There are several types of heating apparatus, many of which use fuel to produce the heat. Scientists are constantly reinventing apparatus in order to promote the efficient use of fuel so as to produce heat and light energy.

Examples of Heating Apparatus

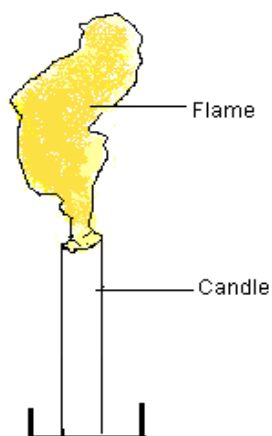
There are different heating apparatus. Some of them are manufactured while others can be made using locally available materials. Below are some examples:



Hand-made spirit burner



Bunsen burner



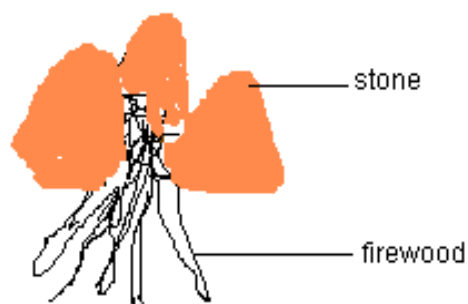
Candle



Charcoal stove



Kerosene stove



three stone stove

Figure 2.1: Different heating apparatus

Activity 2.1: Making a spirit burner

By the end of this activity, you should be able to:

1. make a spirit burner.
2. use the spirit burner for heating.

What you need

- a wick (ply cotton thread)
- glass bottle with a metallic lid
- a nail (size 4")
- hammer
- pin
- spirit (crude waragi or ethanol)
- wooden block

What to do

1. Remove the metallic lid of the bottle.
2. Place the lid on the wooden block and hammer the nail into the centre of the lid as shown in Fig. 2.2.

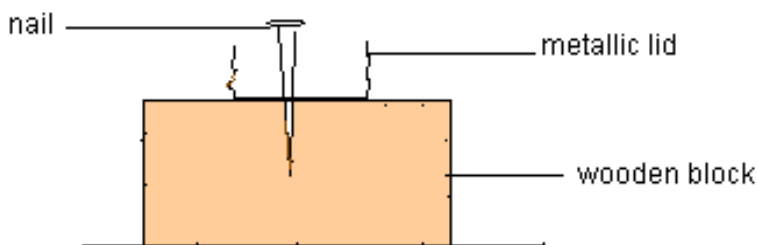


Figure 2.2: Showing one stage in the process of spirit burner making

3. Using the “forked” end of the hammer, carefully remove the nail from the metallic lid.
4. Pass the wick through the hole in the lid. What do you think is the use of the wick? Why is dry ply cotton used instead of nylon?
5. Push the pin through the wick on the upper side of the lid as shown in Figure 2.3. What do you think is the use of the pin?
6. Pour the spirit into the glass bottle. (**Caution! DO NOT** fill the glass bottle to the brim with spirit. Always leave some space between the lid and the level of the spirit in the bottle - why? **Spirit is inflammable.** Ensure that it does not come into contact with a naked flame except through the wick).
7. What is the use of the spirit? Can you name one other substance that can be used instead of spirit?
8. Put the lid back onto the bottle. Your burner should be similar to that in Figure 2.3.

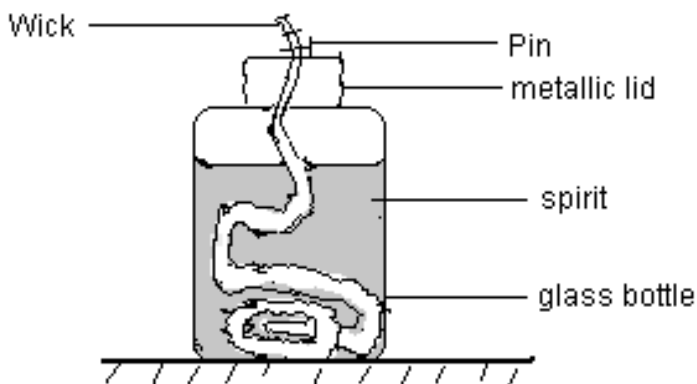


Figure 2.3: Spirit burner

How can you make the spirit burner in Figure 2.3 more efficient as a heating apparatus?

Follow-up activity

Visit a place in your community where heating apparatus are locally manufactured. Write a detailed report describing how the given apparatus are manufactured.

Types of Flames

A flame is burning gas(es) that produces heat and light. Flames are categorised as either non-luminous or luminous. Each type of flame has unique characteristics that determine its uses. Each of the various heating apparatus produces a type of flame depending on the amount of air and the type of fuel used. The luminous flame is mainly used for lighting while the non-luminous flame is mainly used for heating.

Activity 2.2a: Investigating the properties of a luminous flame and a non-luminous flame

By the end of this activity, you should be able to identify:

1. a luminous flame and non luminous flame.
2. the type of flame that does not produce soot.

What you need

- Bunsen burner
- a candle
- spirit burner
- a pair of tongs
- pipe clay triangle
- tripod stand
- 2 glass beakers(250mls)
- broken porcelain
- matchbox and match sticks
- water
- measuring cylinder(100mls)
- timer

What to do

1. Pick the candle which your teacher has provided.

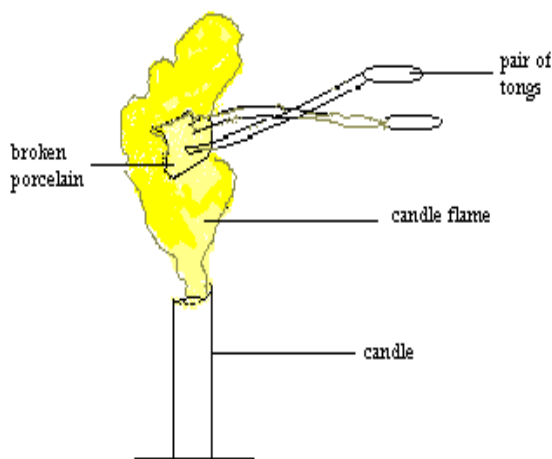


Figure 2.4: Flame of a burning candle

2. Using a match stick, light the candle and carefully observe its flame. Draw a diagram showing the candle flame and clearly indicate the different zones of the flame.

3. Use the pair of tongs to hold a piece of porcelain in the candle flame. Record what you observe. Then blow out the candle flame.
4. Repeat step 1-3, but using a spirit burner. Record your observations. How many zones of the flame produced do you see?

Activity 2.2b: Investigating Bunsen burner flames

By the end of this activity, you should be able to:

1. differentiate the types of flames produced by the Bunsen burner when the air holes are closed and when they are open.
2. operate a Bunsen burner.
3. identify which of the two flames is hotter.

What to do

1. Connect the Bunsen burner to the gas and then light it with the air holes closed. Carefully observe the flame. Draw a diagram of the flame and clearly indicate the different zones.
2. Carefully open the air holes of the lit Bunsen burner. Observe what happens to the flame. How many zones of the flame do you see? Draw the flame and clearly indicate the different zones.
3. Measure 100 ml of water using the measuring cylinder and transfer this volume into a beaker as shown in Figure 2.5. Close the air holes and measure the duration it takes to boil the water using the flame obtained.

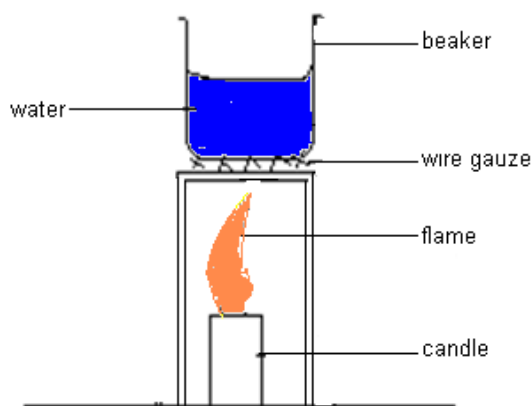


Figure 2.5: Showing water boiling using a Bunsen flame

- Repeat step 3, but this time use the flame obtained when the air holes of the Bunsen burner are open.
- Considering steps 3 and 4 in the procedure, which of the flames took a longer time to boil the water? What did you observe on the bottom surface of the beakers after carrying out steps 3 and 4 in the procedure?
- Can you now list the differences between the flames obtained in activities

2.2a and 2.2b in the procedure as in the following table?

<i>Source of heat</i>	<i>Flame zones</i>	<i>Effect on porcelain/ beaker</i>	<i>Time to boil 100 mL water</i>
<i>Candle</i>			
<i>Spirit burner</i>			
<i>Bunsen burner with air hole closed</i>			
<i>Bunsen burner with air hole open</i>			

What happens when different types of flames are used for heating?

Follow-up activity

Identify the types of fuels used for cooking and lighting in your community and the types of flames they produce. Record the advantages and disadvantages of using each.

Chapter 3: Matter

Inquiry question

- ***What is matter?***

Matter is anything that occupies space and has got mass. It can neither be created nor destroyed in any chemical reaction. Matter is made up of tiny particles. Matter exists in **three** different states. These states are: solids, liquids and gases. Matter can be changed from one state to another. This interchange of states of matter is applied in chemistry for preparing different products, such as ice, cheese, ice cream, distilled water, candles, etc. In this chapter, the knowledge of changes in the states of matter shall be used to produce such products.

States of Matter

Activity 3.1: Investigating the properties of the states of matter

By the end of this activity, you should be able to;

1. identify the different states of matter.
2. list the properties of each state of matter.

What you need

- plastic beaker
- evaporating dish
- wire gauze
- tripod stand
- Bunsen burner
- ice/candle wax
- weighing scale

What to do

1. Weigh the evaporating dish and record its weight.
2. Place about 5g of ice in the evaporating dish. Record the weight of both the evaporating dish and the ice.
3. Arrange the apparatus as shown in Figure 3.1 and gently heat the evaporating dish. Stop heating when all the ice has **just** melted (turned to liquid).

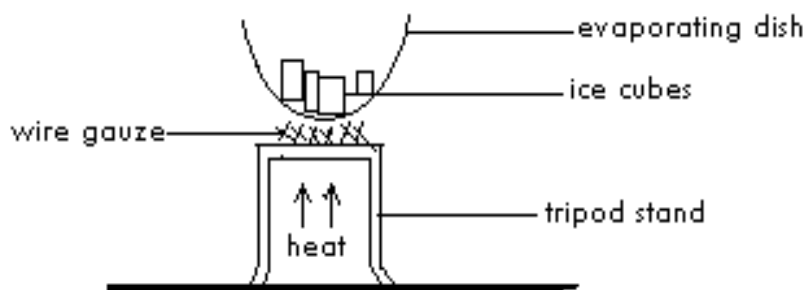


Figure 3.1: Showing the set up of the experiment

4. Allow the evaporating dish to cool. Weigh and record the weight of the evaporating dish and the water in it.
5. Gently heat the evaporating dish containing the water until the water level has reduced significantly. Weigh the evaporating dish and its contents. Record all the observations. Explain why there is change in mass of the water.

Caution

Handle the hot heating apparatus with care!

Learning points

Matter is physical substance that occupies space, has mass, is composed of atoms—or, in the case of subatomic particles, is part of an atom—and is convertible to energy. On Earth, matter appears in three clearly defined forms—solid, liquid, and gas—whose varying structural characteristics are a function of the speeds at

which its molecules move in relation to one another. A single substance may exist in any of the three phases: liquid water, for instance, can be heated to become steam, a vapour; or, when sufficient heat is removed from it, it becomes ice, a solid. These are merely physical changes, which do not affect the basic composition of the substance itself: it is still water. Matter, however, can and does undergo chemical changes, which (as with the various states or phases of matter) are an outcome of activity at the atomic and molecular level.

Knowledge of the conditions for the change of state can be applied in different technologies.

Follow-up activity

Identify examples in real life situations that are a result of change of given substances from one state of matter to another.

Activity 3.2a: Making a candle

By the end of this activity, you should be able to make a candle out of candle wax

What you need

- refined paraffin wax
- stearic acid
- candle mould
- dyes and mosquito repellent
- water bath
- saucepan/beaker
- ply cotton thread

What to do

1. Put 1kg of wax in the saucepan and melt it over a water bath as in Figure 3.2.
2. Add 40% (40 g in 100 mL of water) of stearic acid into the melted wax and stir. What is the use of the stearic acid?
3. Once the acid has dissolved, mosquito repellent or dye may be added if one desires. Can you explain how you think the candle would work as a mosquito repellent if you added mosquito repellent to the melted wax?

Caution: Handle the melted wax and stearic acid with caution. They can be harmful to your skin)

4. Suspend the cotton thread into the centre of the mould and carefully pour the hot wax into the mould as in Figure 3.2.

Note: Don't fill the candle mould with wax to the brim.

5. When the wax has solidified, open the mould and remove the candle.

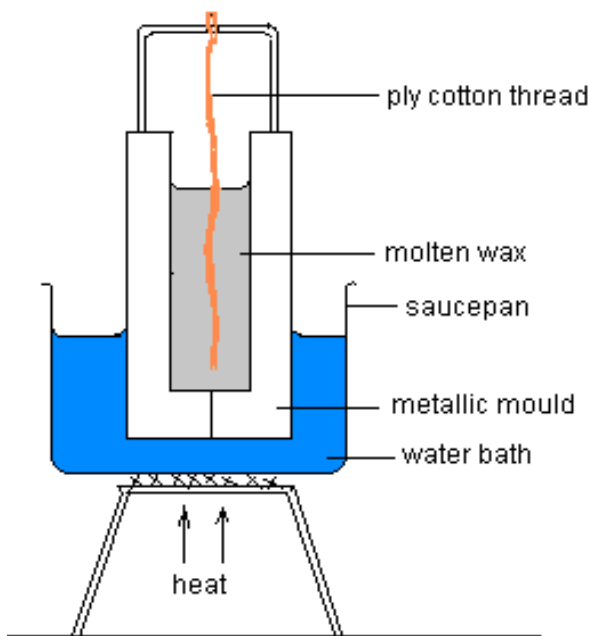


Figure 3.2: Making a candle

Follow-up activity

- Candle making can be a money generating activity. Try to calculate how many candles can be produced from 1 kg of wax so as to determine the selling price in order to make profit.
- Explore your locality to identify other types of candles you can make.
- Make another type of mould.

Activity 3.2b: Making beeswax candles

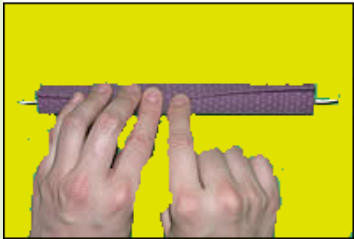
By the end of this activity, you should be able to make a candle from beeswax.

What you need

- beeswax
- scissors
- piece of wick
- work surface

What to do

Step	Product
1. Fold one sheet of the beeswax in half and press along the fold with your hands to flatten it. This makes the edge easier to follow when you are cutting it. Using scissors cut the sheet of beeswax in half.	
2. Lay one of the cut pieces of the beeswax sheet on your clean work surface. Lay the piece of wick on the edge of the piece of beeswax directly in front of you. At least a half an inch of the piece of wick should extend past each side of the piece of beeswax.	
3. Slowly and carefully start to roll the edge of the piece of beeswax closest to you over the wick.	
4. Tuck the edge of the piece of beeswax in around the wick. You want to seal it in so when you continue rolling your candle it will roll up neatly and evenly.	
5. Roll up your candle slowly and carefully with your fingers pushing the beeswax with both hands exerting even pressure at the same time. While you are rolling, keep an eye on your edges to make sure you are rolling straight so your candle will be straight. You want to roll a nice straight, tight candle so it will look nice and burn well.	

Step	Product
6. Continue rolling at the same pace you have been until you are about two inches from the end then slow down. You want to make sure the end of your candle is rolled as tight as you did the beginning and that the edges are still straight while you are rolling up the last of your candle.	
7. When you have the end of the piece of beeswax rolled up around to finish off your candle, you need to seal the edge to the roll so it won't come undone. Once again you can use your body heat to warm up the wax. Starting in the middle of the roll of beeswax, press your fingers gently along the edge of the piece of beeswax. You only need to do this lightly and briefly. You don't want to push down on the edge too hard or you will crush your candle. You are only trying to adhere the edge to the rest of the candle. Continue this until the entire edge is molded to the body of the candle.	
8. Trim the wick at one end of the beeswax candle so that it flushes with the base of the candle. Trim the wick at the other end of the candle to 1/4" long.	

Learning points

The following steps can guide on how to make candles in a safer manner in homes for commercial purposes.

The first thing you will need is **something to melt your wax in**. The double boiler method is the safest way to melt wax but it is also the slowest. However, there are

a few ways to melt your wax for making candles, which have not been discussed here.

One thing you should be aware of is to never melt wax on your stove unless you are using the double boiler method. **Never melt wax on or in anything that does not have some form of temperature control because wax does have a flash point and will burst into flames without warning once it reaches that point. Depending on the wax, the flash point may be between 290 - 380 degrees.** Always be safe when you make candles.

A possible **wax melter** recommended for good work is a Presto “Kitchen Kettle” **which** can be purchased from your local hardware or Walmart store. The “Kitchen Kettle” you get should be that one which comes with a numerical temperature gauge. Never melt wax in anything that does not have a **numerical temperature gauge**. Do not ever use anything with just high, medium and low settings.

You will also need a **good thermometer**. You need to use this with your kitchen kettle to ensure a proper candle wax temperature at all times.

You will need a **“pour pot”** to pour your melted candle wax into your molds. You can even start out using an old, clean coffee can to pour your candles.

Next, you will **need an accurate and reliable scale for weighing your wax to put in your melter**. A scale that weighs up to 4.5 kg should be sufficient. When we weigh our wax we break it up with a hammer and put the chunks on the digital scale until we reach the amount we are going to melt.

So far you will need the following items to begin your candle making:

- Melting Pot
- Thermometer
- Pour Pot
- Scale to weigh your wax
- Hammer to break up wax

Activity 3.3: Preparing ice from liquid water

By the end of the activity, you should be able to:

1. prepare ice from liquid water
2. identify uses of ice
3. suggest ways of marketing ice

What you need

- cup/bucket/beaker/saucepan/bowl cover
- refrigerator
- water
- polythene papers
- ice block cutter

What to do

1. Pour boiled and clean boiled water into a clean cup/bucket/saucepan/bowl, and cover it.
2. Place the container in a refrigerator/deep freezer.
3. Leave it to stay overnight.
4. Cut ice blocks formed and pack them in polythene papers ready for sale.

Follow-up activity

Prepare ice from home, package and market it.

Learning points

A solid such as ice or wax melts to form a liquid when it is heated. The liquid then boils to form a gas when heated more strongly. Changes like these are called temporary changes (*physical changes*) because they can be reversed very easily. The gas can be changed to liquid and the liquid can be changed to solid simply by cooling.

When wood is burned, completely different substances are formed, that is;

charcoal, ash plus some liquids and a gas. A change like this is called a permanent change (or a *chemical change*).

A physical change is one in which no new substance is formed. In physical changes, the chemical properties of the products and reactants are the same.

A chemical change is one in which a new substance is formed. Here, the chemical properties of the products are different from those of the reactants.

Matter can never be created nor destroyed. However, it can be converted into different states. The process of inter-converting matter between any two states is a physical change (no new substances are formed). This principle can be used to make different products like ice cream, ice cubes, and cool-cool bar and candles.

Chapter 4: Mixtures and Compounds

Inquiry questions

- *What are mixtures?*
- *How important are mixtures in everyday life?*

A mixture is a substance consisting of two or more substances NOT chemically combined together. We encounter mixtures of various forms every day. A mixture can be made up of solid and solid, solid and liquid, liquid and liquid, liquid and gas, gas and gas. For example, air is a mixture of gases; milk, ink, soda, fruit juice, bread, cakes and toothpaste are all mixtures. A compound is a substance made of two or more elements chemically combined. For example, water, sugar, salt, baking powder, carbon dioxide and iron sulphide.

Many substances used in everyday life are mixtures.

Preparation of Mixtures

Activity 4.1: Investigating a mixture of iron and sulphur, and a compound of iron and sulphur

By the end of this activity, you should be able to:

1. Prepare a mixture and compound of iron and sulphur.
2. State the differences between a compound of sulphur and a mixture of sulphur.

What you need

- hard test-tube
- bar magnet
- Bunsen burner/stove

- stand and clamp
- iron filings
- powdered sulphur
- water
- dilute hydrochloric acid
- carbon disulphide

What to do

1. Put two spatulafuls/spoonfuls of powdered sulphur on a manila paper and two spatulafuls of iron filings on another manila paper.
2. Move a magnet towards each of the manila papers. What do you observe?
3. Mix the two substances on the manila sheet and move the magnet towards the mixture. What do you observe?
4. Divide the mixture into two parts. Place one part of the mixture into a hard test tube and clamp it as shown below:

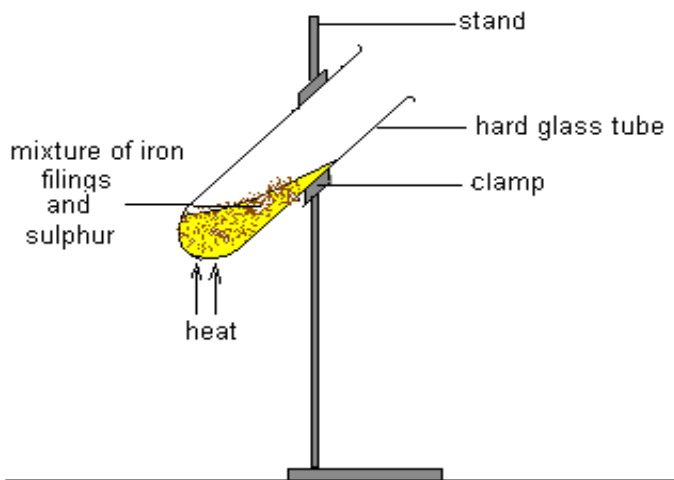


Figure 4.1: Preparation of iron(II) sulphide

5. Heat the mixture and remove the burner/ stove when the mixture starts to glow. What do you observe?
6. Remove the mixture from the test tube. A black compound of Iron(II)

Sulphide is formed.

7. Move a magnet towards the compound formed. What do you observe?

Activity 4.2: Preparing dyes from coloured flowers

By the end of this activity, you should be able to:

1. prepare dyes of different colours.
2. package and market the dyes.

What you need

- pestle and mortar
- glass rod/ wooden rod
- bottles
- test tubes
- ethanol (98%) or propanone or crude waragi
- coloured flower petals (red cabbages, bougainvillea, hibiscus etc)

What to do

1. Pick coloured flowers and pluck off the petals.
2. Crush the petals in a mortar with a pestle.
3. Add either ethanol or propanone or crude waragi to the mortar and stir with the glass rod or wood rod or pestle.
4. Filter the mixture to obtain a clear coloured mixture which is the dye prepared.
5. Pour the dye in a clean bottle.
6. Different flower extracts can also be mixed to get varying dyes.

Follow-up activity

Use different flowers to make a variety of dyes and try to market the dyes.

Activity 4.3: Preparing cough mixture

By the end of this activity, you should be able to:

1. prepare a homemade cough mixture.
2. use, package and market the product.

What you need

- saucepan
- glass rod/wooden rod
- burner/stove
- packing bottles
- knife
- lemon fruits
- ginger
- honey
- mango tree bark and leaves
- water

What to do

1. Cut fresh/green 3 lemon fruits and 200g of ginger into small pieces.
2. Place the cut parts in a saucepan.
3. Add the crushed pieces of mango bark or mango leaves (100g) to the saucepan.
4. Measure distilled/ rain/ tap clean water to the saucepan.

5. Heat on the burner/ stove to boiling, till most water has evaporated.
6. Allow to cool then add honey. Stir and cover with a net/ cover.
7. Leave the mixture to cool further.
8. Filter the mixture and measure the required volumes and package in bottles.
9. You have made a cough mixture. Now calculate the unit cost and selling price. Try to market the products.

Follow-up activity

Get other substances that are locally used to prepare such cough mixture.

Activity 4.4: Making toothpaste (mouth cleanser)

By the end of this activity, you should be able to:

1. make a mouth cleanser.
2. use and promote the use of mouth cleansers by members of the community.

What you need

- measuring cups of the following sizes; 1 cup, $\frac{1}{2}$ cup, $\frac{1}{4}$ cup (the 1 cup size has a capacity of 250 ml)
- measuring cylinder
- spoon of capacity 5 ml
- small buckets(2l)
- a table spoon
- cold boiled water
- glycerine
- peppermint oil
- baking soda (sodium bicarbonate)
- containers with lid(50ml)
- table salt

What to do

1. Place 1 cup of baking soda in the bucket followed by $\frac{1}{3}$ cup of a mixture table salt and glycerine. Mix these thoroughly well using the table spoon.
2. Measure out $\frac{1}{4}$ cup of dry mixture into another bucket containing $\frac{1}{4}$ cup of cold boiled water to the dry mixture. Add more water to make a thick paste. What is the role of the glycerine in the mixture?
3. To the thick paste, add a spoonful of peppermint oil. What is the function of peppermint oil? What is the colour of the paste you have made?
4. Pack the toothpaste in the container with a lid.

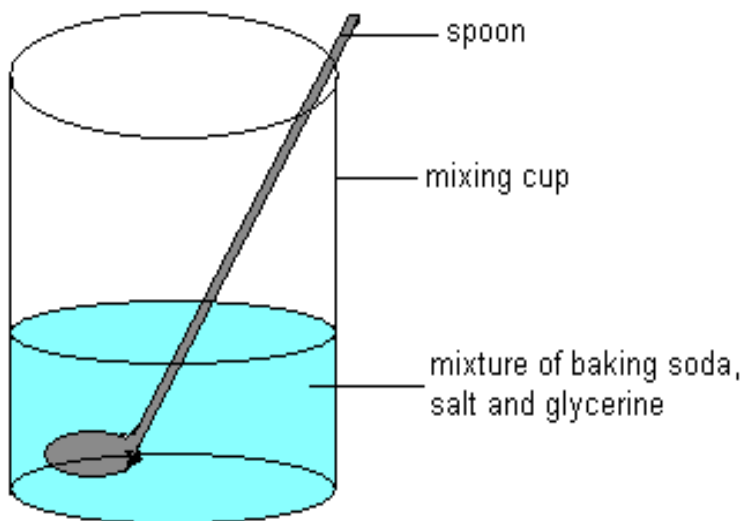


Figure 4.2: Showing preparation of toothpaste

5. What is the “active ingredient” in toothpaste?
6. What is the role of the “active ingredient” in toothpaste?

Follow-up activity

- Brush your teeth using your toothpaste and make a comparison between your “homemade” toothpaste and the commercially manufactured tooth paste.
- Carry out health education in communities on why it is important to practise good oral hygiene.

Learning points

When iron and sulphur are mixed, a mixture is formed with properties which are average of both components. However, when the mixture is heated in a container, a compound (iron sulphide) is formed. The properties of the compound are different from those of the mixture or the components.

Many of the products we use in everyday life are mixtures. Each component in the product serves a particular function and retains its chemical properties in the mixture. An example of a product which is a mixture is toothpaste. It is used to clean the teeth and prevent tooth decay.

The environment in the mouth is conducive for growth and multiplication of bacteria. Bacterial activity in the mouth produces acids which cause tooth decay. Therefore, it is necessary to practise proper oral hygiene by frequently cleaning our teeth.

Chapter 5: The Atmosphere and Air

Inquiry questions

- *What is the active part of air?*
- *Of what significance is the active part of air?*

Air is a mixture of gases that surrounds the earth. All living organisms need air to survive. Air is not visible but it is felt as wind. The percentage composition of the components of air varies from place to place. Several substances undergo chemical changes when they come in contact with air under given conditions. However, not all the components of air take part in these chemical reactions.

Activity 5.1: Estimating the active part of air

By the end of this activity, you should be able to determine the fraction of air which is active.

What you need

- 2 glass gas syringes of a capacity of at least 100 cm³
- 2 rubber bungs with a hole to fit the glass tube
- heat source
- hard glass tube (silica tube/combustion tube). (this should be at least 15 cm long and have a diameter of at least 1.5 cm)
- glass wool
- 2 glass tubes
- 2 retort stands
- rubber tubing
- a pair of tongs
- spatula
- zinc powder

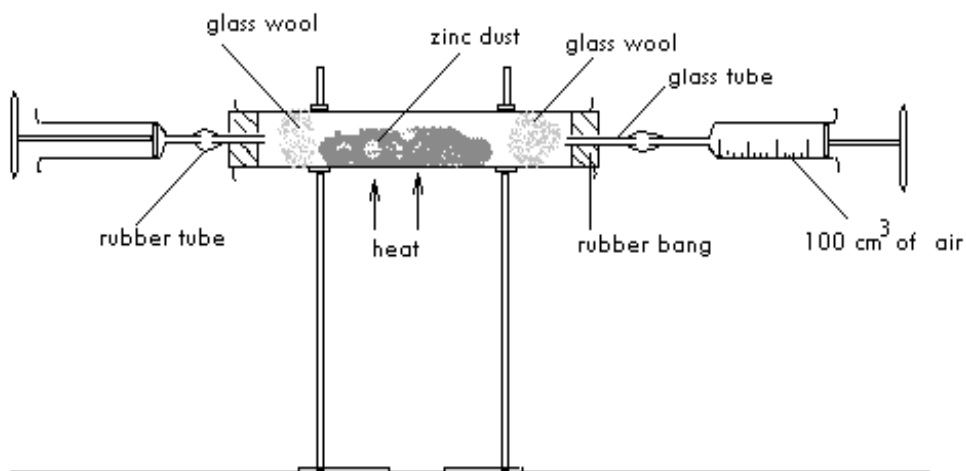


Figure 5.1: Estimating the active part of air

Caution

1. Use tongs to plug the combustion tube with glass wool and put on a nose mask.
2. Handle hot apparatus with care!

What to do

1. Place about 5 spatula end-fulls of zinc dust in a combustion tube and plug each end loosely with glass wool.
2. Connect the two syringes to the combustion tube as shown in Figure 5.1. The piston in one of the syringes should be at the 100cm^3 mark while that of the other syringe should be at the 0cm^3 mark.
3. Heat the tube containing the zinc dust strongly as you pass the air back and forth over the zinc dust. Do this by pushing in the piston alternately. What do you observe in the combustion tube as you heat?
4. Continue heating while observing the combustion tube and its contents. When there is no further change in the volume of air in the syringe, stop heating. What kind of change occurred in the combustion tube? Give a reason for your answer.
5. Allow the tube to cool and observe what happens to the contents of the

combustion tube.

6. Make sure one syringe is at the 0cm^3 mark and read the volume of air in the other syringe.
7. Record all your observations. What volume of air took part in the reaction? Calculate the percentage of the component of air used up in the reaction.
8. From your knowledge of the components of air and percentage composition, which component of air took part in the reaction?

Follow-up activities

- Identify a process in nature that uses the same active component of air.
- Describe one similar activity that can be carried out to estimate the percentage of the active component of air.

The Active Part of Air

Why do you think oxygen is important in our everyday life?

Oxygen is a gas that occupies approximately 21% of air by volume. It is vital in several processes. These include respiration and combustion. Most living things need oxygen for respiration. Oxygen is also necessary for burning and rusting.

Activity 5.2: Preparing oxygen in the laboratory

By the end of the activity, you should be able to:

1. prepare oxygen gas in the laboratory.
2. state the properties of oxygen.
3. state the uses of oxygen in everyday life.

What you need

- dropping funnel
- flat-bottomed flask

- water trough
- bee hive shelf
- 2 gas jars
- 2 gas jar lids
- spatula
- matchbox/lighter
- hydrogen peroxide solution
- manganese(IV) oxide
- wooden splint
- blue and red litmus papers.

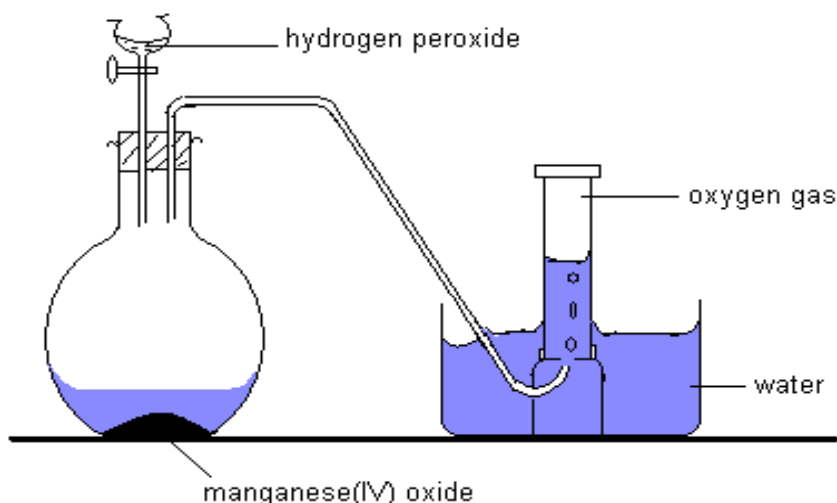


Figure 5.2: Preparation of oxygen gas

What to do

1. Place four spatula end-fulls of manganese(IV) oxide in the flat-bottomed flask.
2. Arrange the apparatus as shown in Figure 5.2
3. Introduce the hydrogen peroxide solution drop wise into the flask through the dropping funnel by slightly opening the tap. What do you observe? Which property of oxygen allows you to collect it in the way you do in the

experiment. Write a word equation for the reaction that takes place.

4. Collect two gas jars of oxygen and cover these with the gas jar lids. What is the colour of oxygen?
5. To the first jar of oxygen, introduce a glowing splint. What happens to the glowing splint? Record what you observe.
6. To the second jar, introduce moist blue litmus paper and red litmus paper. What do you observe? Record your observations.
7. Describe the smell of oxygen.
8. Name this method of gas collection.

Follow-up activity

1. Describe another method that can be used for the laboratory preparation of oxygen.
2. Describe how oxygen is manufactured
3. Why is it important to prepare oxygen on a large scale?

Combustion of Elements in Oxygen

As with any type of chemical reaction, combustion takes place when chemical bonds are broken and new bonds are formed. It so happens that combustion is a particularly dramatic type of oxidation-reduction reaction. Whereas we cannot watch iron rust, combustion is a noticeable event. Even more dramatic is combustion that takes place at a rate so rapid that it results in an explosion.

Coal is almost pure carbon, and its combustion in air is a textbook example of oxidation-reduction. Although there is far more nitrogen than oxygen in air (which is a mixture rather than a compound), nitrogen is very un-reactive at low temperatures. For this reason, it can be used to clean empty fuel tanks, a situation in which the presence of pure oxygen is extremely dangerous. In any case, when a substance burns, it is reacting with the oxygen in air.

As one might expect from what has already been said about oxidation-reduction, the oxygen is reduced while the carbon is oxidised. In terms of oxidation numbers,

the oxidation number of carbon jumps from 0 to 4, while that of oxygen is reduced to -2 . As they burn, these two form carbon dioxide or CO_2 , in which the two -2 charges of the oxygen atoms cancel out the $+4$ charge of the carbon atom to yield a compound that is electrically neutral.

Activity 5.3: Use of oxygen in gas welding

By the end of this activity, you should be able to:

1. light and use a gas welding torch.
2. explain the role of oxygen in gas welding.

What you need

- gas welding machine
- piece of steel

What to do

1. You will be instructed by the teacher on the rules and regulations of a field trip.
2. The teacher will then lead you in a group to visit a nearby workshop where gas welding is done.
3. The technician will demonstrate the welding process. What have you observed during the process of cutting steel and welding? Can you explain how the two pieces of metal get joined together?
4. You will be given an opportunity to operate a gas welding torch. How does the amount of oxygen affect the torch flame?
5. The teacher will explain to you how the oxygen gas works in the welding process.

Activity 5.4: Finding out what happens when metallic and non-metallic elements burn in oxygen

By the end of this activity, you should be able to:

1. name the products formed when elements burn in oxygen.
2. describe what happens when elements burn in oxygen.
3. write word equations for the reactions that take place when elements burn in oxygen.
4. classify the products of the reactions as acidic, basic or neutral.

What you need

- Bunsen burner
- magnesium ribbon
- crucible
- pair of tongs
- 2 deflagrating spoons
- gas jar of oxygen
- white phosphorus
- powdered carbon
- sulphur
- sodium
- litmus paper

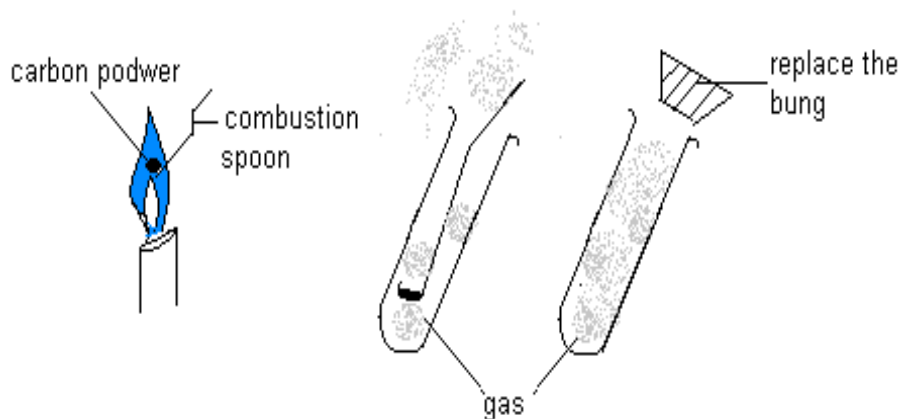


Figure 5.3: Burning powdered carbon in air

What to do

1. Heat some powdered carbon on a deflagrating spoon in a hot Bunsen flame as shown in Figure 5.3. What do you observe?
2. Lower the burning carbon in the gas jar of oxygen and dissolve the product in about 10cm^3 of water. Test the solution with litmus paper. What do you observe?
3. Hold a piece of magnesium ribbon using a pair of tongs and ignite it in a Bunsen flame. What do you observe?
4. Lower the burning magnesium in a gas jar of oxygen and dissolve the product in about 10cm^3 of water. Test the solution with litmus paper. What do you observe?
5. Repeat steps 1 and 2 using sodium.

Caution: Sodium and white phosphorus are inflammable and should only be handled by the teacher.

<i>Element</i>	<i>Appearance before heating</i>	<i>Appearance after heating</i>	<i>Effect of the solution formed on litmus</i>
<i>Copper foil using tongs</i>			
<i>Powdered carbon using combustion</i>			
<i>White phosphorus using tongs</i>			
<i>Sodium using tongs</i>			
<i>Magnesium ribbon using tongs</i>			

- Repeat steps 1 and 2 using Phosphorus.
- Record all your results as in the table above.
- Write word equations for the reactions that take place in steps 1- 6.

Learning points

Combustion is the burning of substances in air. A French scientist called Lavoisier carried out accurate experiments involving measurements of mass to prove that combustion involves a substance combining with something in the air. He found that it is oxygen in the air that combines with the burning substance.

Thus, when elements burn in oxygen, they form products which may be gases, liquids or solids. The gaseous products may not be visible and “disappear” leaving no trace behind. The formation of some other products is accompanied by changes in appearance, colour and texture. This change is referred to as a chemical change. The compounds formed are called **oxides**.

The nature of the oxide formed will depend on the element that has burned in the

oxygen. Most metals burn in air to form basic oxides for example, magnesium oxide, and Calcium oxide. Those basic oxides that dissolve in water form solutions that turn litmus paper blue. Non-metals burn in air to produce acidic **oxides** (carbon dioxide, sulphur dioxide); or **neutral oxides** (carbon monoxide, water).

Chapter 6: Rusting of Iron

Inquiry questions

- *How does rusting of iron come about?*
- *Can any other metals apart from iron rust?*

When materials containing iron are exposed to the atmosphere, they tend to react forming a brown substance. The brown substance is called rust. The formation of rust makes the objects lose their strength and their original appearance which decreases their quality and usefulness.





Figure 6.1: Showing rusted old cars and a house roof

Factors Affecting Rusting

When reactive metals are exposed to the atmosphere, they undergo a chemical process called corrosion. In the case of materials containing iron, this corrosion process is called **rusting**. Many materials are made up of iron for example razor blades, nails, knives, forks, spoons, doors, bridges and vehicle parts. When these materials are exposed to the atmosphere, they react with oxygen and moisture to form a reddish brown substance called rust. The chemical name for rust is hydrated iron(III) oxide, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (where x is a whole number). Rusting has a negative effect on the properties materials. For this reason, methods of preventing rusting are very crucial. In order to prevent rusting, one must control the conditions that are necessary for rusting to take place. The conditions responsible for rusting are water and oxygen. Impurities in water such as salts, carbon dioxide and acids accelerate the process of rusting.

Activity 6.1: Investigating the conditions necessary for rusting to take place

By the end of this activity, you should be able to:

1. state the conditions necessary for rusting to take place.
2. control rusting of objects containing iron.

What you need

- cotton wool
- anhydrous calcium chloride
- steel wool
- distilled water
- oil
- test tube rack
- test tubes
- freshly boiled water
- porcelain tile/flat piece of wood
- knife
- lemon juice
- a small trimming of a galvanised iron sheet

What to do

1. Place small pieces of steel wool in each of the four test tubes and label them *a*, *b*, *c* and *d* as in Figure 6.2.

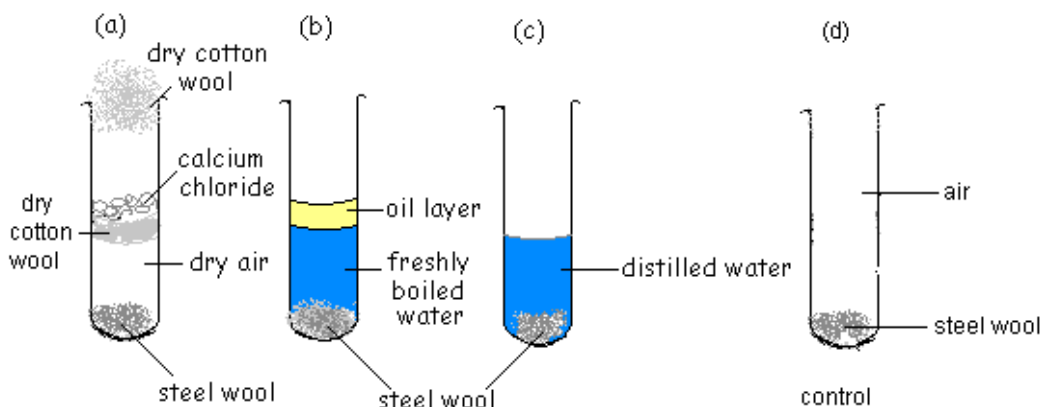


Figure 6.2: Showing investigation of conditions necessary for rusting to occur

2. In test tube (a), place a dry cotton wool and on it put some anhydrous calcium chloride, then plug it with dry cotton wool as shown in the diagram.
3. In test tube (b), add about 10cm^3 of freshly boiled water, followed by 2cm^3 of oil.
4. In test tube (c), add about 10cm^3 of distilled water.
5. Leave test tube (d) open as in the diagram.
6. Place the trimming of galvanized iron sheet on the porcelain tile/flat piece of wood and add lemon juice as shown in Figure 6.2.

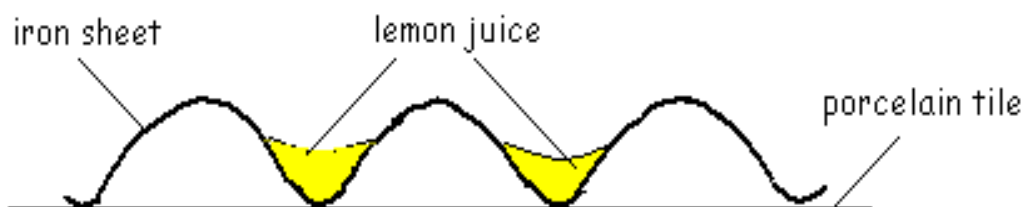


Figure 6.3: Showing rusting of galvanized iron sheet

7. Allow the experiment setups in Figure 6.1 and Figure 6.2 to stand for one week in a place where they will not be disturbed. What do you notice after one week in test tubes (a), (b), (c), (d) and on the iron sheet? How would you explain your observations? Why is it necessary to study the conditions that promote rusting?
8. Why do you think rusting is a chemical change or physical change? Explain your answer.

Activity 6.2: To determine how much air is used in rusting

By the end of this activity, you should be able to:

1. observe the change in volume of air during rusting
2. calculate percentage of oxygen found in air.

What you need

- Iron wool
- Test tube
- Beaker (100 cm³)
- Ruler
- Iron wool, Fe(s)

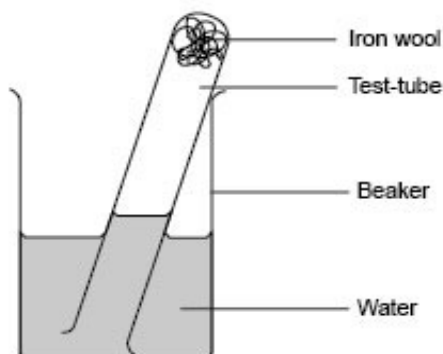


Figure 6.4: Showing volume of oxygen used during rusting

What to do

1. Put about 3 cm depth of iron wool into the test tube and wet it with water. Tip away excess water.
2. Put about 20 cm³ water into the beaker. Invert the test tube and place it in the beaker of water (see diagram). Measure the length of the column of air with the ruler.
3. Leave for at least a week.
4. Measure the new length of the column of air, taking care not to lift the test tube out of the water.

Caution: The test-tubes used in this experiment can get stained by the rust. They can be cleaned with a 'Stain Devil'

Prevention of Rusting

Activity 6.3: Essay competition on methods of preventing rusting

By the end of this activity, you should be able to identify the methods used by the community to prevent rusting.

What to do

1. In this activity, you will be given a topic for essay competition such as, "Discuss the methods of preventing rusting of iron".
2. Do an extensive research from the library about the topic.
3. Go to the nearby community and list the materials on which different methods of preventing rusting have been applied.

Activity 6.4: Making rust-resistant dustbins

By the end of this activity, you will be expected to:

1. provide services to the school or to your community by applying the methods of preventing rusting.
2. make rust-resistant dustbins for sale to the community.

What you need

- metallic tin
- piece of cloth
- large polythene bag
- oil paint

What to do

1. Get a metallic cylindrical container from your teacher.
2. Clean it in and out with a clean cloth or papers.
3. Make small holes at the bottom of the tin.
4. Mix oil paint and paint the inside and outside of the container.
5. Place the large, polythene bag in the painted container.
6. Place it outside the classroom for use as dustbin.

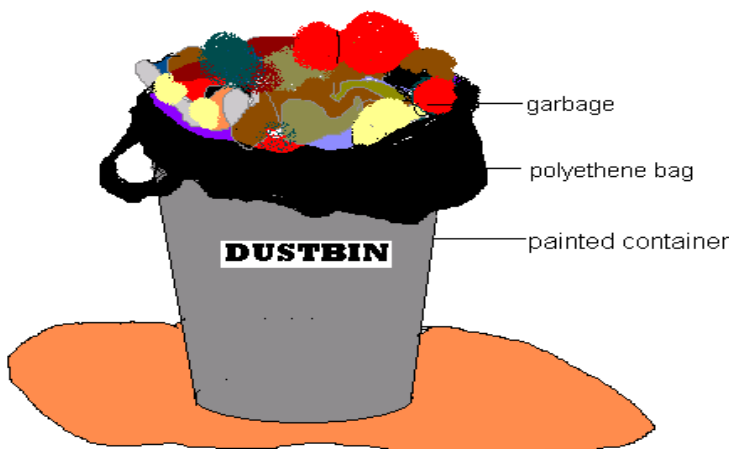


Figure 6.5: *Rust-free dustbin*

Follow-up activity

- Educate the community on how to protect materials containing iron from rusting.
- Describe the different methods that can be used to prevent rusting.
- Explain why iron sheets used for roofing in industrial areas rust much faster than those in far away from the industrial areas.
- Does rusting have any advantages? Explain your answer.

Chapter 7: Acids, Bases, Salts and Indicators

Inquiry question

- *Of what importance are acids and bases in everyday life?*
- *How are salts formed?*

In the previous chapter, we learnt about acidic, basic and neutral oxides. These oxides dissolve in water to form solutions which are acidic, basic or neutral. In this chapter, the behaviour of acids and bases will be investigated.

Reactions of Acids and Bases

Acids are compounds that break into hydrogen (H^+) ions and another compound when placed in an aqueous solution. Bases are compounds that break up into hydroxide (OH^-) ions and another compound when placed in an aqueous solution.

That pH scale we talked about is actually a measure of the number of H^+ ions in a solution. If there are a lot of H^+ ions, the pH is very low. If there are a lot of OH^- ions that means the number of H^+ ions is very low, so the pH is high.

When the acids react with bases they form a salt and water only. The reaction is called neutralisation.

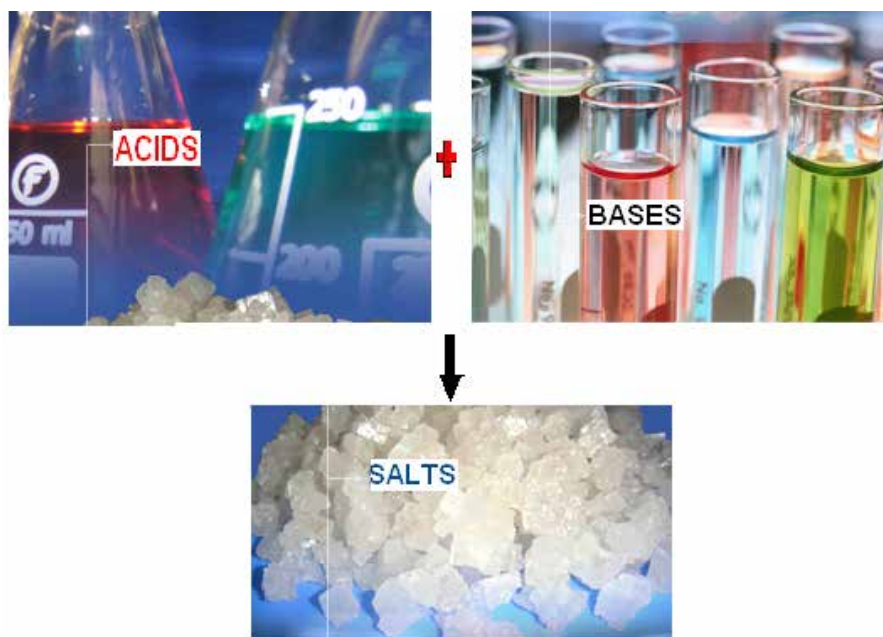


Figure 7.1: Acids reacting with bases to form salts

Activity 7.1: Preparation of copper(II) sulphate fungicide

By the end of the activity you should be able to:

1. Prepare copper (II) sulphate from copper (II) oxide.
2. Use copper II sulphate to control fungi.

What you need

- 2 glass beakers (250cm³)
- filter funnel
- filter papers
- stand and clamp
- glass rod
- spatula
- Bunsen burner (heating apparatus)
- tripod stand and wire gauze
- dilute sulphuric acid

- copper(ii) oxide
- desiccator
- water

What to do

1. Measure out 50cm³ of dilute sulphuric acid, pour into the glass beaker and warm it for 5 minutes.

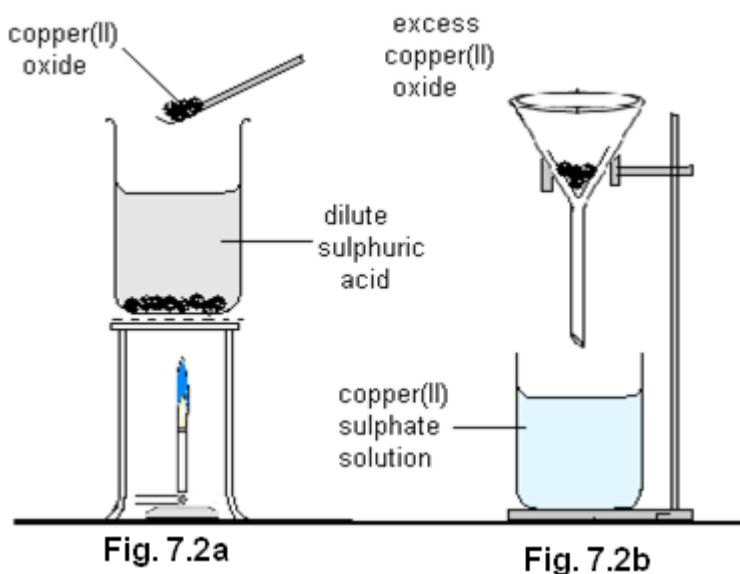


Figure 7.2: Showing preparation of copper(II) sulphate crystals

2. Using a spatula, add copper(II) oxide to the hot acid a little at a time with constant stirring until the solid can no longer dissolve. What do you observe?
3. Filter off the excess copper(II) oxide and collect the filtrate in a separate beaker shown in Figure 7.2 (b). What is the colour of the filtrate?
4. Evaporate the filtrate and at intervals check for the formation of crystals on a glass rod after dipping it in the hot solution. Once crystals begin to form, remove Bunsen flame and leave the hot saturated solution to cool.
5. Filter off the crystals and dry them between filter papers or under sunshine or in a desiccator.

6. Make a 50% copper(II) sulphate solution and use it for dressing seeds.

Follow-up activity

Use the principle of preparing salt by evaporation and crystallisation to prepare local salt by burning grass, leaves and branches of trees to form ash; obtaining the ash filtrate and then evaporating it to get the salt.

Activity 7.2: Preparation of barium sulphate paint

By the end of this activity, you should be able to:

1. prepare a sample of barium sulphate from barium nitrate.
2. use barium sulphate to prepare a paint.

What you need

- 2 glass beakers (250 cm³)
- filter funnel
- filter papers
- stand and clamp
- zinc sulphide
- glass rod
- dilute sulphuric acid
- barium nitrate solution
- water

What to do

1. Measure 50cm³ of 0.5M barium nitrate solution and pour it into a glass beaker.
2. Add about 60cm³ of dilute sulphuric acid into a beaker containing barium nitrate solution as in Figure 7.2.

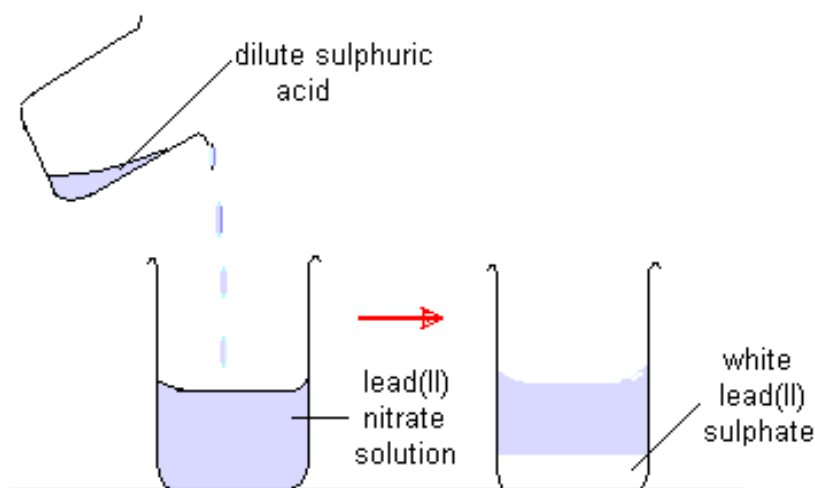


Figure 7.3: Showing preparation of barium sulphate paint

3. Stir the mixture with a glass rod and then filter.
4. Wash the residue with cold distilled water and spread it on a filter paper to dry.
5. Mix the product with zinc sulphide in a ratio of 2:1 to obtain water paint.

Follow-up activity

Investigate which other salts can be used as paints.

Learning points

Acids and bases are widely used in our everyday life in agriculture, medicine, industry and in preservation of food. Examples of acids and their uses include: sulphuric acid (for making paints, detergents, polymers and fertilizers); nitric acid (for making fertilizers, explosive substances (such as T.N.T), dyes, and plastics; benzoic acid (to preserve food); carbonic acid (to make fizzy drinks); ethanoic acid (a component of vinegar); tartaric acid (to make baking powder).

Salts play an important role in agriculture, medical field, food preparation and preservation. Copper(II) sulphate is used as one of the inorganic pesticides in agriculture. Other salts that can be used as pesticides include: iron(II) sulphate

(FeSO_4), sodium arsenate (Na_2HAsO_4) and mercury(I) chloride (HgCl).

Different types of salts such as normal salts, acid salts and basic salts are known. Salts can either be soluble or insoluble. Soluble salts are those that dissolve in water whereas insoluble salts do not dissolve in water.

Salts are formed as a result of action of an acid on:

- a metal
- a base (oxide or hydroxide of a metal)
- a carbonate or hydrogen carbonate
- a salt solution of a metal

Soluble salts are prepared in solution and obtained through crystallisation. Crystallisation involves heating the saturated salt solution to evaporate off water and allowing the hot solution to cool. Crystals formed are filtered from the solution, washed and then dried. Insoluble salts are prepared by double decomposition or precipitation method. This involves mixing two aqueous solutions with the desired ions that will form the insoluble salt. The salt is filtered off, washed and dried. Common salt, (sodium chloride) is an example of a salt which is used in preserving and seasoning food. It is also used in the manufacture of washing and laundry soap such as Mukwano soap.

Follow-up activity

Using the knowledge of preparation of salts in the laboratory, you should go further and prepare salt from basic solutions such as ash obtained by burning grass, leaves and stems of trees.

Indicators

In chemistry, we use indicators when faced with the task of finding out whether a solution in question is an acid, alkali or neutral. Certain substances in solution turn to distinct colours when they come in contact with acids, alkalis (soluble bases) or neutral solutions. They are called indicators of acids and alkalis. When we think of acid-base indicators, we normally think of chemical compounds like litmus and phenolphthalein. However, many natural compounds are indicators. Indicators can be extracted from plant materials such as petals, beet roots and

leaves such as those of red cabbage, among others.

Activity 7.3: Preparation of indicators from plant materials

By the end of this activity, you should be able to make an indicator from plant materials.

What you need

- glass beakers (50cm³)
- filter funnel
- filter papers
- stand and clamp
- glass rod
- mortar and pestle
- test tubes
- measuring cylinder (5 cm³)
- red or blue flower petals
- water
- ethanol or local waragi
- lemon or orange juices
- dilute sulphuric acid
- soap solution
- dilute sodium hydroxide
- solution of ashes (from plants)
- small bottles

What to do

1. Collect blue flowers from the compound and then follow the steps as given in 2 – 8 below.
2. Place the blue flower petals in a mortar.
3. Grind the petals using a pestle.
4. Add a portion of a solvent made by mixing ethanol (Waragi) with water and continue grinding until the extract is as deep in colour as possible.
5. Decant the extract into a test tube until you have about 10cm³ of it.
6. Repeat procedure 1 – 4 with red flower petals.
7. Use the flower extracts to test the solutions: lemon or orange juices, dilute

sulphuric acid, soap solution, dilute sodium hydroxide, solution of ashes (from plants) and soap solution.

- Record your observations in a suitable table and discuss them with your fellow students. Classify the solutions as acids and bases.

Activity 7.4: Preparation of indicators from red cabbage

By the end of this activity, you should be able to:

- prepare an indicator from red cabbage.
- test its colour in acid and base.

What you need

- 250 ml beakers
- hot plates
- beaker tongs
- 2l distilled water
- large screw – top storage bottle
- dropper bottles of 1M HCl
- dropper bottle of 1M NaOH
- red cabbage
- watch glasses
- goggles

What to do

- Pull a few leaves of red cabbage into very small chunks with your fingers.
- Fill a 250 ml beaker halfway with torn cabbage leaves and add water until the beaker is about two-thirds full. Place the beaker on a hot-plate at a medium temperature.
- You should notice that the water turns purple as the cabbage boils. When the water is very purple, remove the beaker from the heat with beaker tongs and let cool. The purple juice you made is acid-base indicator.

4. To find the colour of your indicator in acid, add one drop of indicator to one drop of 1M Hydrochloric acid in a watch glass. Record the colour of the indicator in acid.
5. To find the colour of your indicator in acid, add one drop of indicator to one drop of 1M sodium hydroxide in a watch glass. Record the colour of the indicator in a base.

Follow-up activity

- Prepare different indicators including universal indicators from plant materials of different colours. Test the nature of the various solutions using the different plant material extracts including that of a mixture of red, orange, yellow, green, blue, indigo and violet.
- Study indicator properties of other dyes. Boil other vegetables and determine whether the resulting juices can be used as acid-base indicators.
- Package the various juices in plastic bottles, cost the products and try to market to other schools.

Chapter 8: Water and Life

Inquiry questions

- *Why is water important to everyday life?*
- *Where does the water from water bodies such as lakes come from?*
- *How can you show that water is an oxide?*

We study about water because it is encountered in our lives every day. Water is one of the most important chemical compounds to human beings and all living things. It is commonly said that even the Bishop of Canterbury is 85% water. This means that nobody can do without water in a lifetime. The importance of water lies in its unique properties especially as a universal solvent.

Prior knowledge of solutions, reactivity of metals with water and inter-conversion of states of matter i.e. evaporation, condensation and melting, is important to understanding this chapter.

By the end of this chapter, you should be able to:

1. carry out investigations illustrating that water is an oxide.
2. illustrate that water contains hydrogen.
3. investigate products formed when candle wax burns.
4. conserve wetlands.

The Water Cycle

Activity 8.1: Modelling a water cycle

By the end of this activity, you should be able to advise the community on the importance of wetland conservation.

What you need

- manila papers
- markers

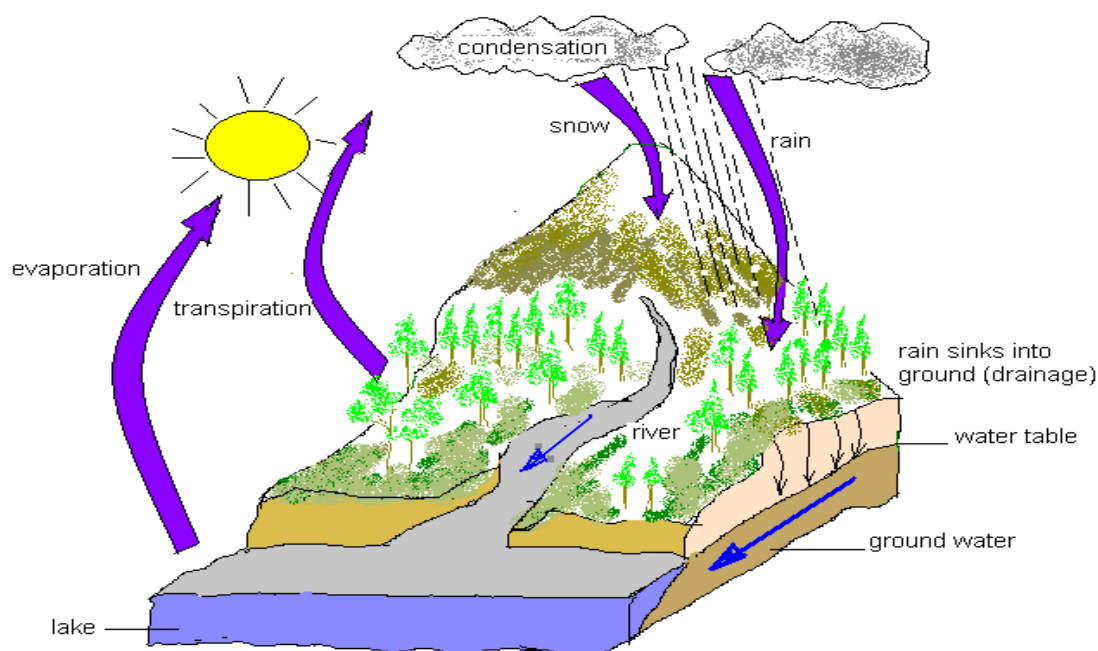


Figure 8.1: Showing water cycle

What to do

1. Using the markers, make an illustration of the water cycle using a scheme diagram.
2. Describe the different processes in the cycle. What is the importance of

evaporation and condensation in a water cycle?

3. Can you explain why sea water is salty while river waters are fresh?
4. Spring water comes from rain water sunk in the ground. Explain why spring water appears to be clean yet rain water running on the surface of the ground is dirty.
5. What is the importance of wetlands in the water cycle?

Learning points

Water is a naturally occurring substance. Water forms through the changes called **water cycle**. What happens is that water vapour forms in the air when the sun shines or winds blow across water bodies. As it rises, the vapour cools. Finally, it condenses into tiny drops of water which float in the sky and form a cloud. The drops may join together and fall as rain.

The rain water passes into the springs, rivers and seas. Once again it evaporates, and the cycle is repeated. Water on the earth's surface is continually changing into vapour, going into the atmosphere and falling as rain.

Water is useful to living organisms and important for domestic use and industrial processes. Water is a chemical compound of hydrogen and oxygen.

Follow-up activity

Design an experiment to show how the water cycle occurs (Hint: evaporation and cooling vapour).

Activity 8.2: Showing that water is an oxide of hydrogen

By the end of this activity, you should be able to demonstrate that water is an oxide of hydrogen.

What you need

- cotton wool or glass wool
- two test tubes
- a trough
- magnesium ribbon
- sand paper

What to do

1. Using a pair of tongs or forceps, loosely pack wet glass wool at the end of a hard test tube.
2. Clamp the test tube horizontally. Clean about 15 cm of magnesium ribbon using sand paper. Make the clean magnesium into compact cylinder. Place the magnesium about 2-3 cm from the wet glass wool. Heat the glass wool gently at first to generate steam. Then heat the magnesium strongly until it is about to melt. Why was the glass wool heated before magnesium?
3. Fit with a coke carrying a glass tube and put a test tube full of water at the other end of the tube as shown in Figure 8.2.

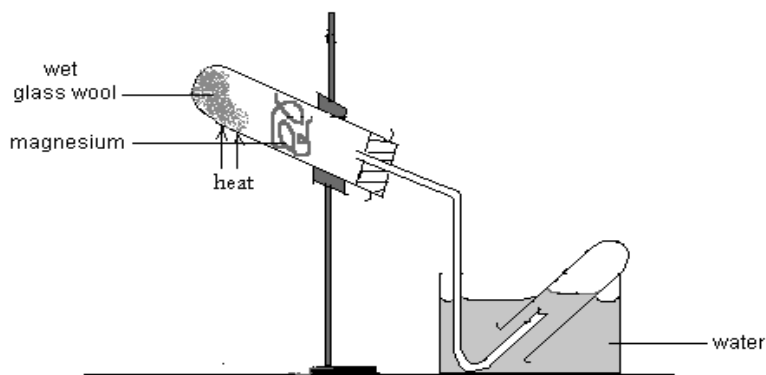


Figure 8.2: Preparation of hydrogen gas

4. Gently touch the end of the glass tube with glass wool to generate steam to pass over the hot magnesium. A reaction will occur and a gas collects in the second test tube. Carefully test the gas with a lighted splint.
5. Continue heating and generating steam until the reaction is complete. When the test tube cools, tip out the contents and carefully examine them.

Follow-up activity

Find other metals that can be used to show that water is an oxide of hydrogen.

Activity 8.3: To Show products formed when a candle burns

By the end of this activity, you should be able to describe the products formed when candle burns in oxygen.

What you need

- Candle
- Watch glass
- Glass funnel
- Delivery tube
- Anhydrous copper(II) sulphate
- Lime water

What to do

1. Pack the U-tube with anhydrous copper(II) sulphate powder. What is the colour of anhydrous copper(II) sulphate?
2. Place concentrated sodium hydroxide solution in a test tube. What happens to the sodium hydroxide solution?
3. Fit the glass funnel, test tube and glass tube as illustrated in Figure 8.3.
4. What effect does the liquid that forms when the candle burns have on anhydrous copper(II) sulphate.

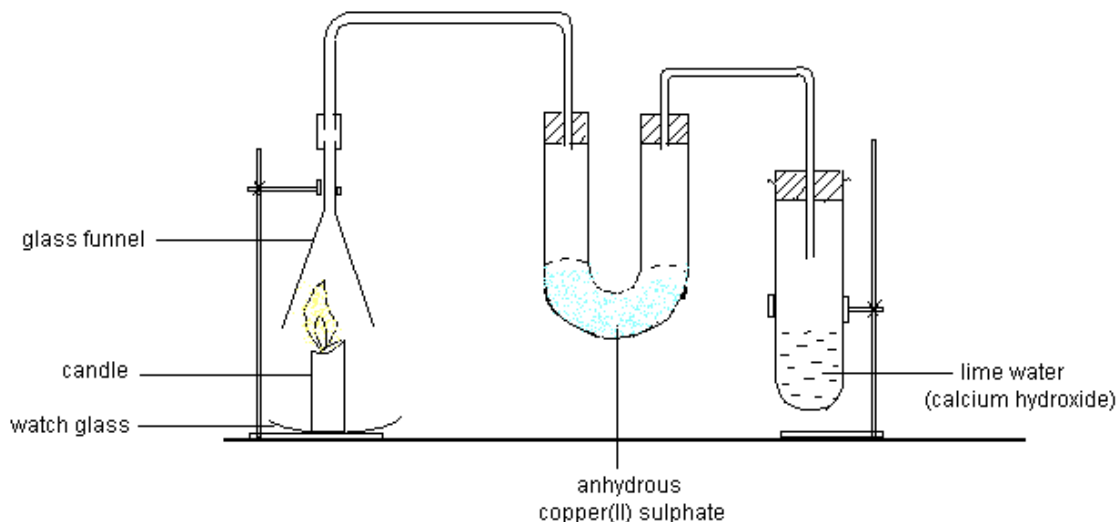


Figure 8.3: Showing products formed when a candle burns

Water Purification

Activity 8.4: Homemade water purifier

By the end of this activity, you should be able to make a homemade water purifier to see how suspended matter can be filtered from water.

What you need

- eight-inch-tall cardboard box
- two bowls
- water
- dirt
- wool yarn

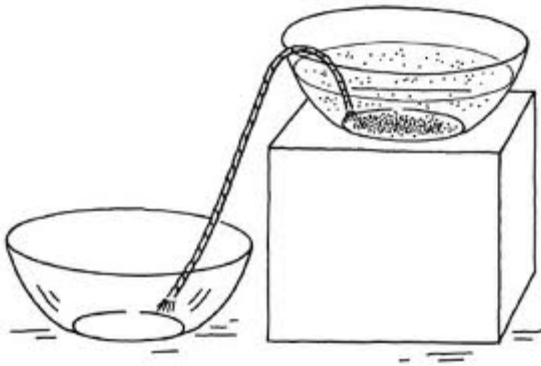


Figure 8.4: Showing apparatus for homemade water purifier

What to do

1. Set an eight-inch-tall cardboard box on a table. Set a bowl of clean water on top of the box.
2. Gently drop a small handful of dirt into the water. Much of the dirt will remain suspended in the water, and the water in the bowl will be discolored.
3. Set an empty bowl on the table right next to the cardboard box.
4. Twist together several one-foot strands of wool yarn to make a rope.
5. Put one end of this rope, or wick, into the bottom of the bowl of dirty water. Place the other end of the wick in the empty bowl. After a while, drops of clear water will drip off of the free end of the wick into the empty bowl.
6. Explain what happened.

Learning points

What Happened? The material in your rope absorbs water and draws it from the bowl. It leaves the dirt behind, however, so the water that drips into the second bowl is clean.

Activity 8.5: Homemade charcoal water filter

By the end of this activity, you should be able to make a homemade water filter easily and inexpensively that will help to serve the immediate clean water needs.

What you need

- 2 litre pop bottle
- straw
- sand - fine and large grain
- gravel- fine and large grain
- Charcoal granules
- cotton batting
- coffee filter
- cup

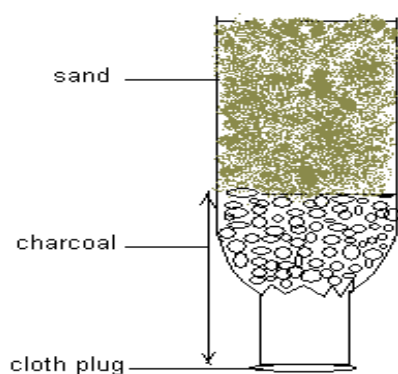


Figure 8.5: showing apparatus for homemade water purifier

Alternatively

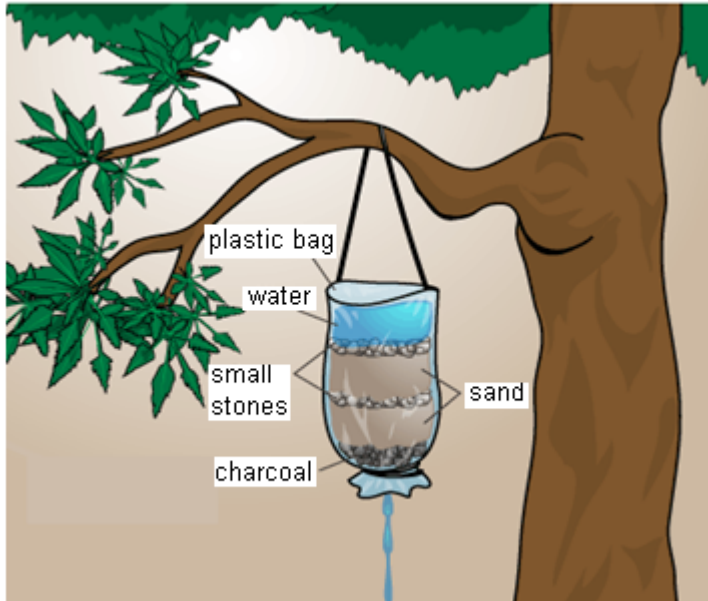


Figure 8.6: Showing apparatus for homemade water purifier

What to do

1. Cut the bottom off of the two litre bottle. Cut a hole in the lid and insert the straw into the hole so the filtered water will flow out of the straw.
2. Insert the cotton batting at the bottom of the bottle. If you have charcoal granules, layer those next. The next layer is fine grain sand followed by larger grain sand. After the sand layers, add the gravel layers. Start with the fine grain gravel followed by the larger grain gravel. Alternate these layers until you reach the top of the bottle. Top the layers off with the coffee filter.
3. Once you have everything in place, pour the water through the coffee filter. The water will work through the different layers to remove the impurities. The cotton batting at the bottom works as the final layer to catch particles from the sediment. The water will flow out of the straw and into the cup.

Learning points

When making a homemade water filter, it is important to have the right materials readily available. A homemade water filter can easily and inexpensively be made, and it can help to serve your immediate clean water needs. There are filters using sand and pebbles that are really effective to remove impurities from the water. Many use an added coffee filter to get rid of the big impurities before the water goes through the sand and pebbles.

You have to keep in mind what the purpose of your homemade water filter is. If the water has a lot of impurities, but the impurities are mainly sand and gravel and there are no harmful bacteria in the water, a very simple water filter is sufficient.

If there are a lot of harmful bacteria in the water, simple filtration will not be enough. This is normally where the charcoal comes in. What is better though, is to use activated carbon for removing the bacteria. It is, of course, also possible to use chlorine, but most of us do not approve of the taste and smell it will give the water.

Follow-up activities

- Visit water treatment plants to see how water is treated.
- Find out from nearby shops the different chemicals used for purifying water.
- Find out whether there are any plant materials that can be used for water purification.

Chapter 9: Hydrogen and Some of its Sources

Inquiry question

- *How are some of the sources of hydrogen important?*

Hydrogen, like carbon, is an element that is present in all living things. Hydrogen is a component of water, dilute acids and organic substances, for example; wood, paraffin candle wax. Its properties and uses will be discussed for example, hydrogen balloons floating in air at parties.



Figure 9.1: Products containing the element hydrogen

By the end of the chapter, you should be able to utilise the physical and chemical

properties of hydrogen.

Reactions of Hydrogen with other Substances

Activity 9.1: Reaction of hydrogen with hot copper(II) oxide

By the end of this activity, the learner should be able to demonstrate the reaction between hydrogen and copper(II) oxide.

What you need

- combustion tube
- porcelain boat
- copper(ii) oxide
- anhydrous copper sulphate
- test tube
- a beaker with cold water

What to do

1. Place copper(II) oxide in a porcelain boat carefully.
2. Arrange the apparatus as in Figure 9.2.
3. Prepare hydrogen as in the previous experiment but this time instead of collecting the gas; pass it over the hot copper(II) oxide.
4. Heat the copper(II) oxide strongly. What change occurs when copper oxide is strongly heated in a stream of hydrogen? Write an equation for the reaction between hydrogen and copper oxide.
5. When the reaction is complete, leave the combustion tube to cool but keep the hydrogen flowing.
6. After cooling the combustion tube, examine the contents in the porcelain tube. What are the substances left after hydrogen is passed over the hot copper?
7. Test the liquid that has collected in the test tube in the beaker with anhydrous copper(II) sulphate. What can you observe?

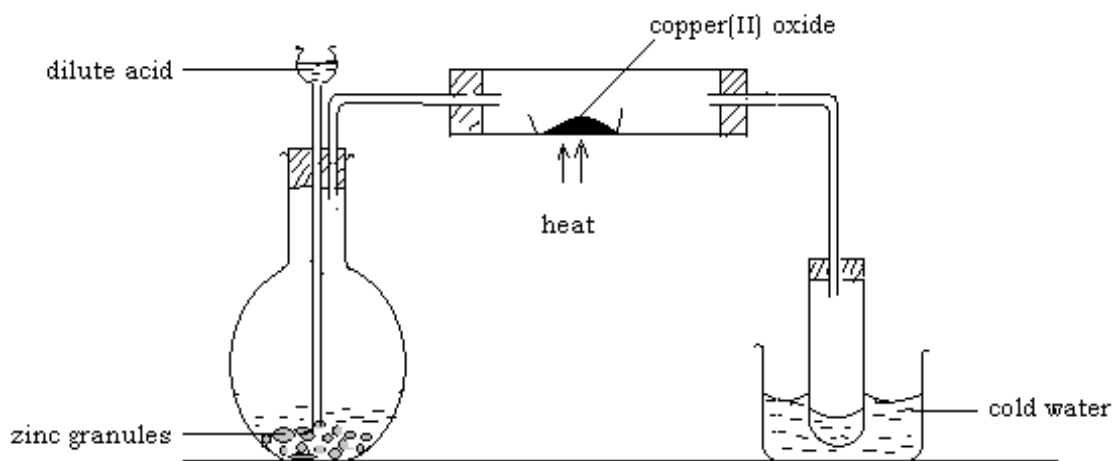


Figure 9.2: Showing reducing action of hydrogen

Activity 9.2: Finding what product forms when hydrogen burns in air

By the end of this activity, you should be able to test for the product formed when hydrogen burns in air.

What you need

- glass funnel
- delivering tube
- test tube
- beaker of cold water

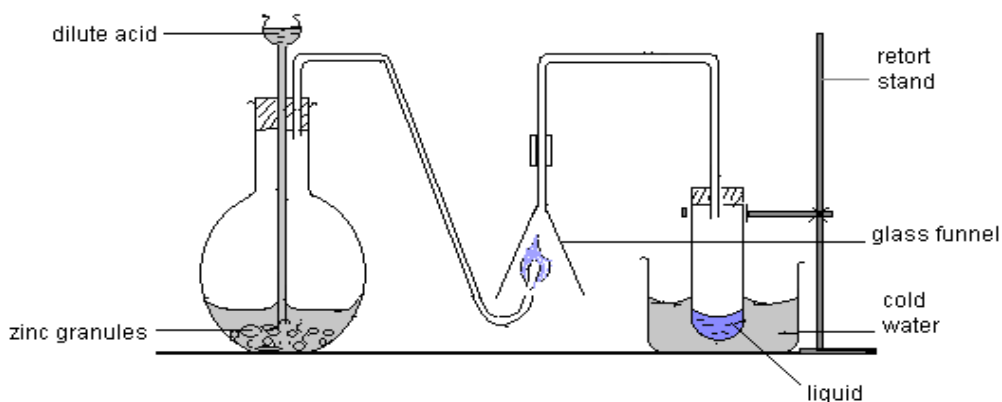


Figure 9.1: Showing products formed when hydrogen burns

What to do

1. Slide zinc granules in a flat bottomed flask.
2. Cork the flask with a bang carrying a thistle funnel and a delivery tube with a clamped glass funnel over it.
3. Connect the glass tube to a test tube placed in cold water.
4. Drop dilute sulphuric acid or dilute hydrochloric acid onto the zinc through the thistle funnel. What did you observe?
5. Allow the hydrogen to pass through the apparatus for about a minute to expel air that would cause an explosion.
6. Light the hydrogen at the end of the tube, note the colour of the flame. What is the identity of the product formed when hydrogen burns in air?
7. After the reaction, remove the condensate, test it with litmus and anhydrous copper(II) sulphate.
8. From the reaction above, can you state a use to which hydrogen can be put?

Chapter 10: Atomic Structure

Inquiry question

- ***How do solids acquire charges when rubbed together?***

The study of this topic will help one to explain why: solids acquire charges when rubbed together; metals are able to conduct electricity while non-metals do not, some substances are good conductors of electricity while others are insulators, some substances are able to dissolve in water while others are insoluble, why some elements are very reactive while others are less reactive or not reactive at all, to mention but a few.

Matter is particulate in nature. Each substance being is composed of tiny particles called atoms. Atoms are the tiny particles that make up elements and participate in reaction. Atoms are not indivisible as was believed before; they are made up of sub atomic units.

Electrical Charges

The particles that make up atoms have electrical charges. Scientists say that each proton has a positive charge, and each electron has a negative charge. We can show the existence of the electrons by using two inflated balloons. Two inflated balloons are tied together with a thin thread and hung. They are then rubbed simultaneously on hair and then hung as before.

When the balloons are rubbed on the hair, electrons are transferred from one surface to another. Both balloons acquire similar charges and therefore repel each other.

Activity 10.1a: Making objects attract or repel

By the end of this activity, you should be able to show that objects can either attract or repel each other depending on their charges.

What you need

- string
- plastic spoons
- ruler
- table
- sandwich wrap

What to do

1. Tie an 8-inch length of string around a plastic spoon, and label the spoon #1. Suspend it from a ruler or stick secured on a table so that it hangs on its side.
2. Label a second plastic spoon #2, and bring it close to the first spoon until the two spoons almost touch.
3. Now rub both sides of the spoon #1 several times with a piece of sandwich wrap, then rebalance the spoon.
4. Rub #2 the same way, and then bring both spoons close together. (They should move apart.). Why is this?

Activity 10.1b: To show the existence of the electrons by using two inflated balloons

By the end of this activity, you should be able to show that objects can either attract or repel each other depending on their charges.

What you need

- Balloon
- ruler with a string
- piece of wool

What to do

1. Number two balloons as #1 and #2,
2. Hang balloons from a ruler with a string. What has happened?

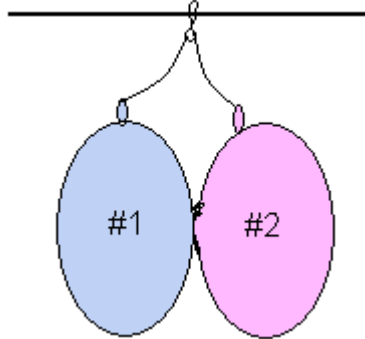


Figure 10.1a: Balloons before rubbing

3. Rub balloon #1 with a piece of wool and hang it on the string with a ruler. (Try not to make too much wind near the balloon, because wind blowing around the balloon can make it difficult to observe what happens.)
4. Rub balloon #2 in the same way, and bring the balloons together. What happened?

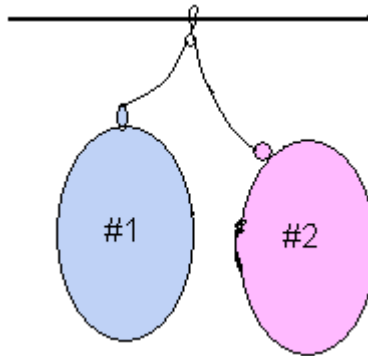


Figure 10.1b: Balloons after rubbing

Learning points

From these tests, you will realise that when their electrical charges are alike, objects repel each other; when their electrical charges are different, objects attract each other. In the atom, the electron is attracted to the proton because their charges are not alike. Is this what people mean when they say that “opposites attract?”

Materials acquire charges due to the sub atomic units they are composed of. Conduction of electricity by materials is due to the presence of charged particles in matter due to the presence of sub atomic units. Electrons are negatively charged sub atomic particles. They are found surrounding the nucleus of an atom. Electrons are very small and easy to move. When a piece of any material is rubbed with a cloth or hair, electrons go from the material onto the cloth or the hair depending on the material.

When plastic spoons are rubbed with sandwich wrap, electrons from the atoms in the spoons are transferred to the wrap, leaving the spoons with more protons (positive charges) and giving the wrap more electrons (negative charges).

Therefore, when the two positively charged spoons are brought together, they repel each other.

Activity 10.2a: Building model of the atom

By the end of this activity, you should be able to make a model of an atom showing the fundamental particles.

What you need

- Wooden spherical balls
- Wire

What to do

5. Mould spherical wooden balls to represent electrons
6. Drill a small hole through the centre of each ball
7. Weld, pass the balls through the wire according to their electronic arrangement.
8. Weld the wires into a circular shape and distribute the balls on the circular wires.
9. Connect the wires perpendicularly by welding them as shown.

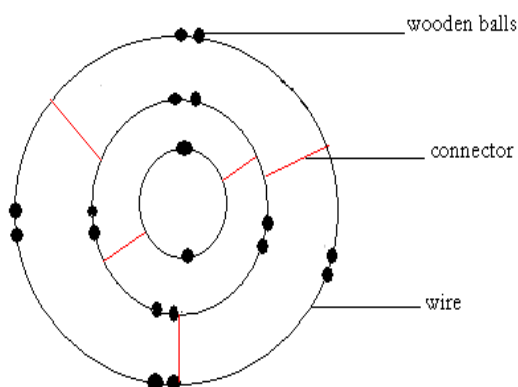


Figure 10.2: Showing arrangement of electrons in the shells

Activity 10.2b: To Make a sodium atom model

By the end of this activity, you should be able to construct a model of the sodium atom.

What you need

- 23 small balls, such as table tennis balls or styrofoam, one to two inches in diameter
- two colours of paint
- paint brush
- glue
- 11 beads, all of the same colour
- wire or pipe cleaners
- wire cutter

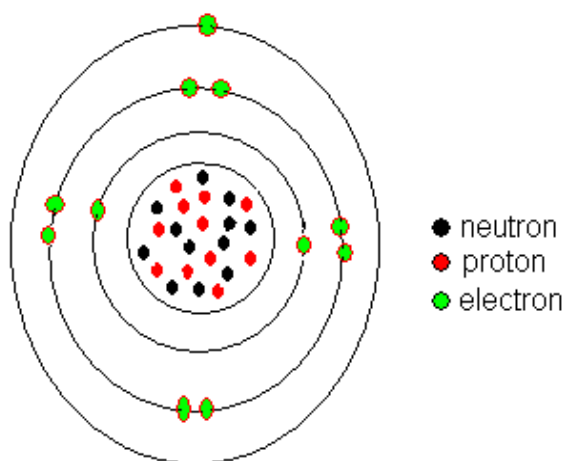


Figure 10.2: Showing arrangement of electrons around the nucleus of sodium atom

What to do

1. Divide the balls into two groups, one containing 11 balls (the protons) and the other with 12 (the neutrons). Paint all the balls in one group the same colour. Paint all the balls in the other group with the second colour. Allow the paint to dry thoroughly.
2. Glue all the balls into one cluster to represent the atom's nucleus, making sure to distribute the protons and neutrons evenly throughout the cluster. Set the nucleus aside until the glue has set.
3. Cut a length of wire or combine several pipe cleaners to be long enough to form a circle around the nucleus. Repeat this step three more times, making each wire or set of pipe cleaners long enough to form a slightly larger circle than the previous one. These circles are the electron shells.
4. Glue two beads to represent electrons onto the first circle, positioned so they are across the circle from each other. The second circle also has two beads, again positioned opposite each other on the circle. On the third shell, attach six beads spread evenly around the circle. The last shell has only one bead, or electron.

Learning points

Sodium, which has the chemical symbol Na, is one of the most commonly found elements, ranking sixth in abundance on Earth. It is highly reactive, meaning it is never found by itself in nature but is always found in combination with other elements. A single sodium atom contains 11 protons, 23 neutrons and 11 electrons. A model of this atom is not difficult to make, but it must include the correct number of protons, neutrons and electrons to be considered accurate.

Chapter 11 : Structure and Bonding

Inquiry questions

- *What is a chemical bond?*
- *How do atoms combine to form a bond?*
- *Does the way the atoms arrange themselves affect the properties of the compounds formed, for example solubility in water?*

A bond is a strong force between atoms that holds them together as molecules. When two or more atoms are close enough, an attractive force between the electrons of individual atoms and the nuclei of one or more of the other atoms can result. Thus the force that keeps the atoms together is called a **chemical bond**. To understand bonding, it is believed that the electrons surrounding the nucleus of the atoms in the outermost shell or energy level are the ones involved. The number of electrons in the outermost shell determines the Group of the element in the Periodic table and its valency. When atoms combine, they always strive to attain the noble gas electronic configuration which is very stable e.g. helium (2); neon (2.8); argon (2.8.8), etc. They can attain this stable electronic configuration by either transferring or sharing of one or more of the outermost electrons.

Element Molecules and Compound Molecules

When two or more atoms join together, they form a molecule. There are two basic types of molecules: elements and compounds.

Here is one way to construct models of each of the two types of molecules:

Activity 11.1a: To construct model of element molecule

By the end of this activity, you should be able to construct a model of an element

molecule like oxygen.

What you need

- 2 styrofoam balls
- 2-inch length heavy wire



Figure 11.1: A molecule of oxygen

What to do

1. Get two medium-sized styrofoam balls. (Each ball represents an oxygen atom.)
2. Join the styrofoam balls together with 2-inch lengths of heavy wire.
3. When two oxygen atoms join together in this way, they form one molecule of oxygen.

Activity 11.1b: To construct model of compound molecule

What you need

- styrofoam balls (small for sodium)
- styrofoam balls (big for chlorine)
- 2-inch length heavy wire

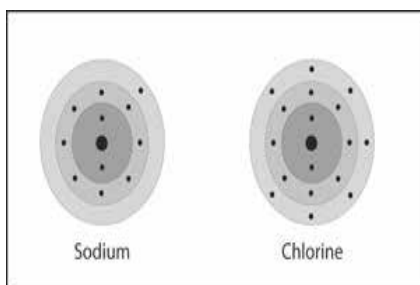
What to do

1. Obtain one small and one big Styrofoam balls. Construct a model of a salt molecule using styrofoam balls (a small one for sodium and a large one for chlorine).
2. Put the balls together with wire so that a sodium and a chlorine atom are always side by side.
3. Repeat the step above so that the sodium and chlorine atoms are up and down, in front and in back of each other (the molecules will always form a cube).

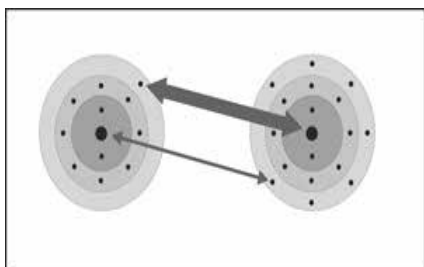
Learning points

Salt is made up of two different kinds of atoms: one sodium atom and one chlorine atom joined side by side make one molecule of salt.

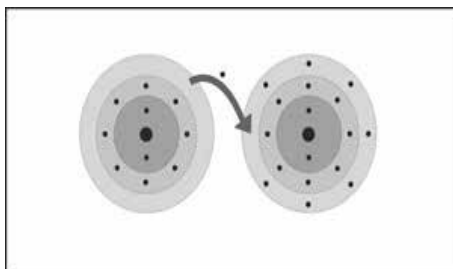
When a small piece of sodium metal is placed in a flask of chlorine gas, and a little water added to help expose the sodium, it can react with the chlorine gas. The reaction releases a lot of heat as the ionic compound sodium chloride is formed. Sodium and chlorine atoms are near each other. An electron from each atom feels the attraction from the other atom's nucleus. The attraction by chlorine is stronger than the attraction by sodium. An electron is transferred from sodium to chlorine. Sodium becomes a positive ion and chlorine becomes a negative ion. The positive and negative ions attract each other and form the ionic compound sodium chloride.



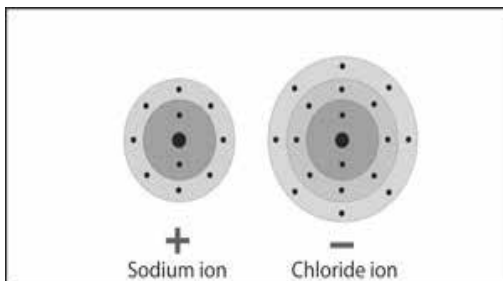
- i) Sodium and a chlorine atom are near each other



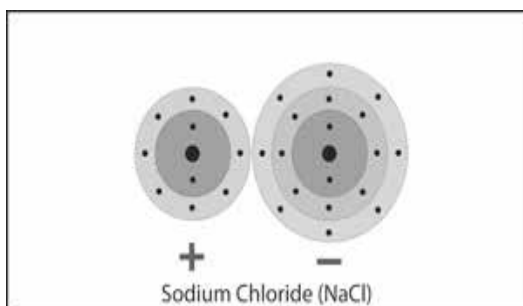
- ii) An electron from each atom feels the attraction from the other atom's nucleus



- iii) The attraction by chlorine is stronger than the attraction by sodium.



- iv) An electron is transferred from sodium to chlorine. Sodium becomes a positive ion and chlorine becomes a negative ion.



- v) Positive and negative ions attract each other and form an ionic bond and the compound sodium chloride. The sodium chloride crystal is made up of alternating sodium and chloride ions.

Figure 11.2: Formation of sodium chloride

Activity 11.2a: To Model Salt (NaCl)

By the end of this activity, you should be able to form a model of sodium chloride.

What you need

- common salt (sodium chloride)
- water
- beaker
- evaporating basin
- microscope
- beaker

What to do

1. Dissolve about 50g of common salt in water in a beaker.
2. Transfer some of the solution into an evaporating basin.
3. Heat the solution to dryness and observe what happens.

Learning points

Forming a crystal

Salt is an essential ingredient of an estuary. Basic science ideas about salts (especially NaCl) can effectively be explored by hands-on modeling of crystals of NaCl (table salt).

Imagine a solution of NaCl, as the H_2O evaporates. With less and less water present, the Na^+ and Cl^- ions come closer and closer together. Opposites attract, likes repel, so which of the following will (will not) happen?



Either of these possible?



Either of these possible?

As the (possible) trios above crowd together, which of the following will happen?



OR

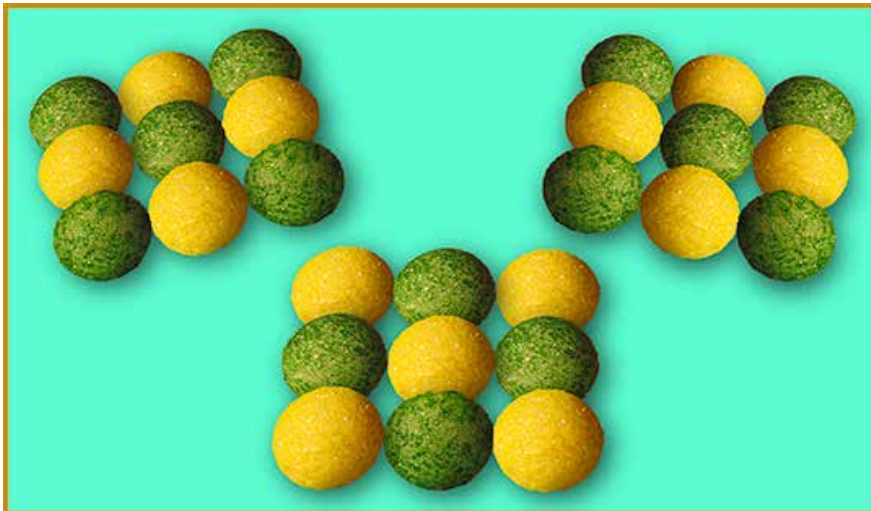


OR



?

The more water evaporates, the more Na's and Cl's stick together:



Yet even more water leaves, and these larger clumps stick to each other...



producing crystals of characteristic shape (as seen through a microscope).

Figure 11.3: Formation of sodium chloride model

Activity 11.2b: To observe actual sodium chloride crystals and relate their shape to the molecular model

By the end of this activity, you should be able see the relationship between the arrangement of ions in a model of a sodium chloride crystal and the cubic shape of real sodium chloride crystals.

What you need

- black paper
- salt
- photograph of salt crystal
- cup with salt from evaporated saltwater
- magnifier
- permanent marker

What to do

1. Place a few grains of salt on a piece of black paper.
2. Use your magnifier to look at the salt crystals in the cup. What do you observe?
3. What do the photograph, molecular model, and your observations of real salt crystals tell you about the structure of salt?

Activity 11.2c: To build a 3-dimensional model of sodium chloride

By the end of this activity, you should be able to make 1 unit of sodium chloride and then put the sodium chloride units together to form a combined model of a sodium chloride crystal.

What you need

- 2 small styrofoam balls
- 2 large styrofoam balls
- 2 toothpicks

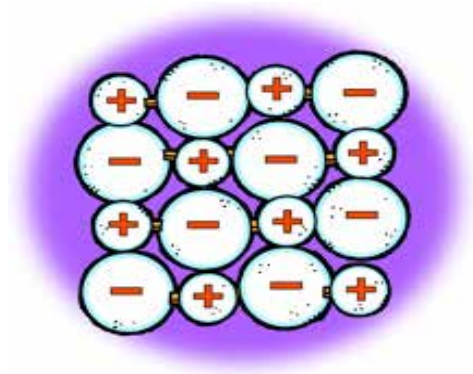
What to do

1. Use the marker to put a “-” on the large balls which represent chloride ions.
2. Use the marker to put a “+” on the small balls, which represent sodium ions.
3. Break two toothpicks in half. Use one of the half-toothpicks to connect the centres of the small and large ions together to make a unit of sodium chloride (NaCl). Do the same thing with the other small and large ball.
4. Use another half-toothpick to connect the two NaCl units in a straight line as shown.

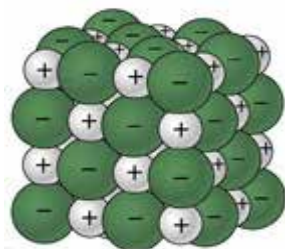


5. Put NaCl ions together to make one layer of ions.

6. Make several of such units and arrange them to make a 4×4 square of ions.
7. Use half-toothpicks to attach the ends of each line to hold the ions together.
You only need to place toothpicks in the balls at the end of each line.



8. Then build a class sodium chloride crystal by stacking these layers one over the other.



Chapter 12: Carbon and Its Compounds

Inquiry questions

- *Why is carbon and its compounds important to the natural environment and industry?*
- *Can an element, like carbon exist in different forms? If so, how does this arise?*

Carbon is used by all living things. It continuously circulates in the earth's ecosystem. In the atmosphere, it exists as the gas carbon dioxide. This is used by plants in the process of photosynthesis (manufacturing of food from water, carbon dioxide and light), where they make their own food. Animals acquire the carbon stored in plant tissue when they eat and exhale carbon dioxide as a by-product of metabolism. Although carbon is removed from circulation temporarily as coal, petroleum, fossil fuels, gas and limestone deposits, respiration and photosynthesis balance to keep the amount of carbon.

Forms of Carbon, their Structure and Properties

Activity 12.1: Manufacture of animal charcoal

What you need

- bones
- heat source

What to do

1. In a group of about five students, dig a pit which is about one square metre wide and half metre long.
2. Pile the bones collected in the pit in such a way that a small opening is left at the bottom. Carefully cover the bones with soil. Using the opening left at the bottom, a fire is lit and the bones left to burn in limited supply of oxygen until no further change occurs.
3. Leave the black spongy product formed to cool and collect in bags as animal charcoal.
4. Crush the black spongy bones into fine powder and mix it with oil and candle wax in small containers to obtain shoe polish.

Activity 12.2: Manufacture of wood charcoal

What you need

- A hoe
- an axe
- trees and dry logs

What to do

1. As a group, cut a tree (planted) and chop into pieces of about one metre in length.
2. Using the hoe, dig a pit in the ground of about half a metre deep and two metres in diameter.
3. Pile the logs inside the hole in such a way that a small opening at the bottom is left to allow for firing the logs.
4. Cover completely the logs with soil, then fire and allow it to continue burning until no more further change occurs.

5. After cooling, remove the charcoal and process them for packing in bags ready for use or sale.

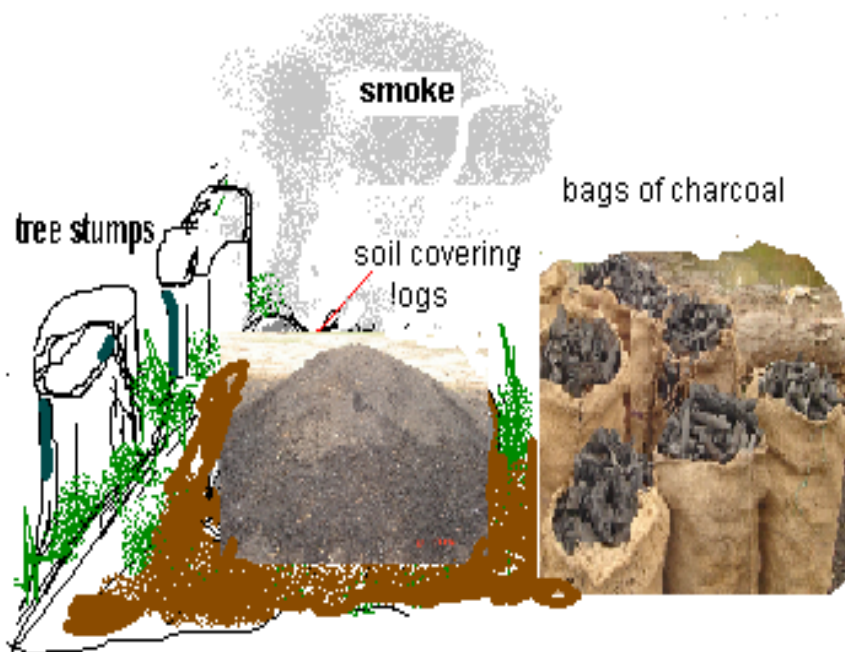


Figure 12.1: Making of charcoal

Activity 12.3a: Making shoe polish

What you need

- oil (olive)
- lemon juice

What to do

1. Combine one part lemon juice to two parts olive oil to form polish.
2. Use a clean piece of cloth to rub a small amount of the polish onto your shoes, and then allow it to soak for a few minutes.
3. Buff with another piece of cloth and enjoy your shiny shoes!

Note

- Apply to clean dry shoes
- Repeat monthly or as needed to extend the life of your leather shoes

Activity 12.3b: Making shoe polish

By the end of this activity, you should be able to make shoe polish and use it.

What you need

- pot
- bee wax
- bar of soap
- stove
- potassium carbonate
- sugar
- charcoal powder
- gun powder

What to do

1. Put some water in a pot and place in a stove. Bring it to boil on high heat and add bee wax. Melt it completely, stir in your bar soap, which should have graded into flakes.
2. Add potassium carbonate to the water-wax mixture. You may see more foaming and this is normal. Keep boiling the mixture until it begins to

form a paste.

3. Turn your stove off and remove the pot. Add the gun powder, sugar and charcoal into the pot and mix.

Learning Points

Carbon, symbol C, is the element that is crucial to the existence of living organisms. It has many important applications. Carbon is found to occur naturally in form of a free element in different forms. Carbon can also be obtained by burning wood, bones or oils in limited supply of air. The products of this incomplete combustion are used for cooking, as an adsorbent (charcoal), for poison adsorption and making of shoe polish. Coal which occurs naturally and can be extracted is used in some countries such as China to generate energy to run industries or produce electricity. In this chapter, the oxides (i.e. carbon monoxide, CO, and carbon dioxide, CO₂) will particularly be investigated.

Carbon exists in different forms as diamond, graphite and amorphous carbon (charcoal, lamp black, soot, coke). Diamond and graphite are crystalline and amorphous carbon is non-crystalline. The existence of an element in more than one form without change of state is called **allotropy**. The various forms are called the **allotropes**. Other allotropic elements include: sulphur and phosphorus. Allotropes have similar chemical properties, but different physical properties. To understand how the different forms of carbon can arise, you will construct different models to show the arrangement of the atoms in the different forms.

*Information

Graphite crystals consist of layers of carbon atoms, and each carbon atom is joined to three others by means of covalent bonds. In between the covalent bonds are free electrons. Graphite conducts electricity and heat because it has delocalised electrons (free and mobile electrons). It is used as a lubricant because the layers can slide over each other.

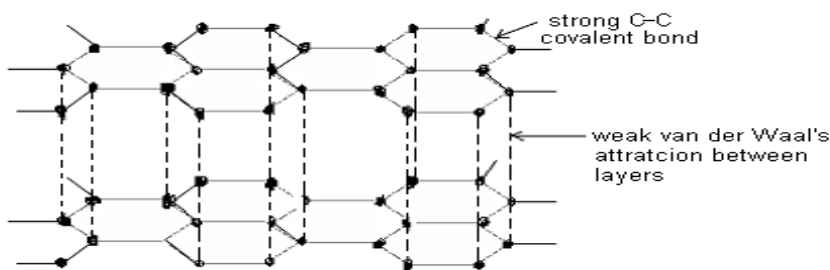


Figure 12.2: Carbon graphite

Diamond has a tetrahedral structure in which each carbon atom is joined to four others by strong covalent bonds. The tetrahedral arrangement of carbon atoms gives diamond a very strong three-dimensional structure which is difficult to break. This arrangement explains why diamond is the hardest substance. All the valence electrons of carbon are used in diamond, and therefore, has no delocalised electrons. This again explains why diamond is a poor conductor of heat and electricity.

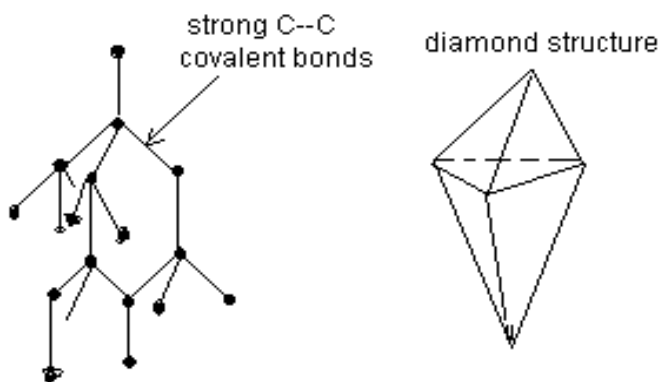


Figure 12.3: Showing carbon and diamond structures

Follow-up activity

Visit a shop or a super market in the vicinity or in your town and list as many products as possible which you think are made of carbon.

Chemical Properties of Carbon

By the end of this section, the learner should be able:

1. to prepare carbon dioxide in the laboratory.
2. to state the uses of carbon dioxide and carbon monoxide.

Activity 12.4: Burning carbon in oxygen

By the end of this activity, you should be able to test for the product formed when carbon burns in oxygen.

What you need

- carbon powder
- Bunsen burner
- 2 test tubes
- combustion spoon
- rubber bung or cork

What to do

1. Obtain carbon powder and place it on a combustion spoon.
2. Heat the carbon powder on the combustion spoon until it glows, and then hold it in the boiling tube containing oxygen. What have you observed?
3. When it has burned completely, take out the combustion spoon and replace the bung so that you can later test it for the gas formed.
4. Which elements have you used to make the gas?
5. Name the compound you have made?
6. How could you test for the gas to prove that it is really the gas you have named?

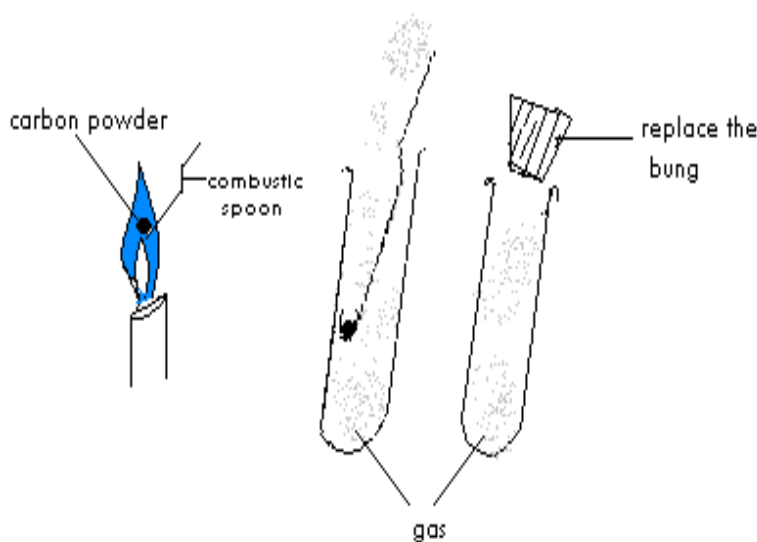


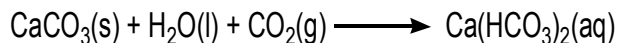
Figure 12.4: Burning of carbon in air

Note: Carbon monoxide is very poisonous. The experiment **MUST** be carried out in plenty of air or in the open.

Try to carry out the test yourself.

Write the word equation and a chemical equation for the reaction between carbon and oxygen.

Note: Lime water (calcium hydroxide solution) reacts with carbon dioxide to form the substance called calcium carbonate which is insoluble and turns lime water milky.



Carbon dioxide is formed when carbon burns in plenty of oxygen and carbon monoxide is formed when carbon burns in limited supply of oxygen.

Activity 12.5: Preparing carbon dioxide in the laboratory

By the end of this activity, you should be able to carbon dioxide in the laboratory.

What you need

- Test tubes (2)
- Delivery tube
- Cork/rubber bung
- Calcium carbonate (marble chips)
- Hydrochloric acid
- Lime water

What to do

1. Obtain a test tube and add a few marble chips into it.
2. Obtain another test tube, and add about 2 mL of lime water into it.
3. Add about 10 ml of dilute hydrochloric acid onto the marble chips in the test tube and immediately close it with a rubber bung fitted with a delivery tube. What did you observe? Sulphuric acid is not a suitable mineral acid to use in the preparation of carbon dioxide. With the aid of an equation, explain this statement.
4. Direct the delivery tube to the other test tube containing lime water and observe. Write equations of the reaction.
5. Carbon dioxide is tested by passing the gas through lime water which turns milky due to the formation of calcium carbonate. With the aid of equation(s), explain what is observed.
6. State at least four uses of carbon dioxide.

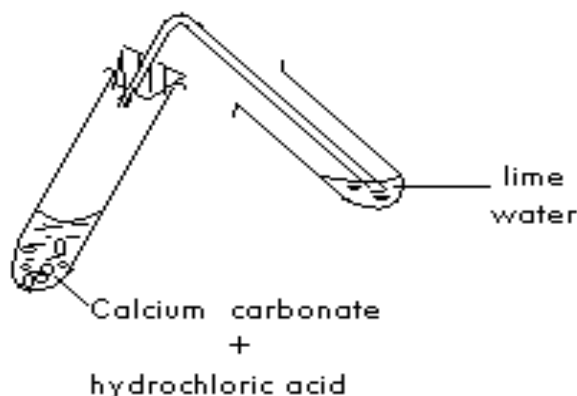
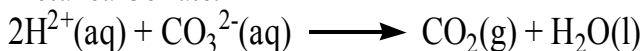


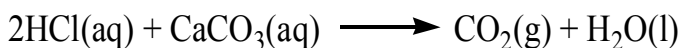
Figure 12.5: Preparation of carbon dioxide

Learning Points

Carbon dioxide is a colourless gas and forms 0.03% of the earth's atmosphere. It is produced by living things during respiration and when it burns in plenty of air. In the laboratory, carbon dioxide is prepared by the action of a mineral acid on a metal carbonate.



For example,



* Information

Burning charcoal needs a plentiful supply of air so that the charcoal can oxidize completely to harmless carbon dioxide. If the air supply is limited, for example in a narrow and small kitchen with tiny or no ventilation, incomplete combustion occurs and leads to the formation of carbon monoxide.



Carbon monoxide is a very poisonous gas. Blood contains haemoglobin, which combines with oxygen to form oxyhaemoglobin. This moves in the blood to all parts of the body and supplies them with oxygen. The haemoglobin returns to the lungs and combines with more oxygen. Carbon monoxide combines with

haemoglobin to form bright red carboxyhaemoglobin. In fact, that is why the meat of animals appears bright red on slaughter. This stops the blood from absorbing more oxygen. A person who inhales carbon monoxide becomes ill or dies by suffocation. The gas is more dangerous than other poisonous gases because it has no colour and smell.

Carbonates and Hydrogen Carbonates

Activity 12.6: Action of heat on carbonates

By the end of this activity, you should be able to describe the action of heat on carbonates.

What you need

- calcium carbonate powder
- zinc carbonate powder
- lead carbonate powder
- copper(II) carbonate crystals
- sodium carbonate powder
- five (5) test tubes
- spatula
- source of heat
- splint
- lime water
- litmus papers

What to do

1. Obtain a spatula end full of calcium carbonate and place it in a test tube.
2. Using a test tube holder, hold the test tube with the carbonate in a source of heat, heat gently and then strongly until there is no further change. What did you observe?

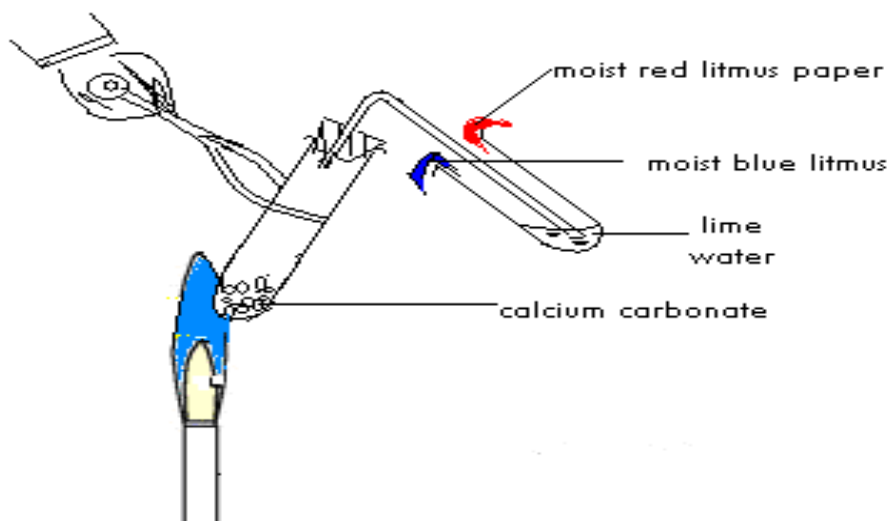


Figure 12.6: Showing action of heat on carbonates and hydrogen carbonates

- Test any gases evolved with litmus paper and lime water as in Figure 10.4. What is the nature of the gases evolved?
- Repeat procedures 1-3 using zinc carbonate, lead(II) carbonate, copper(II) carbonate and sodium carbonate. What observations do you make?
- Tabulate your results as in the table below:

Test	Observation(s)	Equation of reaction
CaCO_3		
ZnCO_3		
PbCO_3		
CuCO_3		
Na_2CO_3		
$(\text{NH}_4)_2\text{CO}_3$		

Explain what happens when ammonium carbonate was heated?

Activity 12.7: Making of bread (mandazi)

By the end of this activity, you should be able to make use of the carbonates and hydrogen carbonates to make mandazis.

What you need

- wheat flour (baking flour)
- baking powder (ekitumbise)
- sugar
- water
- cooking oil
- source of heat



Figure 12.7: Mandazi

What to do

1. One cup (about $\frac{1}{2}$ kg) of wheat flour is mixed with three table spoonfuls of baking powder in an open container e.g. a saucepan. What is the function of the baking powder in making bread?
2. To the mixture formed, add four table spoonfuls of sugar and stir the mixture thoroughly.
3. Add water a little at a time to the mixture with stirring to form a thick paste. What did you observe when water was added to the baking powder?
4. Then cut the paste into required portions or sizes and put them into relatively hot cooking oil on a heat source. Keep turning the pieces until

the whole surface turns brown-red. What happened when the dough was put in the boiling cooking oil?

5. Package the product for marketing.

Follow-up activity

1. Prepare bread from a mixture of wheat flour, baking powder and water using a locally made oven.
2. Make a report on packaging and marketing of your product.
3. Find one other domestic use of baking powder in cooking.

Learning points

Carbonates can be regarded as salts derived from carbonic acid, which is formed when carbon dioxide dissolves in water (rain water). Generally, all carbonates are insoluble in water except those of sodium, potassium and ammonium. Aluminium forms no carbonate. Many rocks in the earth's crust are carbonates. Three examples are **chalk**, **limestone** and **marble**. The shells of sea-animals consist of calcium carbonate. Coral, pearls and egg-shells are other carbonates which contain calcium carbonate. The carbonates of zinc, iron and copper occur naturally.

Hydrogen carbonates are regarded as salts derived from carbonic acid by partial replacement of the hydrogen by a metal or a metal radical. The commonest hydrogen carbonates are sodium hydrogen carbonate, NaHCO_3 and calcium hydrogen carbonate, $\text{Ca}(\text{HCO}_3)_2$. The presence of $\text{Ca}(\text{HCO}_3)_2$ is responsible for temporary hardness of water.

The Science Behind Bread Making

Bread is the product of baking a mixture of flour, water, salt, yeast and other ingredients. The basic process involves mixing of ingredients until the flour is converted into a stiff paste or dough, followed by baking the dough into a loaf.

The aims of the bread making processes are to produce dough that will rise easily

and have properties required to make good bread for the consumer.

To make good bread, dough made by any process must be extensible enough for it to relax and to expand while it is rising. Good dough is extensible if it will stretch out when pulled. It also must be elastic, that is, have the strength to hold the gases produced while rising, and stable enough to hold its shape and cell structure.

Two proteins present in flour (gliadin and glutenin) form gluten when mixed with water. It is gluten that gives dough these special properties. Gluten is essential for bread making and influences the mixing, kneading and baking properties of dough. When you first start to bake bread, learning to mix the ingredients is very important.

Bread making involves the following basic steps:

a) Mixing of ingredients

Mixing has two functions:

- i) To evenly distribute the various ingredients.
- ii) To allow the development of a protein (gluten) network to give the best bread possible.

Dough has an optimum mixing time, depending on the flour and mixing method used. Too much mixing produces dough that is very extensible with reduced elastic properties. Under-mixing may cause small unmixed patches which will remain unrisen in the bread. This will give a final loaf with a poor appearance inside.

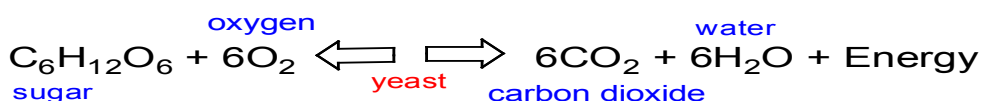
b) Rising (fermentation)

Once the bread is mixed, it is then left to rise (ferment). As fermentation takes place, the dough slowly changes from a rough dense mass lacking extensibility and with poor gas holding properties, into a smooth, extensible dough with good gas holding properties.

The yeast cells grow, the gluten protein pieces stick together to form networks, and alcohol and carbon dioxide are formed from the breakdown of carbohydrates (starch, sugars) that are found naturally in the flour.

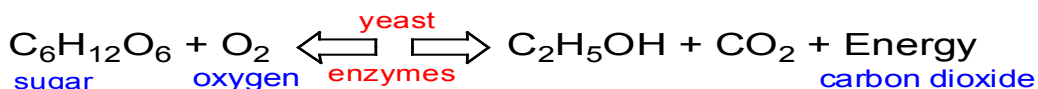
The yeast uses sugars in much the same way as we do, i.e. it breaks sugar down into carbon dioxide and water. Enzymes present in yeast and flour also help to speed up this reaction.

When there is plenty of oxygen present, the following reaction occurs:



The energy which is released is used by the yeast for growth and activity.

In dough where the oxygen supply is limited, the yeast can only partially breakdown the sugar. Alcohol and carbon dioxide are produced in this process known as alcoholic fermentation.



The carbon dioxide produced in these reactions causes the dough to rise (ferment or prove), and the alcohol produced mostly evaporates from the dough during the baking process. During fermentation, each yeast cell forms a centre around which carbon dioxide bubbles form. Thousands of tiny bubbles, each surrounded by a thin film of gluten, form cells inside the dough piece. The increase in dough size occurs as these cells fill with gas.

c) Kneading/moulding into loaf shapes

Any large gas holes that may have formed during rising are released by kneading. A more even distribution of both gas bubbles and temperature also results. The dough is then allowed to rise again and is kneaded if required by the particular

production process being used.

During the final rising (proving), the dough again fills with more bubbles of gas, and once this has proceeded far enough, the dough is then transferred to the oven for baking.

d) Rising of dough

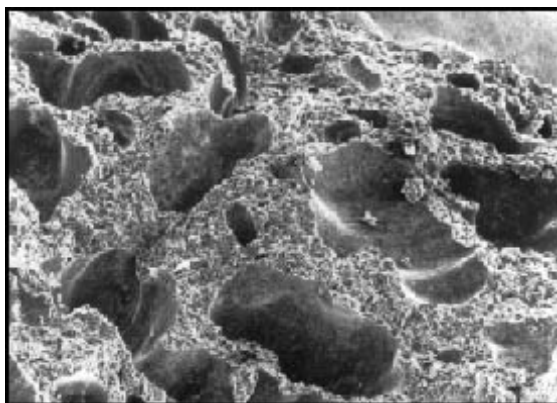


Figure 12.8a: Showing a rising dough

General appearance - large gas holes lined with gluten with smaller holes and ingredients in between these. (Electron microscope view)

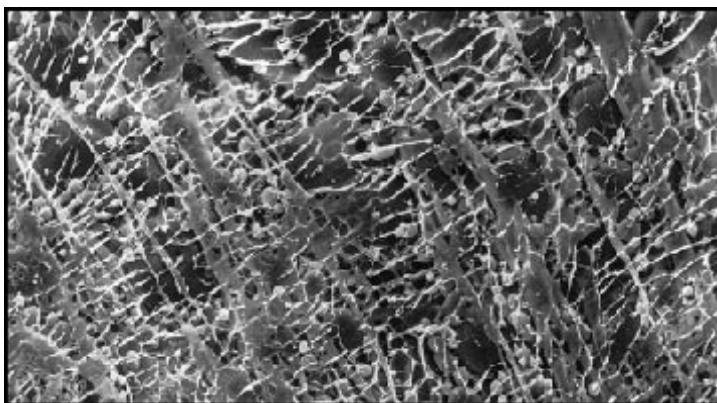


Figure 12.8b: Electron microscope view of rising dough after two hours

e) Baking

The baking process transforms the unpalatable dough into a light, readily digestible, porous, flavourful product. The physical activities involved in this conversion are complex but the fundamentals of these are explained.

- As the intense oven heat penetrates the dough, the gases inside the dough expand, rapidly increasing the size of the dough. This is called “oven spring” and is caused by a series of reactions: Gas + heat = increased volume or increased pressure. Gas pressure inside the thousands of tiny gas cells increases with the heat and the cells become bigger.
- A considerable proportion of the carbon dioxide produced by the yeast is present in solution in the dough. As the dough temperature rises to about 40°C, carbon dioxide held in solution turns into a gas, and moves into existing gas cells. This expands these cells and overall the solubility of the gases is reduced.
- The oven heat changes liquids into gases by the process of evaporation and thus the alcohol produced evaporates.
- Heat also has an effect on the rate of yeast activity. As the temperature rises, the rate of fermentation increases, and so does the production of gas cells, until the dough reaches the temperature at which yeast dies (approximately 46°C).

From about 60°C onwards, stabilisation of the crumb begins. Starch granules swell at about 60°C, and in the presence of water released from the gluten, the outer wall of the starch granule cell bursts and the starch inside forms a thick gel-like paste, which helps form the structure of the dough.

From 74°C upwards, the gluten strands surrounding the individual gas cells are transformed into the semi-rigid structure commonly associated with bread crumb strength.

The natural enzymes present in the dough die at different temperatures during baking. One important enzyme, alpha-amylase, the enzyme which breaks starch into sugars, keeps on performing its job until the dough reaches about 75°C.

During baking, the yeast dies at 46°C, and so does not use the extra sugars produced between 46-75°C for food. These sugars are then available to sweeten the breadcrumb and produce the attractive brown crust colour.

As baking continues, the internal loaf temperature increases to reach approximately 98°C. The loaf is not completely baked until this internal temperature is reached. Weight is lost by evaporation of moisture and alcohol from the crust and interior of the loaf. Steam is produced because the loaf surface reaches 100°C. As the moisture is driven off, the crust heats up and eventually reaches the same temperature as the oven.

Sugars and other products, some formed by breakdown of some of the proteins present, blend to form the attractive colour of the crust. These are known as “browning” reactions, and occur at a very fast rate above 160°C. They are the principal causes of the crust colour formation.

f) Cooling – slicing and wrapping

In bakeries, bread is cooled quickly when it leaves the oven. The crust temperature is over 200°C and the internal temperature of the crumb about 98°C. The loaf is full of saturated steam which also must be given time to evaporate. The whole loaf is cooled to about 35°C before slicing and wrapping can occur without damaging the loaf.

A moist substance like bread loses heat through evaporation of water from its surface. The rate of evaporation is affected by air temperature and the movement of cool air around the loaf.

In a bakery, there are special cooling areas to ensure efficient cooling takes place before the bread is sliced and wrapped.

Chapter 13: Organic Chemistry

Inquiry question

- ***What are organic compounds?***

In Chapter 9, we studied hydrogen, some of its sources, for example, water, acids, and some of its properties. We also learnt about carbon in Chapter 12. We found that carbon is found in non-living substances like carbon dioxide, carbon monoxide, carbonates and hydrogen carbonates. But the majority of the substances containing carbon are found in living things (plants and animals). These substances are formed from carbon, hydrogen and other elements (oxygen, nitrogen, etc). Those which consist of carbon and hydrogen only are called hydrocarbons. If another substance is attached to a hydrocarbon, the substance formed is a hydrocarbon product or derivative.

The branch of chemistry in which hydrocarbons and their derivatives are studied is called organic chemistry. Hydrocarbons include all gas and liquid fuels, oils, greases obtained from fossils. Hydrocarbon derivatives include starch, sugar, alcohols, proteins and vitamins, among others.

Activity 13.1: Manufacturing ethanol from starch (e.g. corn)

By the end of this activity, the learner should be able to prepare ethanol from local materials such as corn, cassava and others.

What you need

- distillation flask
- retort stand
- Liebig condenser
- thermometer
- beaker

- rubber tubing
- corn
- yeast
- water

What to do

1. Grind dry corn and mix with water.
2. Cover the mixture with banana leaves in a bucket or big saucepan and allow it to stand for about a week.
3. After that, roast the mixture and spread it out to dry.
4. Mix the roasted product with yeast (germinated millet or sorghum) and water and allow it to stand for at least 3 days.
5. Filter the mixture.
6. Arrange the apparatus as in Figure 13.1. Carry out distillation of the brown filtrate (which contains ethanol). What is the colour of the distillate?

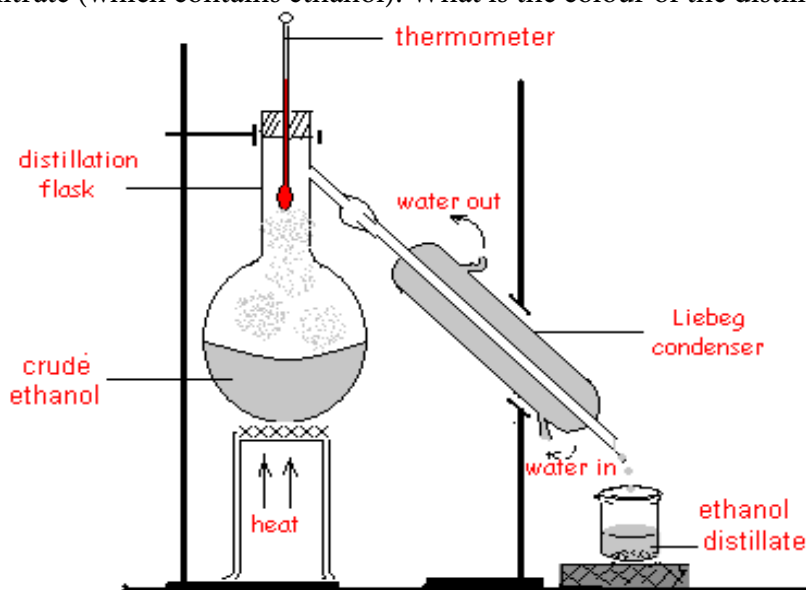


Figure 13.1: Distillation of crude ethanol

7. Purify the product by fractional distillation to remove any harmful substances (e.g. methanol).
8. Try to package and market it.

Follow-up activity

- Calculate how much ethanol (waragi) you have obtained from the amounts of the raw materials you used.
- Project how much of the raw materials you can use to obtain larger quantities of ethanol.

Activity 13.2: Manufacturing ethanol from molasses

By the end of this activity, you should be able to prepare ethanol from molasses.

What you need

- distillation flask
- retort stand
- Liebig condenser
- thermometer
- beaker
- rubber tubing
- molasses
- water
- yeast
- drum

What to do

1. Molasses, which is a by-product of cane sugar manufacture, is diluted with large volume of water and mixed with yeast and allowed to stand for 3 – 4 days.
2. The mixture is then filtered locally using a piece of clean cloth sewn as a bag.
3. The filtrate is then distilled locally to produce crude waragi using arrangement of apparatus as in Figure 13.1.
4. Purify the product to remove any harmful substances (e.g. methanol).
5. Try to package and market the product.

Follow-up activity

- Prepare ethanol from other local sources of carbohydrates e.g. bananas, millet, etc. Try to find out which one of them produces the largest volume of ethanol from the same initial amounts used.

Soaps and Detergents

One importance of chemistry in life is to help keep us clean and healthy and live longer. Soaps and detergents are cleansing agents which are widely used in life. Soaps are obtained by boiling vegetable or animal (goat) fat or oil with alkali. Detergents are, however, better cleansing agents as they can be used with any type of water and also widely used to make shampoos and shaving creams all of which make us healthy with an increased life span.

Activity 13.3a: Making of soap locally

By the end of this activity, you should be able to manufacture soap and detergents and use them.

What you need

- common salt
- banana peelings
- vegetable oil (sunflower)
- kettle
- heat source

What to do

1. Burn two basinfults of banana peelings to obtain ash.
2. Collect the ash formed, add two litres of water, shake and filter with a piece of cloth.
3. Boil the filtrate (solution obtained) with one litre of vegetable oil for about 40 minutes while continuously stirring with a wooden rod and each time adding some little water to replace the one which boils off.
4. Add about half a litre of strong concentrated solution of common salt and then allow cooling. What do you think is the purpose of the common salt?
5. Remove the hard crust at the surface of the liquid and, using wires, process them into bars.
6. Sell your soap

Activity 13.3b: Preparing soap in the laboratory

By the end of this activity, you should be able to prepare soap and market it.

What you need

- Lard (fat)
- sodium hydroxide, NaOH, pellets
- sodium chloride, NaCl, technical grade
- ethanol, C₂H₅OH, 95%
- boiling chips
- Erlenmeyer flask, 125-mL

- beakers, 2 400-mL, 150-mL
- scissors
- graduated cylinders, 25-mL, 125 or 250-mL
- funnel
- spatula
- stirring rod
- cheesecloth
- ring stand
- clamp to hold Erlenmeyer flask
- watch glass

What to do

1. Weigh 5 g of solid sodium hydroxide, NaOH, into a clean 100 or 150-mL beaker. Add 15 ml of ethanol, and 15 ml of distilled water to the beaker. Stir to dissolve the sodium hydroxide.
2. Place about 5 g of lard or shortening into a 125 mL Erlenmeyer flask.
3. Add the solution of sodium hydroxide to the lard or shortening in the flask.
4. Add 2 or 3 boiling chips to the flask, and cover it with a watch glass.
5. Arrange the apparatus as shown in Figure 13.2. If a hot plate is used, support the flask over a beaker of water placed on the hotplate surface.
6. Heat the flask in the steam just above the water surface of a boiling water bath. As the water in the bath evaporates, lower the reaction flask so the bottom almost makes contact with the boiling water surface. Remember to temporarily remove the watch glass while you make this adjustment so that it does not fall and break.
7. Allow the reaction to proceed for at least 30 minutes.
8. While saponification proceeds, prepare a salt solution by dissolving about 25 g of sodium chloride, NaCl, in 150 ml of water contained in a 400 ml beaker.
9. Prepare a mixture of about 25 mL of ice water.

10. After 40 minutes turn off your burner or the hotplate, remove the flask from the heat (you can use the clamp securing the flask as a handle) and pour the hot reaction mixture into the salt solution.

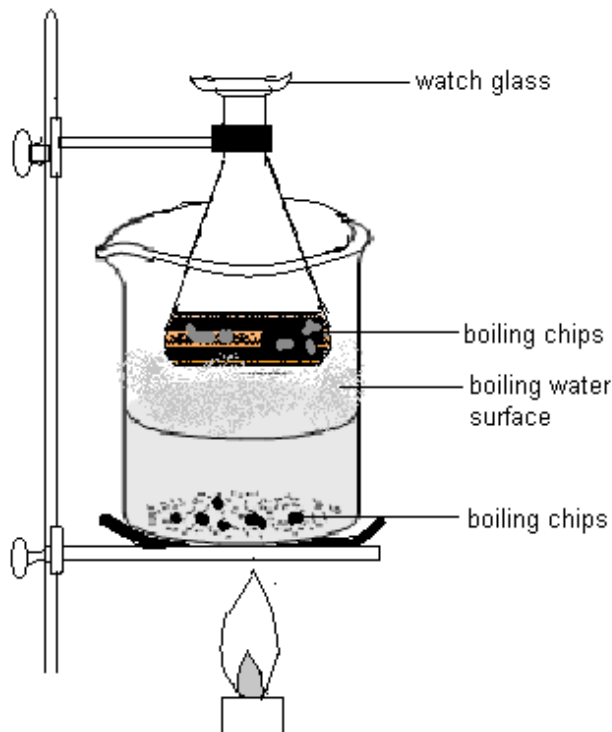


Figure 13.2: Apparatus for the preparation of the soap

11. Pour a small amount of the mixture from the beaker back into the reaction flask to rinse out most of the remaining reaction mixture. Add the rinse material to the beaker.
12. Stir the resulting mixture in the beaker thoroughly for several minutes.
13. Filter the mixture through a funnel containing 2 or 3 layers of cheesecloth.
14. Wash the collected soap twice with 10 mL portions of ice-cold water.
15. Remove the cheesecloth from the funnel, and twist it to remove excess water from the soap. Save the soap for Part C of the experiment.

Activity 13.4: Preparation of a detergent

By the end of this activity, you should be able to prepare a detergent in the laboratory.

What you need

- dodecanol (dodecyl alcohol), $C_{12}H_{25}OH$
- sulfuric acid, H_2SO_4 , concentrated
- sodium hydroxide, NaOH, 6M
- phenolphthalein solution, 1%
- sodium chloride, NaCl, technical grade
- Erlenmeyer flask, 125-mL
- beakers, 400-mL, 150-mL, 100-mL
- graduated cylinders, 10-mL, 25-mL, 125-mL
- funnel
- spatula
- stirring rod
- cheesecloth
- watch glass
- scissors

What to do

1. Place 5 ml of dodecanol into a 100-mL beaker.
2. Measure 2 mL of concentrated sulfuric acid, H_2SO_4 , in a 10-mL graduated cylinder.
3. With stirring, slowly add the 2 mL of concentrated sulfuric acid to the dodecanol in the beaker. Continue to stir for 1 minute after addition of the sulfuric acid is complete. Let the mixture stand for 10 minutes.
4. Fill a 250-mL beaker one-third full of ice, add about 10 g of sodium chloride, NaCl, and mix thoroughly.
5. Add water to bring the total volume of the mixture to 75 ml.
6. Mix 5 ml of 6M sodium hydroxide with 10 mL of water in a small beaker.

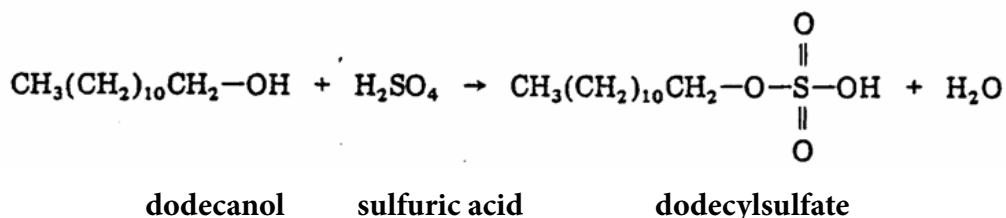
Mix well, and then add 4 drops of phenolphthalein indicator. The pink colour of the phenolphthalein may begin to fade in the strongly basic solution.

7. Prepare a mixture of about 25 mL of ice water.
8. After the 10 minutes, carefully pour the sodium hydroxide solution into the dodecanol-sulfuric acid mixture. Stir until the pink colour disappears. A large amount of solid detergent should form.
9. Pour the detergent mixture into the ice-salt bath. Stir to break up large lumps of detergent.
10. Filter the precipitated detergent mixture through 2-3 layers of cheesecloth in a funnel mounted on a ring stand.
11. Wash the collected detergent twice with 10 ml portions of ice-cold water.
12. Remove the cheesecloth from the funnel, squeeze excess water from the solid detergent, and save the detergent.

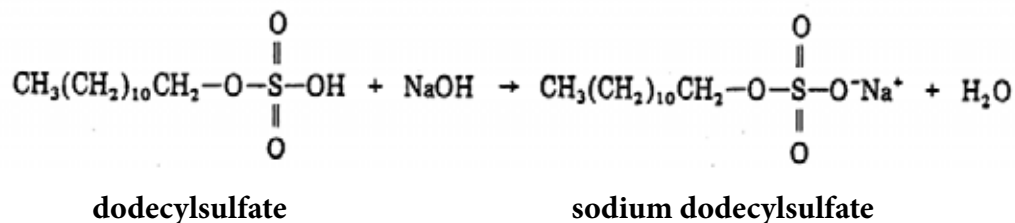
Liquid Soap

Making liquid soap has two basic processes and these include: Making the lye-Water Solution for Your Cold Process Soap and mixing the lye-water with lard (fat).

A synthetic detergent, a sodium alkyl sulfate called sodium dodecylsulfate, will be prepared by reacting dodecyl alcohol (dodecanol) with sulfuric acid.



The resulting dodecylsulfate is converted to the sodium salt by a reaction with sodium hydroxide.



Activity 13.5a: Making lye-water

By the end of this activity, you should be able to:

1. make a lye-water
2. use the lye-water to make liquid soap



Figure 13.3: Making lye-water

What you need

- rain water
- wooden barrel
- cork about 3 inch long
- bricks
- hay or grass
- clean rocks

What to do

1. Start a rain barrel to catch soft water. This is a key step. Depending upon how much lye you want to leach, make sure that you have 2 or 3 gallons of soft water before you proceed.
2. Find a local brewer's supply house and pick up a wooden barrel and a cork about 3" long. You can use a cask-sized or waist-high barrel.
3. Take the barrel home and drill a hole in it approx. 2" above the bottom. Make sure that the cork will fit securely into the hole.
4. Find a place that the barrel will be undisturbed. Lye is caustic. Take the necessary precautions. Put some bricks down and place the barrel on top of them. The brick base must be stable. It raises the barrel up so that you can easily drain off the lye into a container when it is ready. Give yourself room to work.
5. Cover the bottom of the barrel with some palm-sized clean rocks (e.g. river rock). Cover the rocks with approximately 6" of straw (this can be hay or grass). This will filter the ashes and help your lye drain cleanly.
6. Gather branches and/or logs of oak, ash, or fruitwoods. Remember that the best lye is made from hardwoods. Avoid pine, fir, and other evergreens. Palm leaves work well if they are completely dried and brown.
7. Burn it outside in a pile, or better yet, use it in your fireplace or woodstove.
8. Scoop the ashes out and put them in the prepped barrel. (Make sure that the ash is completely cold, or you could set your barrel and anything around it on fire.) You can fill the barrel with ash, but it is not necessary; you can make smaller amounts with less ash.
9. Put a pan under the hole and remove the cork. Pour the soft water in until you see it start to drain into the pan, then put the cork back in tightly. The water level should be about 6" from the top. After a day, the first ash

should settle and you can add more ash.

10. Let it sit for at least 3 days. You can add ash all week and drain it regularly on a specific day of the week.
11. Check to see if your lye is ready. For what purpose are you leaching this lye? Body soap or heavy cleaning? Lye concentration gets stronger with each leaching. For average soap making, you can use these measures: Drop a fist-sized potato or a raw egg into the barrel. If it floats enough for a quarter-size piece to rise above the water, it is ready. If it doesn't, you need to add more ashes or drain all the water and re-leach it (pour it back into the cask and let it set one more cycle).
12. Make sure that you have a wooden crock or glass container to catch your lye when it's ready. Put it under the tap, gently pull the cork, and fill your containers. Leave enough head room so that they will be safe and easy to pour. Make sure that you have tight, fitting lids.
13. Store your lye in a cool dark place until use. (The sooner the better.)

Tips

- Do not start this project until you have collected 2-3 gallons of rain water and have purchased or scavenged all of your supplies.
- Make sure that your lye barrel has a stable foundation and is in a secure place where it cannot be knocked over by, for example, roving children.
- If you run a dehumidifier, its collected water is an alternative to rainwater.
- You can also use electrically distilled water. The purer the water, the more potassium that can be leached from the ashes. Do not use bottled spring water or water from the tap! You can use bottled distilled water that was processed using steam distillation.
- To dispose of old, leached ashes, dig a hole away from everything and pour the muck into it. Don't cover it until the ashes dry thoroughly.

Learning points

Lye, also known as NaOH, sodium hydroxide, or caustic soda, is used in making soap, and also in biodiesel fuel production. Caustic Potash, or Potassium Hydroxide, is also called Lye and can be easily made at home. It can also be used in the production of biodiesel although the recipe will have to be adjusted somewhat. This recipe is for KOH, potassium hydroxide, since hardwoods leach potassium and about 1/10th as much sodium.

Activity 13.5: To make a new bar of soap from used bars of soap



Figure 13.4: Showing used pieces of soap

What you need

- A cheese grater
- a double boiler
- a 450g of used soap shavings
- 355 ml of liquid (milk, water, tea, etc.)
- a mold for soap
- plastic wrap
- non-stick vegetable spray
- stainless steel large spoon
- rubber/latex gloves

What to do

1. Put on your rubber/latex gloves.
2. Prepare your mold(s) with a non-stick vegetable spray.
3. In a double boiler, heat the liquid you chose (milk, water, tea, etc.) to 77 - 82 °C.
4. Add the grated soap while stirring slowly and constantly.
5. Reduce the heat so that the liquid/grated soap comes to a simmer.
6. Stir slowly but not constantly until the grated soap turns into liquid.
7. Add additives if you want (this is not required)- such as exfoliators, essential oils, etc. and stir them in .
8. Immediately pour the mixture into the mold(s) and cover it up in plastic wrap.
9. After 24 hours, remove the plastic wrap and set molds in a draft-free, dry place for three-four weeks.

Learning points

If you have too many leftovers from used soaps, this is a great way to make new soaps using old leftovers. (By the way, this process of making soap is called hand-milling.)

Chapter 14: Mole concept

Inquiry questions

- ***What is a mole?***
- ***Is the mole of any use or application in everyday life?***

The mole is one of the central concepts in Stoichiometry. We will see it come up time and time again. A mole is analogous to a dozen in that both refer to a specific number of items. A dozen of eggs is 12 eggs. A dozen of cars is 12 cars. A dozen of water molecules is 12 molecules.

A mole refers to a specific number of molecules: Avogadro's number or 6.02×10^{23} molecules.

A mole of eggs would be 6.02×10^{23} eggs, which is so many eggs it would never arise in everyday life. But a mole of water molecules, or 6.02×10^{23} water molecules, is about the amount in a small test tube of water. Just as the dozen is a useful quantity for counting eggs, the mole is a useful quantity for counting molecules.

The mole allows us to connect the macroscopic world, where we all live, to the molecular scale, where chemistry occurs.

Chemistry is a qualitative and quantitative science. Generally you have been studying chemistry in a qualitative sense. In this chapter you will look at the quantitative aspect of chemistry and that is stoichiometry. Stoichiometry is the mole to mole relationship in a balanced chemical equation. This unit provides the opportunity to apply chemical principles to everyday life and industry. When studying reactions, you need opportunities to investigate the usefulness of the reactions.

Activity 14.1a: To measure a mole of water

By the end of this activity, you should be able to measure and explain what a mole of water is.

What you need

- plastic syringe
- water
- beaker

What to do

1. Use the syringe to transfer 18 cm^3 of water to the beaker.
2. Estimate the mass the 18 cm^3 water you transferred into the beaker, the density of water at room temperature is approximately 1 g/cm^3 . What mass of water is in the beaker?
3. Estimate mass of one mole of water from the discussion questions below. How many moles of water are in the beaker?

Notes

The density of water at room temperature is approximately 1 g/cm^3 . Therefore, 18 cm^3 of water is 18 g of water. ($1\text{ ml} = 1\text{ cm}^3 = 1\text{ cc}$)

Activity 14.1b: To measure one mole of a gas (by constructing a box holding a mole of gas)

By the end of this activity, you should be able to construct a box holding a mole of a gas.

What you need

- card board,
- knife,
- ruler,
- masking tape
- super glue

What to do

1. Draw three sides of a box each having 28.2 cm long.
2. Cut the pieces to obtain equal size.
3. Fold the edges to obtain the cube.
4. Bind the edges of the box with glue or masking tape.
5. Calculate the volume of the box .
6. Work out the mass of one mole of a gas.

Discussion questions

1. Calculate the volume of the box in cubic centimetres.
2. Convert the volume obtained into cubic decimetres.

Notes

The volume of any gas at s.t.p contains the volume equal to the volume of this box, i e: 22.4 dm^3 . This is called gramme molecular volume (G.M.V.) when the molecular weight of a given gas is expressed in grams. e.g.: 2g of $\text{H}_2(\text{g})$, 32g of $\text{O}_2(\text{g})$, 17g of $\text{NH}_3(\text{g})$ and 44g of $\text{CO}_2(\text{g})$.

Activity 14.2: Making a preservative

By the end of this activity, you should be able to make a preservative based on the mole ratios of the reactants.

What you need

- 0.01M acetic acid
- 0.01M sodium acetate
- beaker
- freshly prepared fruit juice

What to do

1. Mix equal volumes of 0.01M acetic acid and sodium ethanoate.
2. Add the mixture to the freshly prepared fruit juice.

Comment

A mixture of sodium benzoate and benzoic acid or citric acid and sodium citrate can also act as a preservative.

Activity 14.3a: To make barium sulphate as a soil pH tester

By the end of this activity, you should be able to make barium sulphate by reacting equimolar solutions of barium chloride and sulphuric acid.

What you need

- 2M barium chloride/barium nitrate
- 2M sulphuric acid
- 2 beakers (250 ml)
- 1 stirrer
- 1 filter funnel
- 2 measuring cylinders (200 ml)



Figure 14.1: Showing preparation of barium sulphate

What to do

1. Transfer 100 ml of 2M barium chloride into a clean beaker using a measuring cylinder.
2. Using another measuring cylinder, measure out 100 ml of 2M sulphuric acid and transfer it into a clean beaker.
3. Add the sulphuric acid onto the barium chloride while stirring.
4. Using a filter funnel, filter the mixture and retain the residue.
5. Wash the residue several times with distilled water.
6. Dry the sample and pack it for use.

Activity 14.3b: To find the pH of the different soil samples taken from the school botanical garden

By the end of this activity, you should be able to use the barium sulphate made in activity 14.3a to find the pH of different soil samples taken from the school garden

What you need

- boiling tube
- rubber bung
- universal indicator
- distilled water
- soil sample (A: The leaf litter; B: The humus layer; C: The humus mixed with minerals layer and D: The mineral layer mixed with very little humus)
- barium sulphate powder

What to do

1. Set up the apparatus as shown in the diagram below.

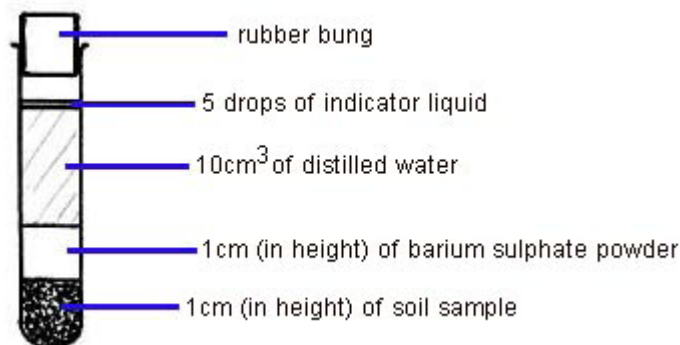
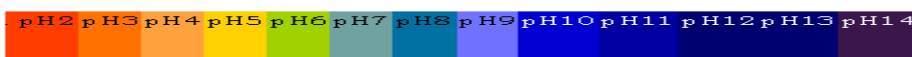


Figure 14.2: Testing pH of soil sample

2. Shake the test tube vigorously for a few seconds. (NB. Place your thumb over the rubber bung to stop it falling out. Shake until the soil sample completely mixes with the rest of the contents of the test tube.)
3. Label your test tube and replace it in the test tube rack. Leave it until the contents have settled. What do you observe?
4. Repeat points 1, 2 and 3 with the three other soil samples.
5. Compare the colours of the liquid in each of the four test tubes with the standards provided. Record your results.
6. Write a conclusion to this experiment. Are there any differences in the pH levels of the four soil layers? Explain the reason for any differences. If your results are not as you might have expected, suggest why.

Learning points

The pH scale measures the acidity or the alkalinity of a substance. The pH scale goes from pH1 to pH14. pH1 is the most acid, pH14 is the most alkaline and pH 7 is neutral (neither acid nor alkaline).



Living things cannot tolerate a pH much above or below pH7 (neutral).

The barium sulphate powder is neutral. It is included in each tube to bind with the particles of soil so that they fall to the bottom of the tube after it has been shaken. This allows you to see the colour of the liquid more easily.

The pH is used to measure the acidity or alkalinity of the growth medium. Plants grown in soils with inappropriate pH will fail. For example, the public are fond of purchasing heathers and other acid-loving species, planting them in alkaline soils and complaining when they grow poorly or die. In addition, the availability of nutrients changes with pH, becoming either deficient or toxic in the extremes. The pH at which most nutrients are freely available for absorption by plant roots is pH 6.5 in soils and pH 5.8 in composts. Similarly, there are many plants grown in commercial crop production that will only tolerate a narrow band of acidity. Accurate pH management is therefore vital to ensure success.

A horticulturalist will need to know:

- the importance of pH.
- methods of measuring soil pH.
- specific plant pH preferences.
- methods to correct soil acidity by liming (raise pH).
- methods to increase soil acidity by sulphur (lower pH).

Chapter 15: Energy Changes in Chemical Reactions

Inquiry question

- *Are chemical energy changes of any use in our daily lives?*

Chemical energy is a concept that is clearly related to every single process of life on earth. Simply put, chemical energy is the potential of a chemical system to undergo a transformation from one system to another; or to impart a transformation on another chemical system. Through simple redox chemistry, energy can be harvested from nearly every molecule on earth. Chemical energy powers the cars that we drive, fuels our bodies, creates glucose, and all living creatures on the planet.

Activity 15.1: Making biogas

By the end of this activity, you should be able to construct a biogas digester plant.

What you need

- hand washing can with tap
- tubing
- Burner
- cow dung

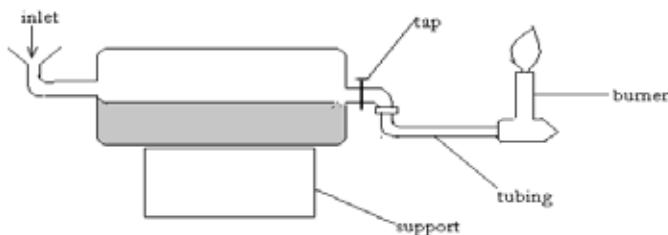


Figure 15.1: Showing a biogas digester

What to do

1. A fabricated hand washing can with tap is placed horizontally on a support.
2. Animal refuse is mixed with water to form slurry and then poured into the can through the inlet.
3. Allow the setup to stay for some days. What do you observe?
4. Biogas is produced and then allowed through the tubing to the burner.
5. Light the biogas formed at the burner. What is the colour of the flame? Is it possible to use the burning flame for cooking or lighting? If so explain your answer.

Follow up activity

Try to design a biogas digester which can produce biogas on a large scale.

Activity 15.2: To Make a homemade biogas plant

By the end of this activity, you should be able to construct a homemade biogas digester.

What you need

- barrel
- large ball-shaped vessel
- pipes
- low-power pump
- organic waste

What to do

1. Set up the barrel as the digester. This is where you will place the daily waste supply, so this vessel must be large enough to contain the waste and ferment it. The barrel should have some room at the top for the gas to form. Temperature is an important element in adequately fermenting the waste into biogas, so you may want to place the unit partially in the ground to keep the temperature around it as constant as possible.
2. Add the large ball-shaped vessel as the central storage unit for the gas. Be sure this storage vessel is rounded so the gas can move safely. It is usually recommended that you place the storage unit at a higher elevation than the digester, so the gas can be pumped upward.
3. Connect the pipes and the low-power pump from the barrel, where the scraps are fermenting, to the storage vessel. The pump will help to move the gas to the storage vessel where you can then connect it to your home as a fuel source.
4. Place scraps and waste into the digester and allow it to ferment for at least eight hours. Smaller pieces of scraps will decompose more quickly, so if possible break the scraps into pieces.

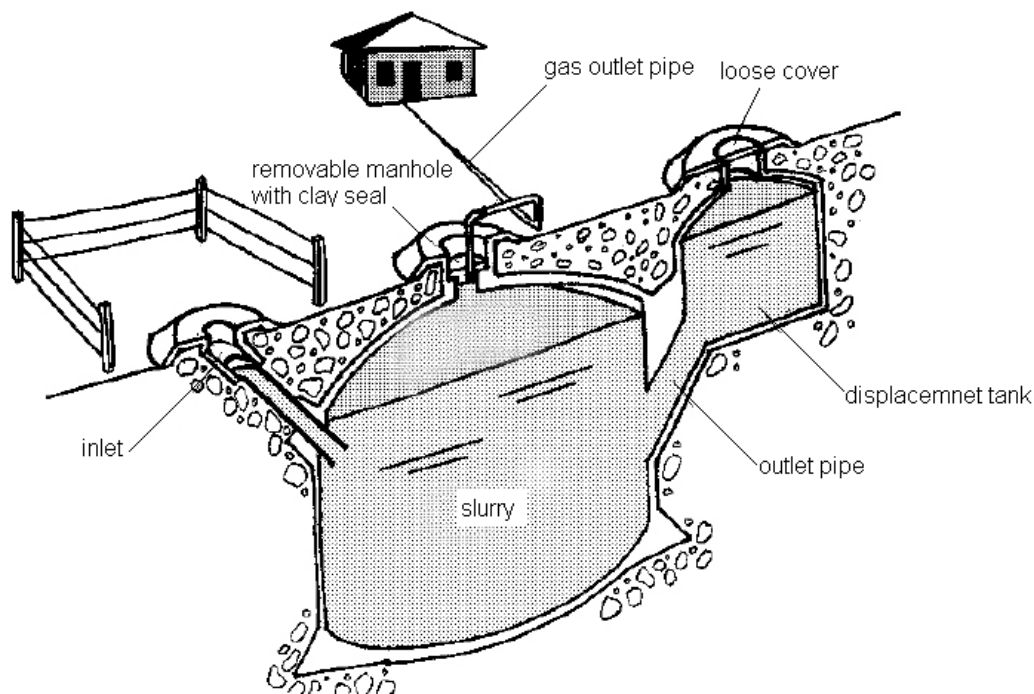


Figure 15.2: A typical biogas plant

Learning points

Biogas is the natural product of bacterial digestion of organic wastes. It is 50 to 65 percent or more methane gas, which is a powerful greenhouse gas when released into the atmosphere, but burned as a fuel for cooking, heating or even running a generator, it is cleaner than gasoline or wood. Biogas can be made either aerobically (in contact with the air) or anaerobically in a sealed chamber, which is the more common approach as it makes collecting the gas simpler and avoids the unpleasant odour. A small-scale batch digester can be constructed for home use.

Chemical Reactions

- *How many objects or products can you think of in everyday life that either give off heat or absorb heat?*
- *Which of these items involve physical processes or chemical processes?*
- *How useful are they in daily lives?*

Chemicals can store energy and release it in the form of heat. When the substances are mixed and new substances are formed, a chemical reaction has occurred.

The change may be either physical or chemical. The changes can be classified as being endothermic or exothermic.

A chemical reaction that releases heat is called an exothermic reaction. But chemical reactions can also absorb heat from the environment and get cold. These reactions are called endothermic reactions. When chemicals are dissolved in water, sometimes heat is released, and sometimes heat is absorbed.

Activity 15.3a: Making hot packs

By the end of this activity, you should be able to construct a hot pack.

What you need

- thermometer
- stirring rod
- 3- 50 ml beakers
- stop watch
- 100 ml graduated cylinders
- stand and clamp
- string and scissors
- weighing machine and spatula
- calcium chloride, CaCl_2
- sodium chloride, NaCl
- distilled water
- large bag
- small bag

What to do

1. Add approximately 117 cm^3 of distilled water to the large bag. Place the bag inside a beaker to keep it upright.
2. Add 4.5 g calcium chloride to the small bag. Partially close the bag leaving some of the seal open.
3. Place the small bag containing the calcium chloride into the large bag.

4. Seal the large bag.
5. Activate the heat pack by squashing the inner bag to release the calcium chloride and shaking the water and calcium chloride together. Observe what happens.

Activity 15.3b: Making Cold Packs

By the end of this activity, you should be able to construct a cold pack.

What you need

- thermometer
- Stirring rod
- 3- 50 ml beakers
- stop watch
- weighing machine and spatula
- 100 ml graduated cylinders
- stands and clamp
- string and scissors
- ammonium nitrate, NH_4NO_3 / ammonium chloride, NH_4Cl

What to do

1. Add approximately 117 cm^3 of distilled water to the large bag. Place the bag inside a beaker to keep it upright.
2. Add 4.5 g ammonium nitrate to the small bag. Partially close the bag leaving some of the seal open.
3. Place the small bag containing the ammonium nitrate into the large bag.
4. Seal the large bag.
5. Activate the heat pack by squashing the inner bag to release the ammonium nitrate and shaking the water and ammonium nitrate together. Observe what happens.

Follow-up activity

a) Try to investigate the best way of making a hot pack as follows:

1. Measure out 100 cm³ distilled water into a beaker.
2. Measure the temperature of the water.
3. Weigh out 10 g of calcium chloride.
4. Add the calcium chloride to the water, stirring occasionally.
5. Record the temperature of the solution.
6. Repeat 1-5 for different masses of calcium chloride.
7. Draw a graph of the highest temperature against the mass of calcium chloride used. Use your graph to decide how much calcium chloride you need to make a heat pack that heats up to 40°C–50°C.

Learning points

Hot and cold packs are used by athletes to minimise swelling of injuries such as muscle and joint sprains. Their construction is based on the phenomenon that when substances react, they either release or absorb energy.

If the dissolving of the chemical in water is an endothermic process and absorbs heat energy, it is a good candidate for making a cold pack because this process will lower the temperature of the content of the pack. If the dissolving of the chemical in water is an exothermic process and releases heat energy, it is a good candidate for making a hot pack because this process will raise the temperature of the content of the pack.

They are constructed of a large pouch containing a dry chemical plus an inner pouch of water. The hot/cold pack is activated by breaking the seal on the pouch of water and shaking the pack vigorously. This action mixes the water with the chemical starting the exothermic or endothermic reaction. Heat is given off as a result of a chemical reaction.

Chapter 16: Electrochemistry

Inquiry question

- *How useful is electrochemistry in everyday life?*

Electrochemistry is the area of chemistry dealing with the interconversion of electrical energy and chemical energy. There are many applications of this in everyday life. Batteries, control of corrosion, metallurgy and electrolysis are just a few examples of the applications of electrochemistry.

Electrochemistry always involves an oxidation-reduction process. Recall that oxidation involves the loss of electrons by a substance and that reduction involves the gain of electrons by a substance. You can't have one process without the other.

Activity 16.1: Making a copper-zinc battery

By the end of this activity, you should be able to construct a simple copper – zinc battery.

What you need

- thin strips of copper and zinc metals
- 250 ml beaker
- two 8 inch test tubes
- voltmeter
- 1 liter 0.5 M copper(ii) sulphate
- 1 liter 0.5 M zinc sulphate
- ring stand and two clamps or a burette clamp.
- voltmeter
- apparatus for setting up a working galvanic cell. This is best done with a beaker and unglazed porcelain cup. You can also use the H-tube apparatus.

If you wish to set up a non-working cell, assemble the equipment shown in Figure 16.1. For the working model, use Figure 16.2 or 16.3.

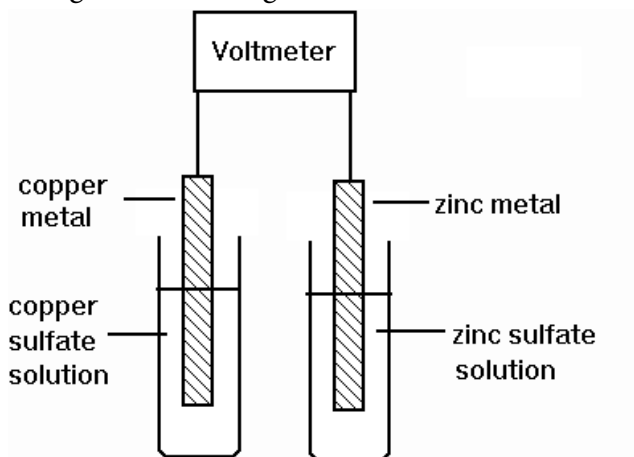


Figure 16.1: First try battery. It won't work.

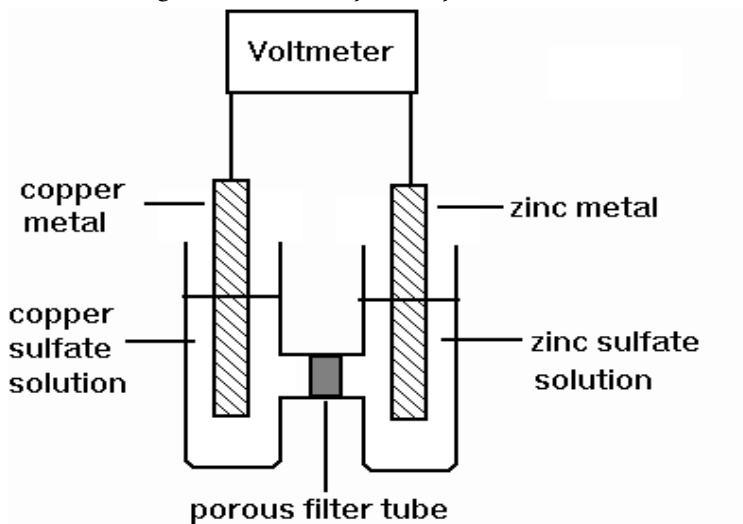


Figure 16.2: H-tube connecting test tubes with a porous partition

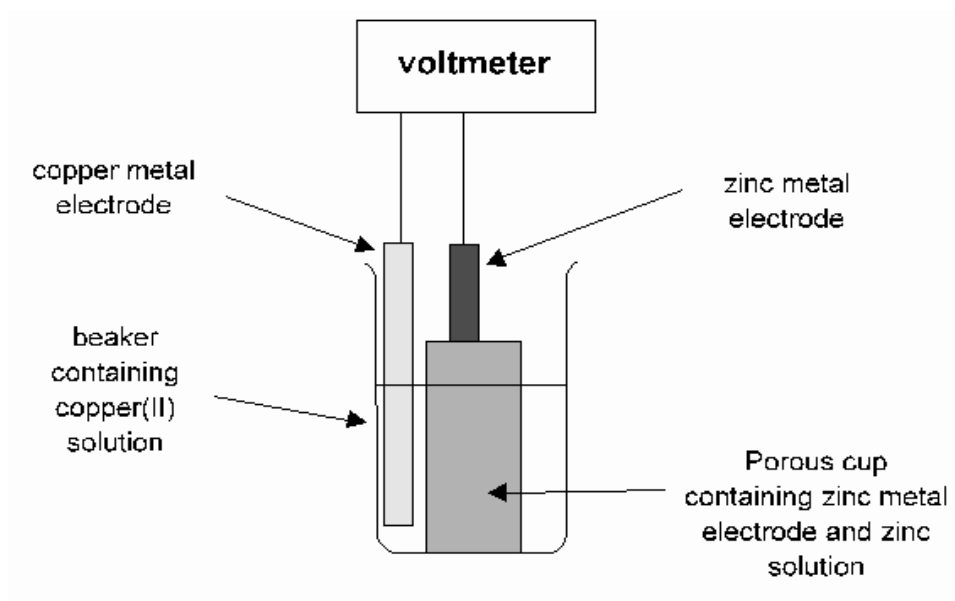


Figure 16.3: Copper-zinc battery made with porous cup separating solutions

What you need

- large glass container
- strip of zinc metal
- strip of copper metal
- solution of zinc sulfate
- solution of copper(II) sulfate
- porous barrier
- two pieces of wire
- small light bulb and socket

What to do

1. Place the porous barrier in the glass container so it divides it into two equal parts.
2. Attach a wire between the copper strip and the light bulb and socket. Attach the second wire between the zinc strip and the light bulb.

3. Place the metal strips in the container on opposite sides of the barrier.
4. Add the copper sulphate solution on the side of the barrier that contains the copper strip, and the zinc sulphate on the side of the barrier with the zinc.

Learning points

Since the zinc and copper have different levels of reactivity, electrons will soon begin to flow from the zinc metal to the copper metal. As the zinc strip begins to lose electrons, it will have a positive charge. As more electrons gather on the copper strip it becomes negatively charged. This forces the negative ions in the copper sulphate to flow through the barrier toward the positive zinc strip, thus replenishing its electrons. The light bulb should light up as the current flows between the two strips.

Activity 16.2: To Generate Electricity Using Citrus Fruits

By the end of this activity, you should be able to construct a lemon battery and use it to generate electricity.

What you need

- wire cutters
- 18 gauge, stiff copper wire
- rough grit sandpaper
- zinc strip
- lemon
- multimeter



Figure 16.4: Showing a lemon

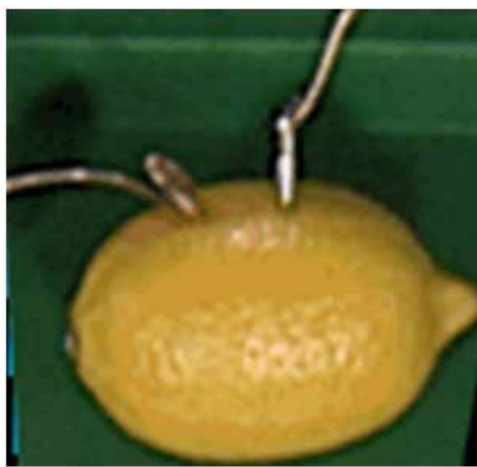


Figure 16.5: Showing copper wires connected through a lemon

What to do

1. Strip the copper wire if it is covered in insulation or any kind of plastic or rubber coating. Use the wire cutters to cut away the insulation from around the wire and peel off. Discard the insulation or plastic that was covering the wire. Cut two inches of copper wire with wire cutters.
2. Cut a two-inch piece from a zinc strip. Alternatively, you can use a straightened steel paperclip instead of the zinc strip.
3. Use the sandpaper to smooth any rough areas of the copper wire or zinc strip. Rub the wires with the sandpaper until they are completely smooth.
4. Squeeze the lemon firmly between your hands, or run it firmly across a tabletop or counter. You want to rupture some of the pockets of lemon juice inside the lemon and make it mushy inside. This is the best way to conduct the electricity. Do not rupture the outside skin of the lemon.
5. Push the copper wire and the zinc strip into the lemon. Place them as close as possible, but do not allow the wires to touch. Place the lemon on a firm surface-do not let it sit in water or any other liquid.

6. Touch the tops of the wires with the tip of a multimeter, which should be able to pick up traces of electricity from the wires and lemon. You can also touch the tip of your tongue to the wires to feel a slight tingle of electricity. The voltage is so mild; it will not hurt or cause any damage to your tongue or body.

Follow up activity

Try an experiment with all the citrus fruits and record which fruit produces the strongest electrical current.

Learning points

Citrus fruits contain the perfect amount of acid to help fuel a battery made from copper and zinc. The acid in the citrus fruit helps the electrodes move from one wire to the other, creating an electrical current. Lemons and limes work better than oranges for this project, because they contain a higher concentration of acid.

Activity 16.3a: Plating metal substance with copper

By the end of this activity, you should be able to electroplate another metal with copper.

What you need

- copper(II) sulphate solution (1M)
- cell holder and 2 cells
- crocodile clips
- copper rod
- switch
- connecting wire
- beaker (500 ml)

What to do

1. Put about 200 cm^3 of 1M copper(II) sulphate in a beaker
2. Arrange the apparatus as shown in Figure 16.6.

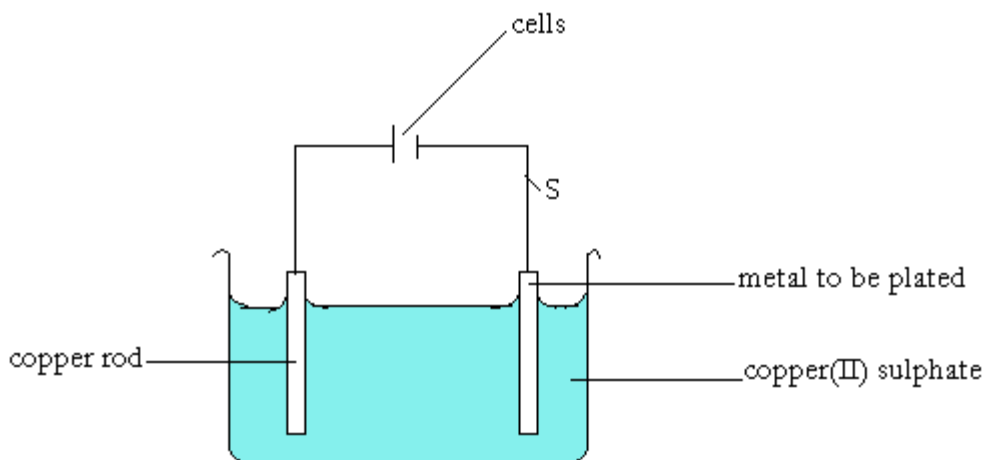


Figure 16.6: Electrolysis of copper(II) sulphate solution using copper electrodes

3. Close the switch (S) to complete the circuit. What do you observe? Why was copper(II) sulphate used as the electrolyte but not any other compound?

Activity 16.3b: Electroplating: Copper-Plated Key



By the end of this activity, you should be able to construct an electrolytic cell for copper plating a brass key.

What you need

- 1.5-V D battery with battery holder
- two crocodile clips or insulated wire
- beaker or glass
- copper sulphate
- copper electrode (or coil of copper wire)
- brass key
- safety equipment

What to do

1. Prepare the key for copper-plating by cleaning it with toothpaste or soap and water. Dry it off on a paper towel.
2. Stir copper sulfate into some hot water in a beaker until no more will dissolve. Your solution should be dark blue. Let it cool.
3. Use one alligator clip to attach the copper electrode to the positive terminal of the battery (this is now the *anode*) and the other to attach the key to the negative terminal (now called the *cathode*).

4. Partially suspend the key in the solution by wrapping the wire lead loosely around a pencil and placing the pencil across the mouth of the beaker. The alligator clip should not touch the solution.
5. Place the copper strip into the solution, making sure it doesn't touch the key and the solution level is below the alligator clip. An electrical circuit has now formed and current is flowing.
6. Leave the circuit running for 20-30 minutes, or until you are happy with the amount of copper on the key.

Learning points

Electroplating uses a form of **electrolysis** in which the electrodes play a bigger role than just conducting the current. Using electricity, you can coat the metal of one electrode with the metal of the other! Jewelry and silverware can be silver- or gold-plated, while zinc is often used to coat iron to protect against rust. Professional electroplating requires specialised chemicals and equipment to make a high-quality coat.

Chapter 17: Nitrogen and some of its Compounds

Inquiry questions

- *Can nitrogen be of importance and yet it is said to be inert?*
- *How can ammonia solution be prepared in the laboratory and what precautions should be observed?*

Nitrogen Gas

Nitrogen is a colourless, odourless, tasteless gas which is slightly lighter than air. Nitrogen is one of the most abundant elements. About 80 percent of the air we breathe is nitrogen. It is found in the cells of all living things and is a major component of proteins. Inorganic nitrogen may exist in the free state as a gas (N_2), or as a nitrate (NO_3^-), nitrite (NO_2^-), or ammonia (NH_3^+).

Nitrogen-containing compounds act as nutrients in streams in rivers. Other nitrogen bearing compounds are found in the soils, for example, ammonium sulphate, $(\text{NH}_4)_2\text{SO}_4$, potassium nitrate, KNO_3 and calcium nitrate, $\text{Ca}(\text{NO}_3)_2$. Nitrate reactions (NO_3^-) in fresh water can cause oxygen depletion. Thus, aquatic organisms depending on the supply of oxygen in the stream will die. The major routes of entry of nitrogen into bodies of water are municipal and industrial wastewater, septic tanks, and animal wastes. Bacteria in water quickly convert nitrites (NO_2^-) to nitrates (NO_3^-).

Activity 17.1: Preparing nitrogen gas in the laboratory

By the end of this activity, you should be able to prepare nitrogen gas in the laboratory given the materials.

What you need

- round-bottomed flask
- delivery tube
- source of heat
- stand and clamp
- 25 mL measuring cylinder
- gas jars
- rubber bung
- sodium nitrite, NaNO_2
- ammonium chloride, NH_4Cl
- distilled water

What to do

1. Mix a spatula-full of sodium nitrate with another of ammonium chloride and then put the mixture in a round-bottomed flask.
 2. Dissolve the solid in about 20 mL of water.
- Arrange the apparatus as shown in the Figure 12.1.
 - Heat the solution gently and collect any gas evolved over water after allowing the first bubbles to escape. What did you observe when the mixture was heated? Why were the first bubbles of the gas produced not collected immediately, but allowed to escape?

Note: Avoid overheating as the reaction is exothermic. Overheating may cause explosion.

- Write the chemical equation for the preparation of nitrogen gas in the experiment above.

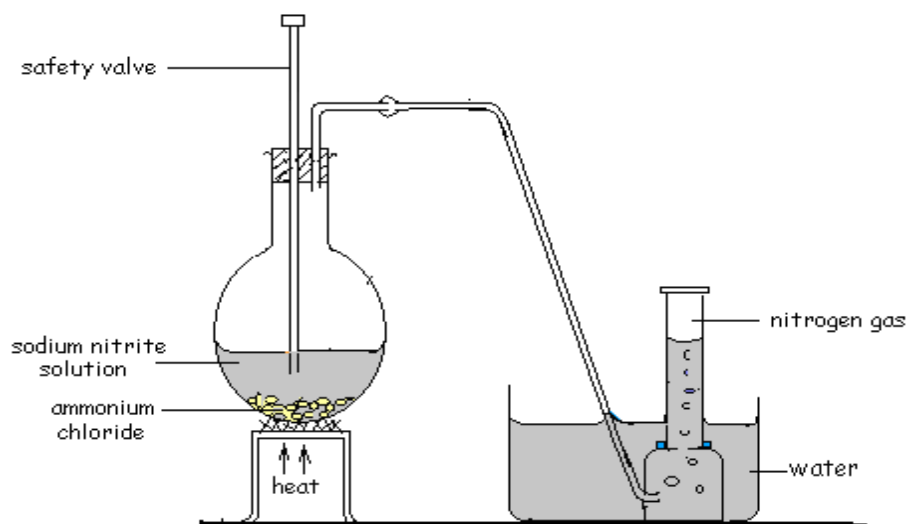


Figure 17.1: Preparation of nitrogen gas

Why is the gas produced collected over water?

Follow-up activities

- Mention at least three other methods of preparing nitrogen in the laboratory.
- Ask your teacher to describe how nitrogen gas can be manufactured industrially.
- List as many uses of nitrogen you know as possible.

Activity 17.2: Making fertilizer by using knowledge of nitrogen chemistry

By the end of this activity, you should be able to make a sample of a nitrogen containing fertilizer.

What you need

- urine (urea) – human/animal
- ash
- dry grass
- cow dung/chicken droppings/goat dung

What to do

1. In groups of five, dig a pit of about one metre deep.
2. Put some dry grass in the pit.
3. Add into the pit cow dung/chicken dung/goat dung.
4. Sprinkle some ash on the dung.
5. Sprinkle urea (urine) on the dung as well.
6. Cover the pit with dry grass.
7. Add soil to cover the pit.
8. Leave the pit for one week to allow the reaction of nitrogenous compounds in urine, and cow dung/chicken dung/goat dung with ash (salts) to form fertilizers, ammonium sulphate, potassium nitrates, etc.

Follow-up activities

- Carry out an experiment by putting fertilizers in soil and planting some crops to observe the effect of fertilizers on growth of plants by comparing the soil where there are no fertilizers.
- Cost your fertilizers.
- Apply the knowledge on fertilizers back in your homes/villages.

Ammonia Gas

Ammonia is a covalent compound. It has a molecular mass of 17.

Activity 17.3: Preparing ammonia gas in the laboratory

By the end of this activity the learner should be able to prepare ammonia gas using the materials given.

What you need

- round-bottomed flask
- gas jars
- lime tower
- source of heat
- wire gauze
- two retort stands
- delivery tube
- rubber tongs
- ammonium chloride
- calcium hydroxide
- quicklime (lumps)

What to do

1. Mix equal amounts of calcium hydroxide and ammonium chloride.
2. Place the mixture in a round-bottomed flask.
3. Arrange the apparatus as shown in the Figure 17.2.
4. Heat the mixture and collect five jars of ammonia by upward delivery.

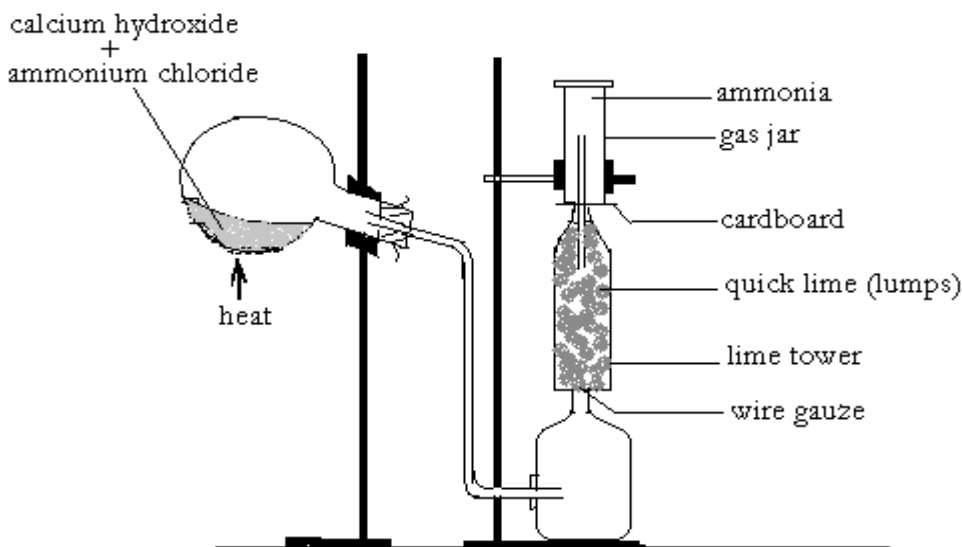


Figure 17.2: Preparation of ammonia gas

Follow-up activity

- Find an alternative arrangement of apparatus for preparation of ammonia.

Learning points

Ammonia, NH_3 , is a colourless, pungent gas, highly soluble in water. A saturated aqueous (water) solution of ammonia contains 45% ammonia by weight at 0° and 30% at ordinary room temperatures. On solution in water, ammonia becomes ammonium hydroxide, NH_4OH , which is strongly basic and similar in chemical behaviour to the hydroxides of the alkali metals.

Ammonia was known to the ancients, who derived both the name and the substance from **sal ammoniac**, which was produced at the Temple of Jupiter Ammon in Libya by the distillation of camel dung. During the Middle Ages in Europe, ammonia was obtained by heating the horns and hoofs of oxen and was called spirits of **hartshorn**. Free ammonia was obtained by the German alchemist Basil Valentine; its composition was determined by the French chemist Comte Claude Berthollet in about 1777.

In the 19th century, the principal source of ammonia was the destructive distillation of coal. It was an important by-product of the manufacture of fuel gases. Today, most ammonia is produced synthetically from hydrogen and nitrogen by the Haber process. Ammonia is an important refrigerant and is widely used in chemical industries, especially in the manufacture of fertilizers, nitric acid, and explosives.

Ammonia melts at -77.7°C , boils at -33.35°C and has a density of 0.68 at its boiling point and 1atm of pressure.

Ammonia, NH_3 , is a compound of nitrogen and hydrogen. It is formed when any ammonium salt is heated with an alkali, either solid or solution. To prepare a solution of ammonia in water (ammonium hydroxide), the gas is directly passed into water via an inverted funnel.

Activity 17.4: Making ammonium sulphate fertilizer

By the end of the activity, you should be able to:

1. explain that fertilizers contain nitrogen which helps plants to grow well.
2. describe how some fertilizers are manufactured.
3. describe what happens if fertilizers are removed.

What you need

- ammonia gas/ammonia solution
- sulphuric acid (dilute 2M)
- Bunsen burner
- glass rod
- litmus papers
- evaporating dish
- measuring cylinder

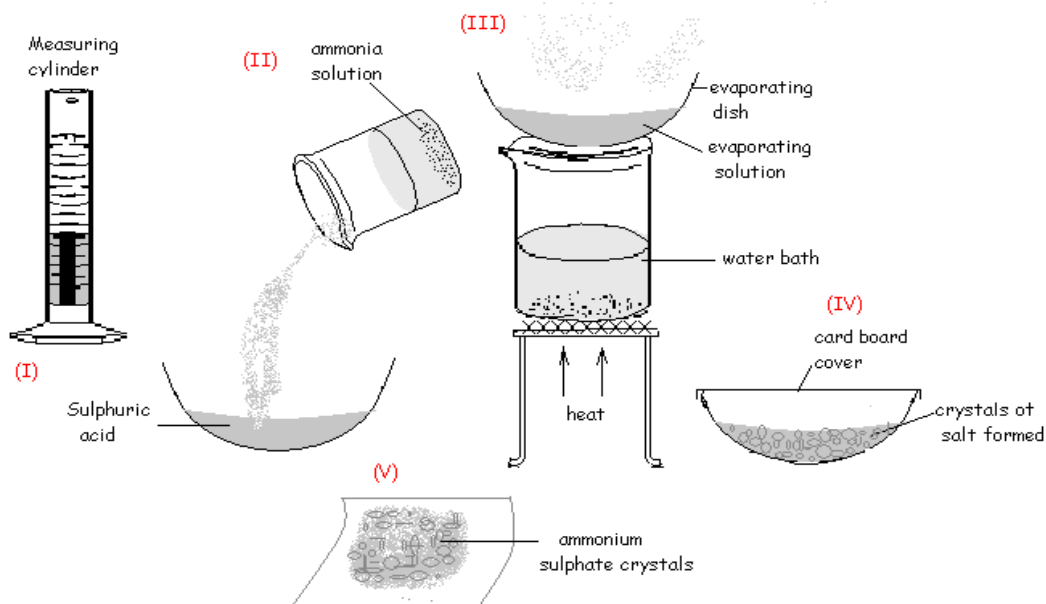


Figure 17.3: Showing preparation of ammonium sulphate

What to do

1. Place 30 cm³ of dilute sulphuric acid (2M) in an evaporating dish.
2. Add ammonia solution until the mixture has a distinctive smell of ammonia. Hold a piece of damp red litmus paper over the basin to confirm the presence of ammonia. What did you notice?
3. Cover the dish and let the solution cool.
4. Put the crystals on a piece of paper and allow them to dry.
5. What is the colour of the product? What is the use of this product?

Note: This is a neutralisation reaction. The sulphuric acid and aqueous ammonia react according to the equation below:



What is the likely explanation of your observation?

Follow-up activity

- Prepare ammonium nitrate and ammonium phosphate fertilizers following the above procedure.
- Apply the prepared fertilizer to some crops in your compound and observe their effect on the growth of crops. Compare them with other plants without fertilizers.

Activity 17.5a: Making ammonium nitrate fertilizers

By the end of this activity, you should be able to prepare a sample of ammonium nitrate fertilizer.

What you need

- ammonia gas/ammonia solution
- nitric acid (dilute 2M)
- Bunsen burner

- litmus papers
- evaporating dish
- measuring cylinder
- glass rod

What to do

1. Follow the steps as laid in Figure 17.3 for preparation of ammonium sulphate fertilizer.
2. Mix ammonia solution with a solution of dilute nitric acid in the molar ratio of 1:1 in the beaker ($2\text{NH}_3(\text{aq}) : \text{HNO}_3(\text{aq})$).
3. Cover the mixture and leave it for ten minutes to allow complete reaction.
4. After ten minutes, concentrate the solution of ammonium nitrate formed to obtain crystals of ammonium nitrate fertilizer. Write a chemical equation for the reaction that takes place.

Follow-up activity

Calculate how much ammonium nitrate you have obtained from the amounts of the chemicals used.

Activity 17.5b: Making ammonium phosphate fertilizers

By the end of this activity, you should be able to prepare a sample of ammonium phosphate fertilizer.

What you need

- ammonia solution
- phosphoric acid (dilute 2M)
- Bunsen burner
- litmus papers
- evaporating dish
- measuring cylinder
- glass rod

What to do

1. Follow the steps as laid in Figure 12.3 for preparation of ammonium sulphate fertilizer.
2. Mix ammonia solution with dilute phosphoric acid in the molar ratio of 3:1 ($3\text{NH}_3(\text{aq}) : \text{H}_3\text{PO}_4(\text{aq})$).
3. Cover the mixture and leave it for ten minutes to allow complete reaction.
4. After ten minutes, concentrate the solution of ammonium phosphate formed to obtain crystals of ammonium phosphate fertilizers.
5. Write an equation for the reaction which took place.

Follow-up activity

- Calculate how ammonium nitrate you have obtained from the amounts of the chemicals used.

Learning points

Plants and animals need nitrogen in their development i.e. building proteins. Animals obtain nitrogen by eating plants, but plants must obtain their nitrogen from the soil or from the air. Plants cannot use atmospheric nitrogen directly, only the leguminous plants “fix” nitrogen with the help of bacteria on their root nodules. Others absorb water soluble nitrogenous compounds from the soil through their roots.

When dead plants and animals decay, they form ammonium salts. Nitrifying bacteria in the soil convert ammonium salts into soluble nitrates which plants can use. Nitrogen is absorbed by plant roots as nitrate (NO_3^-) or as ammonium (NH_4^+). As plants remove nitrogen compounds from the soil, the soil becomes less and less fertile. The nitrogen compounds should be replaced. One way of doing this is to add compounds which contain nitrogen; the nitrogenous fertilizers. We can use natural fertilizers, such as manure, or make use of chemical fertilizers. The natural processes of replacing nitrogen are:

- i) during thunderstorms, some nitric acid is produced which reacts with the soil to form nitrates.

- ii) when plants and animals die, they decay. Their protein is changed into ammonium salts.

The natural processes of replacing nitrogen are too slow and inadequate to balance the natural nitrogen cycle as compared to man's intensive use of land. One way of restoring the balance is to use nitrogen fertilizers to keep a high level of nitrates in the soil. A variety of fertilizers are used, but most contain nitrogen as a compound e.g. NPK fertilizers are a mixture of compounds nitrogen (N), phosphorus (P) and potassium (K) which are all needed by plants to grow well. Although nitrates are important, they can be washed out of soil into rivers and lakes. This can cause eutrophication and excess nitrates in drinking water.

Chapter 18: Sulphur and Some of Its Compounds

Inquiry question

- *How useful is sulphur and its compounds to our lives?*

Sulphur is a bright yellow non-metallic solid. It occurs as free element, mostly in underground deposits in Texas, and Louisiana (USA) where it is deposited in limestone. It also occurs in nature as a wide range of sulphur containing compounds, the most common being; metallic sulphides like iron pyrites (FeS_2), zinc blende (ZnS), copper pyrites (CuFeS_2); metallic sulphates such as; gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) and hydrogen sulphides (H_2S) present in natural gas.

Some compounds of sulphur have unusual or unpleasant odours. For example, sulphur compounds are responsible for the smell of garlic, onion and cabbages. The substance that causes tears from the eyes when cutting onions and the smell associated with rotting eggs are both caused by sulphur compounds.

Most of the sulphur is used in the manufacture of sulphuric acid; which is in turn used for hardening of rubber (vulcanisation), making sulphur drugs, insecticides and fungicides, manufacture of matches and fireworks; and in production of calcium hydrogen sulphite for bleaching of pulp.

Activity 18.1: Preparing sulphur in the laboratory

By the end of this activity, you should be able to prepare sulphur from the materials given.

What you need

- 1M sodium thiosulphate solution
- 2M hydrochloric acid solution

- vaseline cream
- measuring cylinder
- beakers
- heat source (stove or Bunsen burner)
- stirring rod
- funnel
- filter paper
- thermometer

What to do

1. Using 100cm^3 measuring cylinder, measure 50cm^3 of 1M sodium thiosulphate solution into 250cm^3 beaker.
2. Add 100cm^3 of 2M hydrochloric acid to the thiosulphate.
3. Shake the beaker gently for the solutions to mix well.
4. Insert the thermometer in the mixture.
5. Warm the mixture gently until the temperature reaches 60°C on the thermometer.
6. Leave the mixture to cool at room temperature.
7. Using the funnel and filter paper, filter the mixture to obtain the sulphur formed.
8. Write the equation for the reaction that takes place.

Activity 18.2: Making fungicide from sulphur formed in 18.1 above, or sulphur obtained from nature

By the end of this activity, you should be able to prepare a fungicide from sulphur from the materials given.

What you need

- vaseline
- saucepan
- stirring rod
- source of heat

- plastic containers (10 ml, 20 ml)
- sulphur powder

What to do

1. Put Vaseline in the saucepan and heat it to melt.
2. Add the sulphur in the ratio of 1:10 (Sulphur: Vaseline) weight by weight (w/w).
3. Use the stirring rod to mix sulphur with the Vaseline oil.
4. After the sulphur has been mixed thoroughly, let the mixture cool to room temperature.
5. Obtain plastic containers of a particular volume (e.g. 10 mL, 20 mL) and pack the fungicide cream.
6. Cost the fungicide that is prepared and market it.

Follow-up activities

- Try to make fungicides by using the sulphur sold in markets and Vaseline.
- Cost the fungicides and market them to obtain income.

Sulphur dioxide

This section highlights the preparation of sulphur dioxide in the laboratory and on a large industrial scale. It discusses the properties of sulphur dioxide and its uses such as bleaching of straws and sponges, fumigation of our houses to kill insects (i.e. black ants), preservation of some liquids (i.e. orange juice). Sulphur dioxide is also used in the manufacture of sulphuric acid. Sulphuric acid is widely utilized to make paints, plastics, detergents (i.e. Omo, Nomi, Aerial, etc.) and fertilizers.

Sulphur dioxide could be obtained on a large scale from burning sulphur that is itself obtained from natural gas, a component of crude petroleum. In Uganda, the discovery of oil in the western part of the country will in the near future become a source of sulphur.

Sulphur dioxide is a major pollutant of atmospheric air. It is toxic, causes respiratory ailments, damages plant life and corrodes metals. The gas is soluble in water forming what is known as 'acid rain'. Acid rain erodes marble and limestone. In industries where this gas is evolved it is recommended to bubble the gas through an alkali to form sulphites and thus control its polluting effect on the environment.

Sulphites are useful as mild oxidising agents but more often they act as reducing agents. They are oxidised to sulphates by substances such as potassium manganate (VII), potassium dichromate (VI), chlorine, bromine, hydrogen peroxide, oxygen and iron (III) chloride.

A mixture of sodium sulphite and sulphur when heated react to form sodium thiosulphate. A solution of a 'fixer' containing sodium thiosulphate is added to developed films to remove unreduced silver ions to form a soluble complex, which is washed away. This prevents unreduced silver ions from becoming reduced and darkening slowly over time. This is useful in photography, which is a very vital practical application.

Activity 18.3: Preparing sulphur dioxide in the laboratory from sodium sulphite and an acid

By the end of this activity, you should be able to prepare sulphur dioxide from the materials given.

What you need

- dilute hydrochloric acid or dilute sulphuric acid
- sodium sulphite crystals
- litmus paper
- thistle funnel
- round bottomed flask
- delivery tubes

- conical flask
- gas jar
- card board

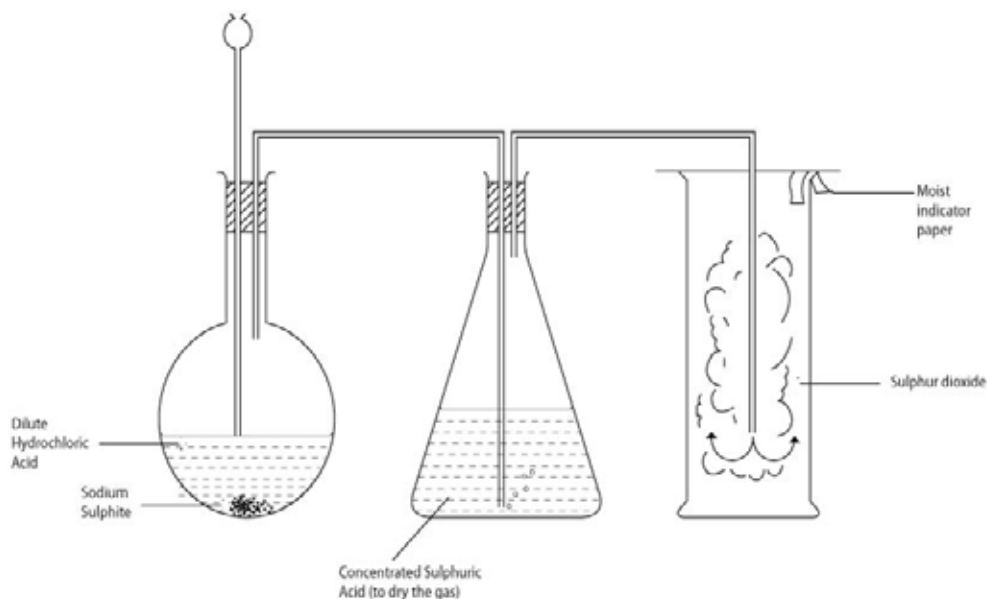


Figure 18.1: Showing preparation of sulphur dioxide

Equation of reaction taking place



Activity 18.3a: Preparing sulphur dioxide in the laboratory by heating copper turning in conc. sulphuric acid

By the end of this activity the learner should be able to prepare sulphur dioxide gas from the materials given.

What you need

- copper turnings
- concentrated sulphuric acid
- source of heat
- litmus paper
- thistle funnel
- round bottomed flask
- delivery tubes
- conical flask
- gas jar
- card board

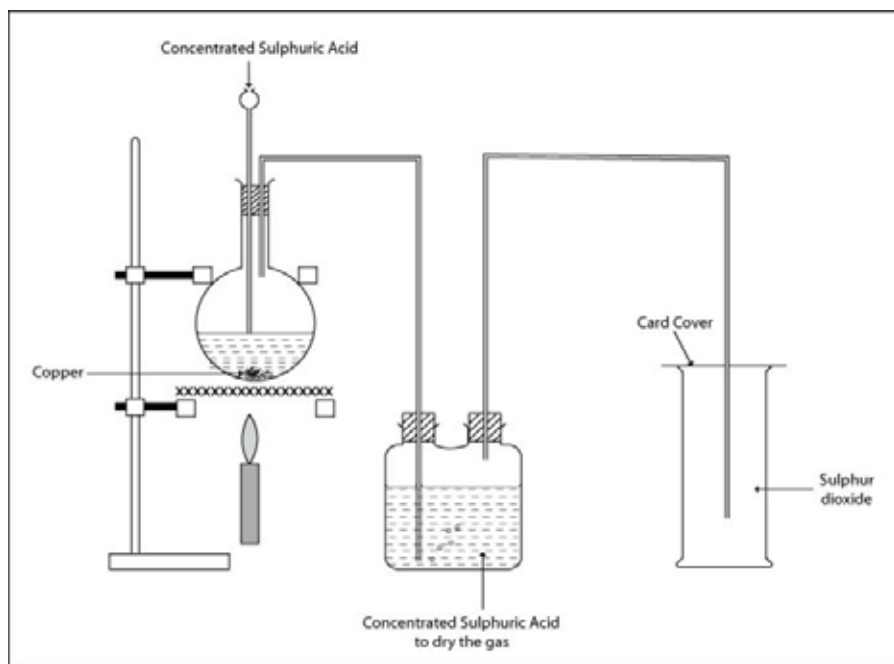
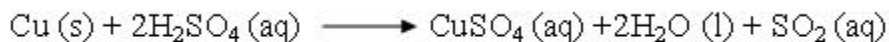


Figure 18.2: Showing an alternative preparation of sulphur dioxide

Equation of reaction taking place



Chapter 19: Chlorine and its Compounds

Inquiry question

- *How useful is chlorine and its compounds to our lives?*

Chlorine is both a useful and dangerous gas. It was used as a poisonous gas in the 1914-18 European war and caused many deaths. It is found in abundance, in the form of sodium chloride dissolved in sea water. There are also vast sources of sodium chloride (rock salt) in the earth's crust. Chlorine itself is used to manufacture a wide range of useful compounds ranging from bleach to plastics, particularly PVC (polyvinyl chloride). Chlorine has a huge variety of uses; as a disinfectant and purifier, in plastics and polymers, solvents, agrochemicals and pharmaceuticals, as well as an intermediate in manufacturing other substances where it is not contained in the final product.

Chlorine also plays a critical role in the productions of thousands of commercial products. Products reliant on chlorine's unique properties include every household item such as bleach and disinfectant to bullet-resistant vests, computer hardware, silicon chips and automotive parts.

This chapter examines the reactions of chlorine and its industrial importance. It also looks at how bleaching agents and disinfectants are prepared from chlorine. Chlorine is used worldwide to purify water supply as the ultimate defense against waterborne microbiological infection.

Activity 19.1: Preparation of chlorine in the laboratory

By the end of this activity, you should be able to prepare chlorine.

What you need

- concentrated hydrochloric acid
- manganese(IV) oxide
- concentrated sulphuric acid
- source of heat
- water
- dropping funnel
- round bottomed flask
- 2 conical flasks
- gas jar
- delivery tubes

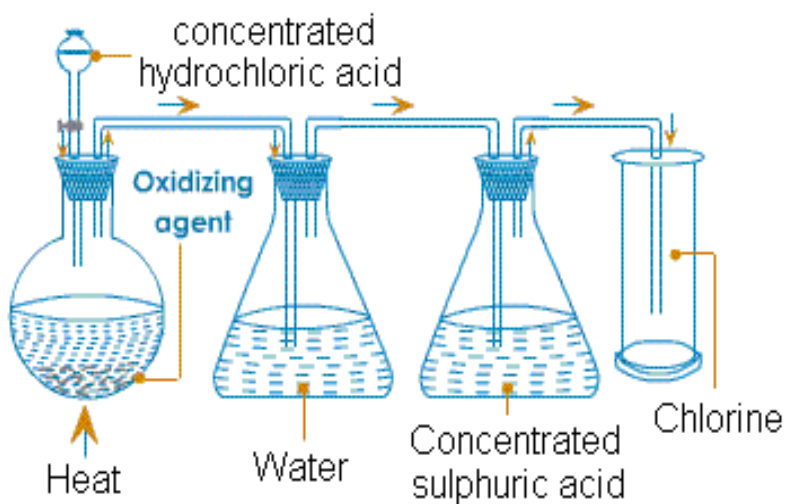


Figure 19.1: Preparation of chlorine from concentrated hydrochloric acid with the help of oxidising agent

What to do

1. Place the oxidising agent in the round bottomed flask.
2. Add the concentrated hydrochloric acid through a thistle funnel onto the oxidising agent.
3. Heat the mixture for some time. Stop heating as soon as the reaction starts.
4. Pass the gas evolved through water in a conical flask, and then through concentrated sulphuric acid. Why do you think the gas was passed through water and the concentrated sulphuric acid?
5. Collect the gas in the gas jar covered with card board paper by downward delivery.

Follow-up activity

Try preparation of chlorine by reacting other oxidising agents apart from manganese(IV) oxide.

Learning points

Chlorine can be prepared by removing the hydrogen from hydrochloric acid using an oxidising agent. Any oxidising agent such as manganese dioxide, lead dioxide, trilead tetroxide, potassium permanganate or potassium dichromate can be used. Firstly, the oxidising agents are taken in the round bottomed flask. Concentrated hydrochloric acid is then added through a thistle funnel. This mixture is then heated. The oxygen of the oxidising agents combines with the hydrogen of the hydrochloric acid leaving behind chlorine i.e. hydrogen is removed from hydrochloric acid. The metallic ions of the oxidising agents combine with part of chlorine to form the respective chlorides.

Activity 19.2: Preparing chlorine disinfectant (sodium hypochlorite), sodium chlorate(I)

By the end of this activity, you should be able to prepare a disinfectant using chlorine.

What you need

- cold sodium hydroxide solution
- concentrated hydrochloric acid
- potassium manganate(vii)
- conc. Sulphuric acid
- sodium, magnesium ribbon
- universal indicator paper
- water
- funnel
- flat-bottomed flask
- delivery bottle
- u-tube
- gas jar
- cardboard cover

What to do

1. Set up the apparatus as shown in the figure below:

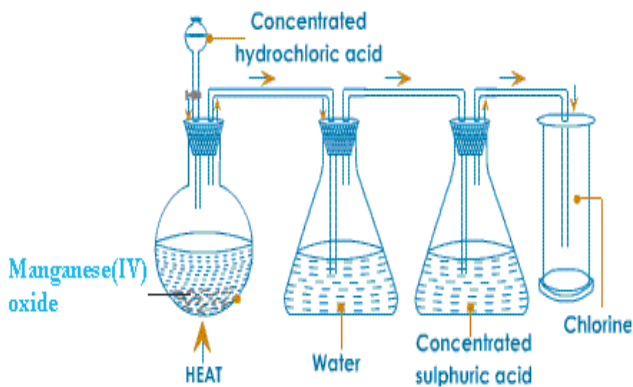


Figure 19.2: Showing preparation of chlorine from manganese(IV) oxide and concentrated hydrochloric acid

2. Pour hot concentrated hydrochloric acid to manganese(IV) oxide in a round-bottomed flask.
3. Add the concentrated hydrochloric acid carefully drop by drop by

adjusting the tap on the funnel. What do you observe?

4. Write the equation for the reaction which took place.

Caution! Chlorine gas is poisonous when inhaled. It is strongly recommended that these experiments be done in a fume cupboard or in the open. Even in a well ventilated room, it can cause problems and the room may have to be evacuated.

Follow-up activities

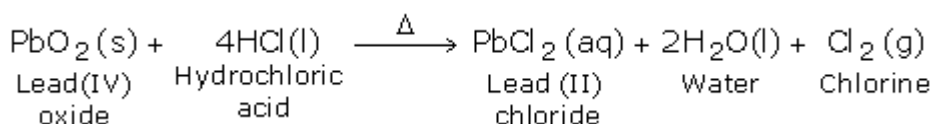
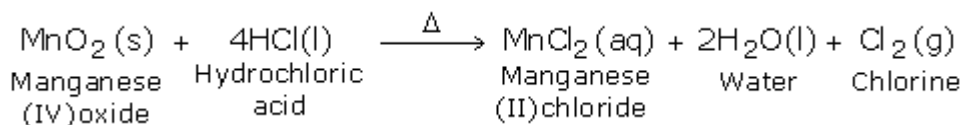
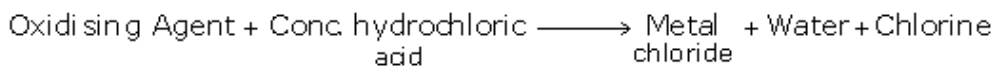
- Make bleaches and disinfectants in groups and then latter individually.
- Package, cost and market the bleaches and disinfectants made.

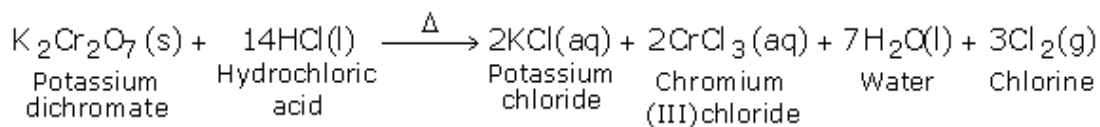
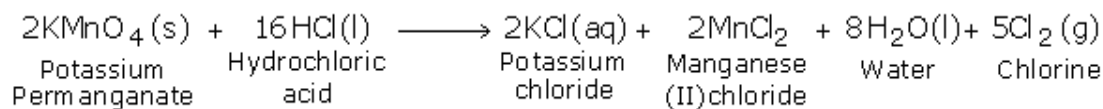
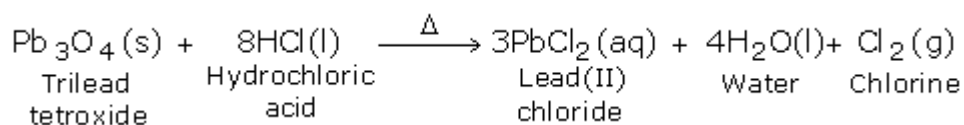
Learning points

Chlorine gas is very soluble in alkalis such as sodium hydroxide. If chlorine is bubbled through dilute sodium hydroxide, the products formed are sodium chloride and sodium chlorate(I) (sodium hypochlorite). The solution is pale yellow which works as a disinfectant because of its germicidal properties.

A bleaching agent, which is the solution of sodium hypochlorite, is manufactured by allowing the chlorine gas released to mix with sodium hydroxide. The bleaching agent is used as bleach for making papers and clothes (cotton).

Equations of reactions





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